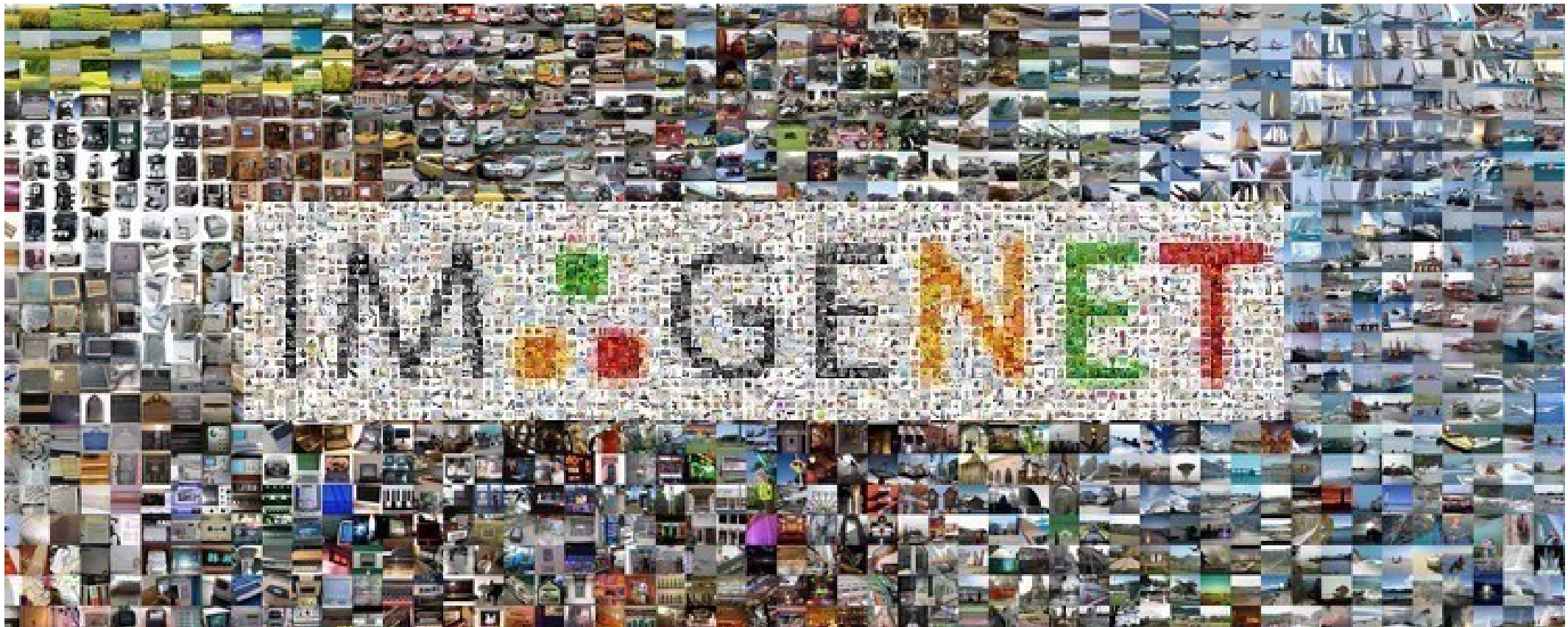


# Lecture 11: Self-Supervised Learning

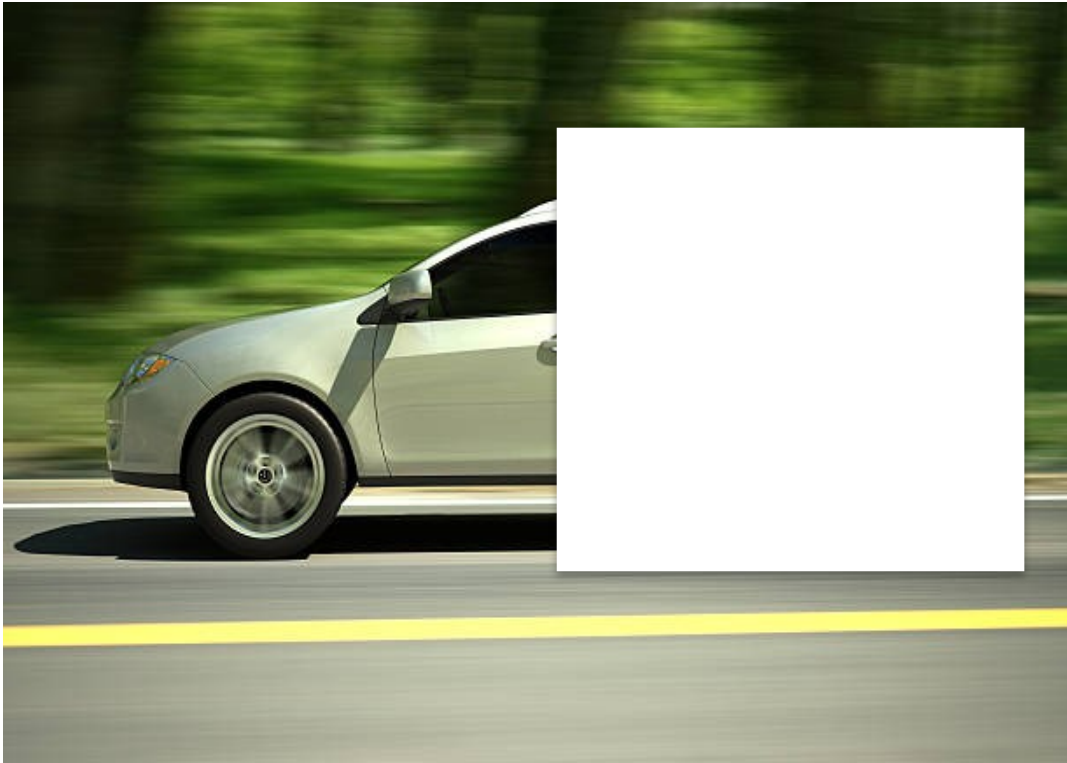


# Is supervised learning sustainable?

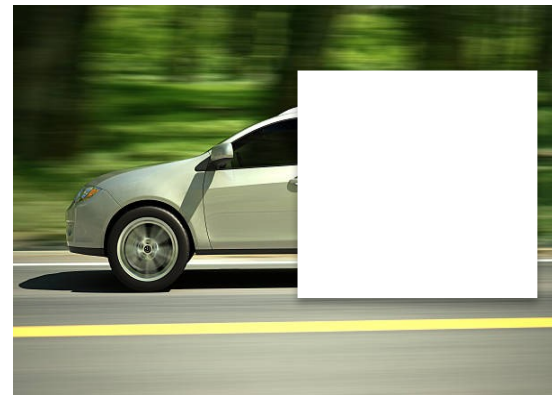
ImageNet: 14M images, **22 human-years** of labeling.



# Context



- What's in the missing area?
- How did you know?
- How much does it cost to get pairs like this?





## How Much Information is the Machine Given during Learning?

### ► “Pure” Reinforcement Learning (**cherry**)

- The machine predicts a scalar reward given once in a while.

### ► **A few bits for some samples**

### ► Supervised Learning (**icing**)

- The machine predicts a category or a few numbers for each input
- Predicting human-supplied data
- **10→10,000 bits per sample**

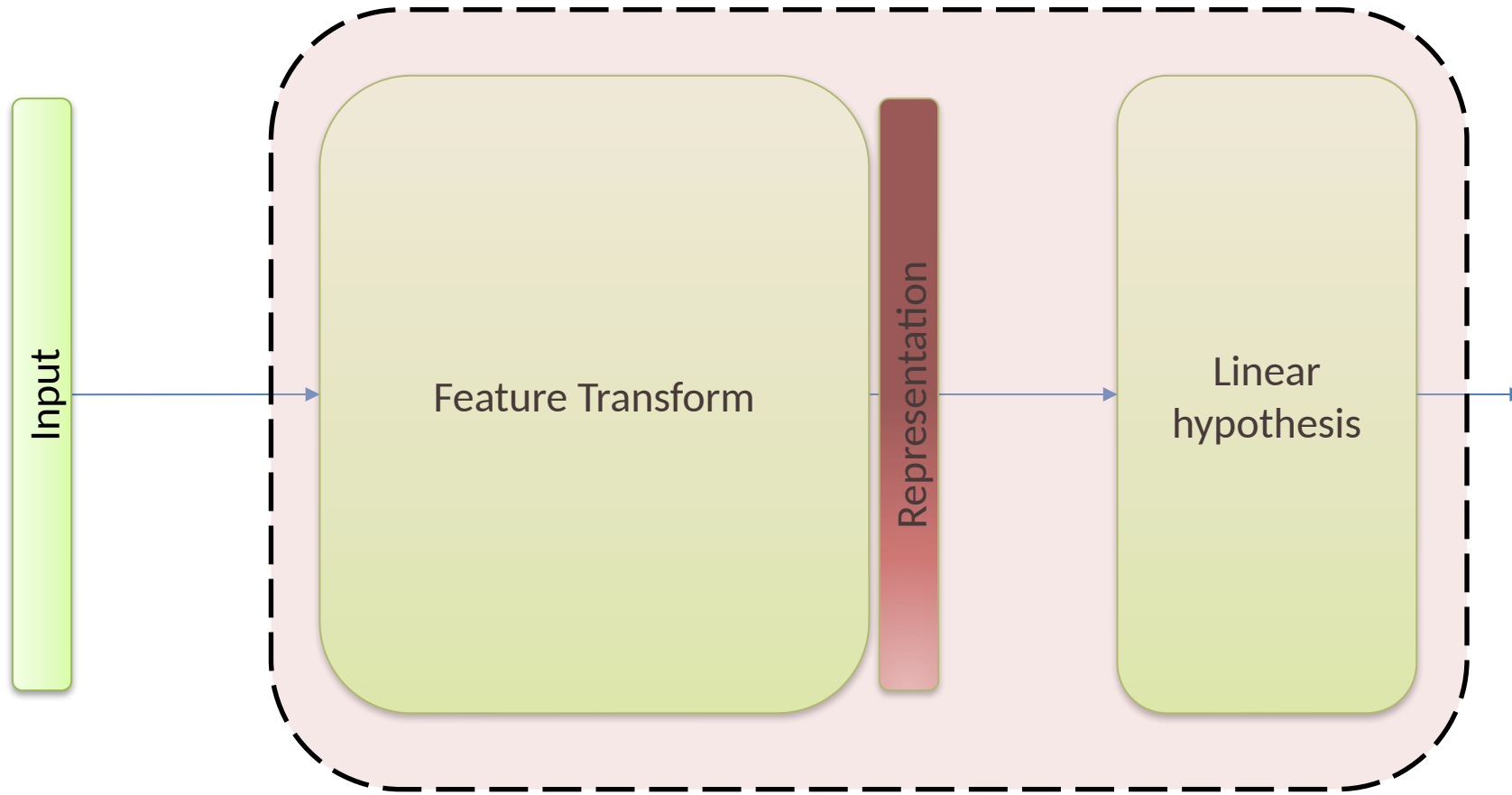
### ► Self-Supervised Learning (**cake génoise**)

- The machine predicts any part of its input for any observed part.
- Predicts future frames in videos
- **Millions of bits per sample**

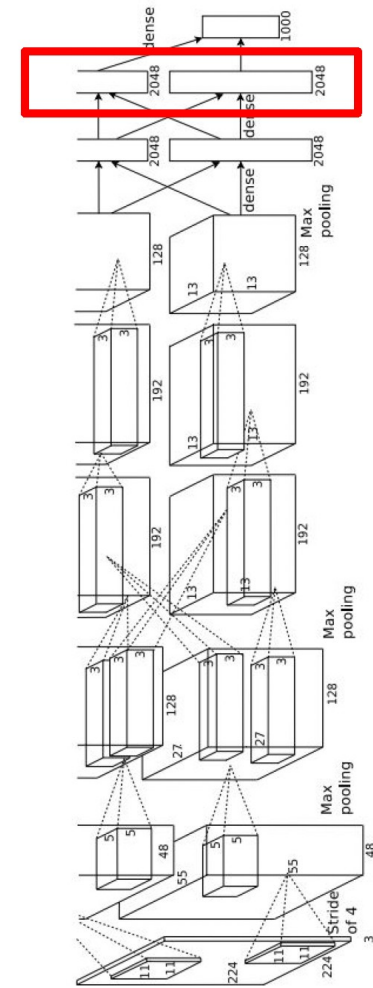




# Representation learning



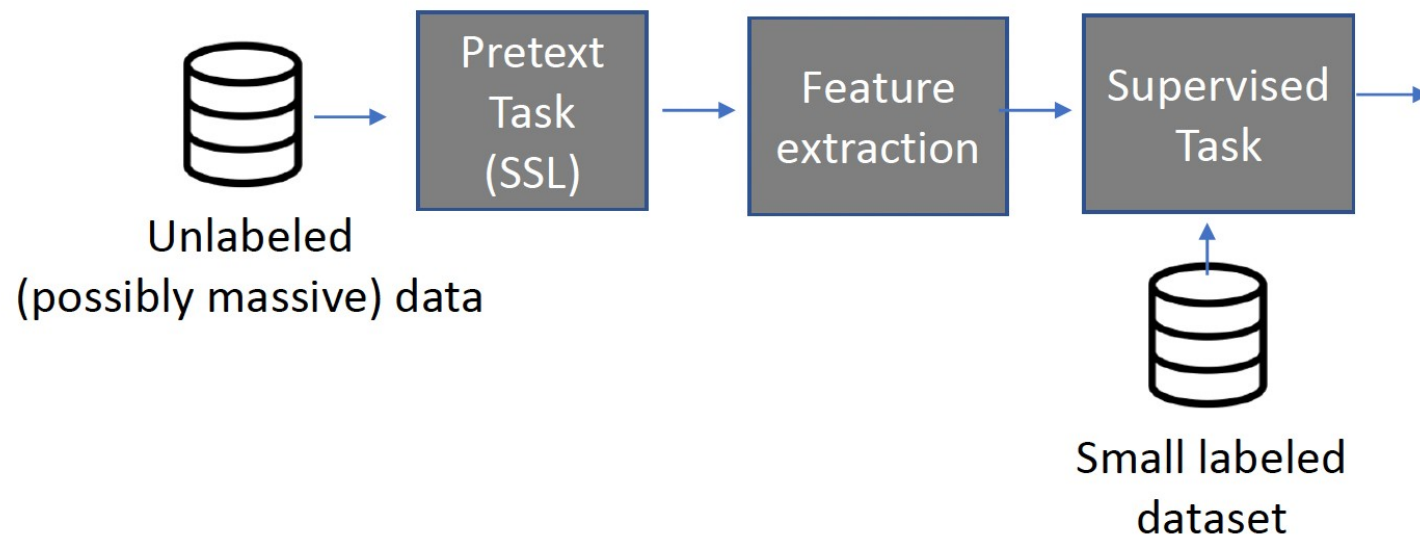
## (Almost) last layer representations



Figures from cs231n-Stanford Fei-Fei, Johnson, Yeung

# The pipeline

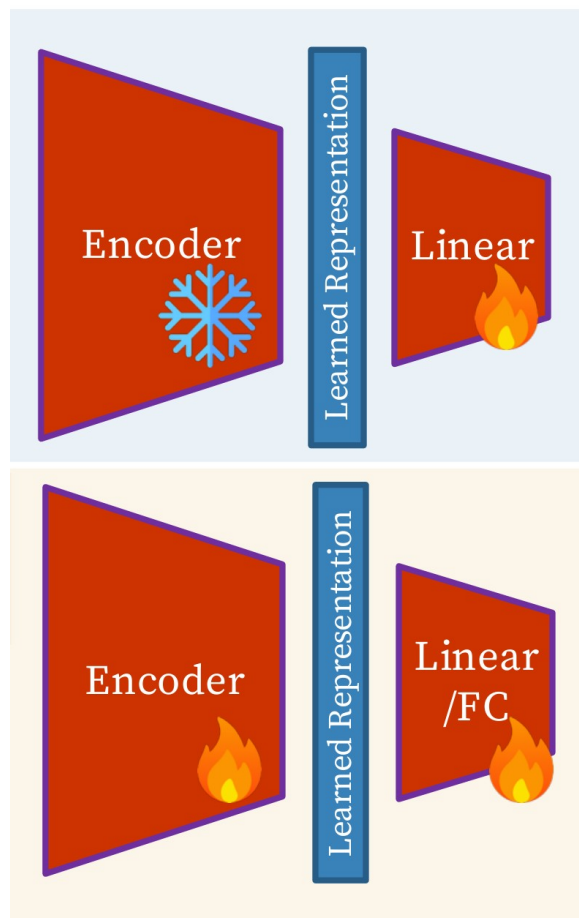
Pretext task on unlabeled data. Keep the encoder, discard the head, transfer.



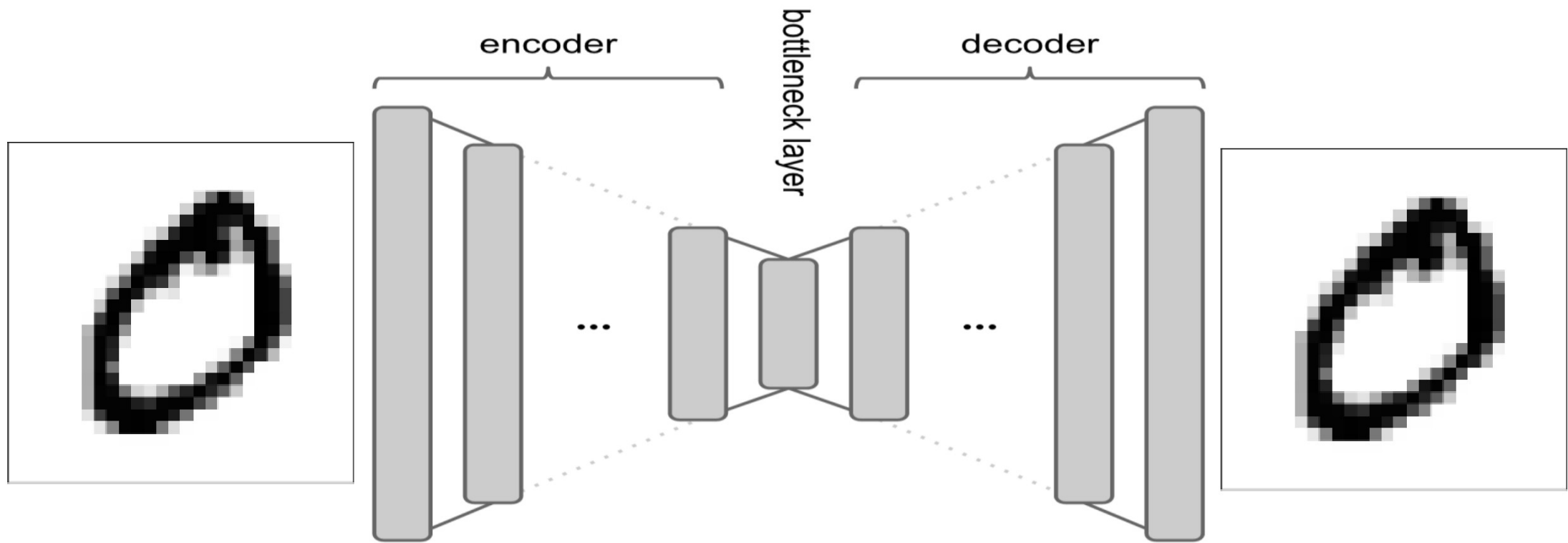


# Evaluation

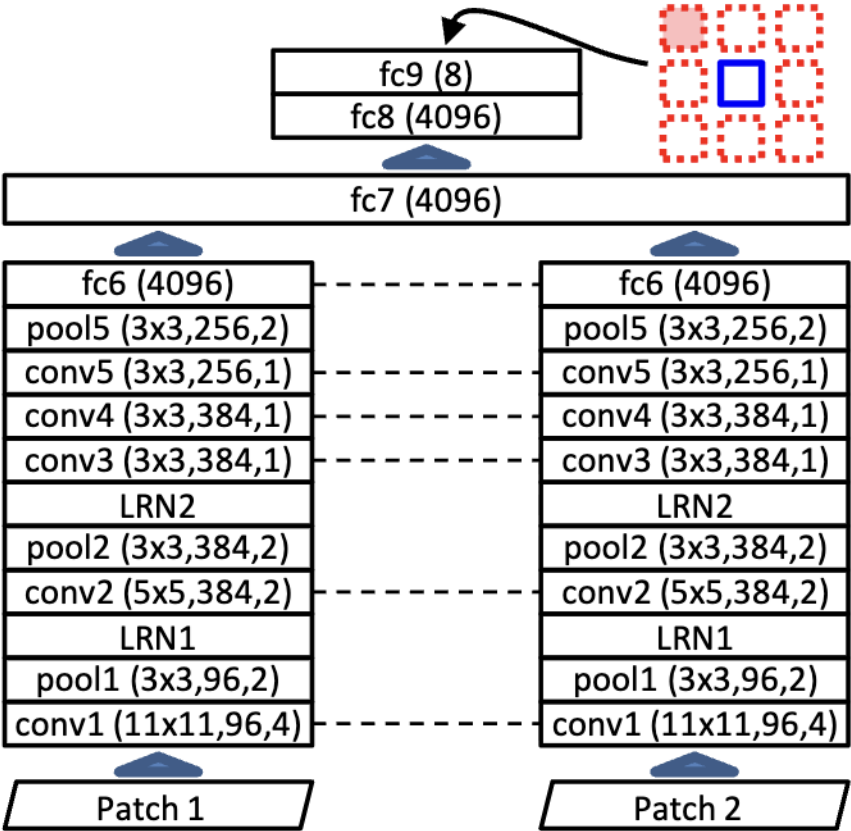
Freeze the encoder and fit a linear probe, or fine-tune everything.



# Autoencoders

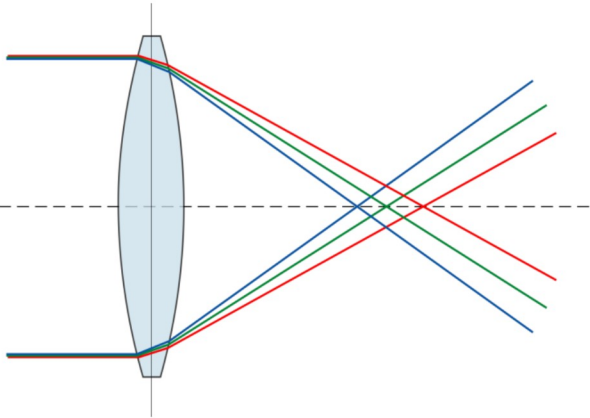
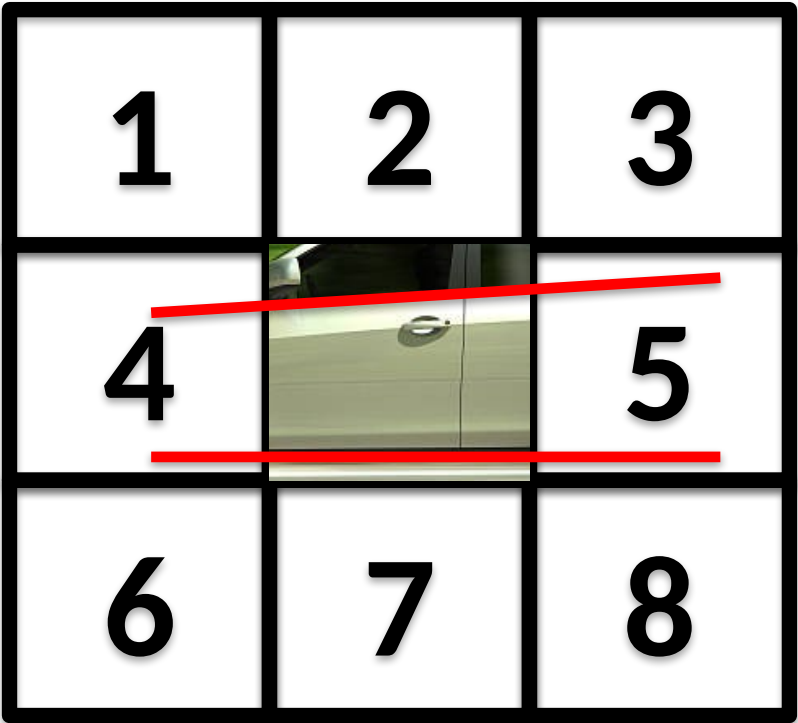


# Context prediction



$$X = \left( \begin{array}{c} \text{Image 1} \\ \text{Image 2} \end{array} \right),$$

$$Y = 5$$



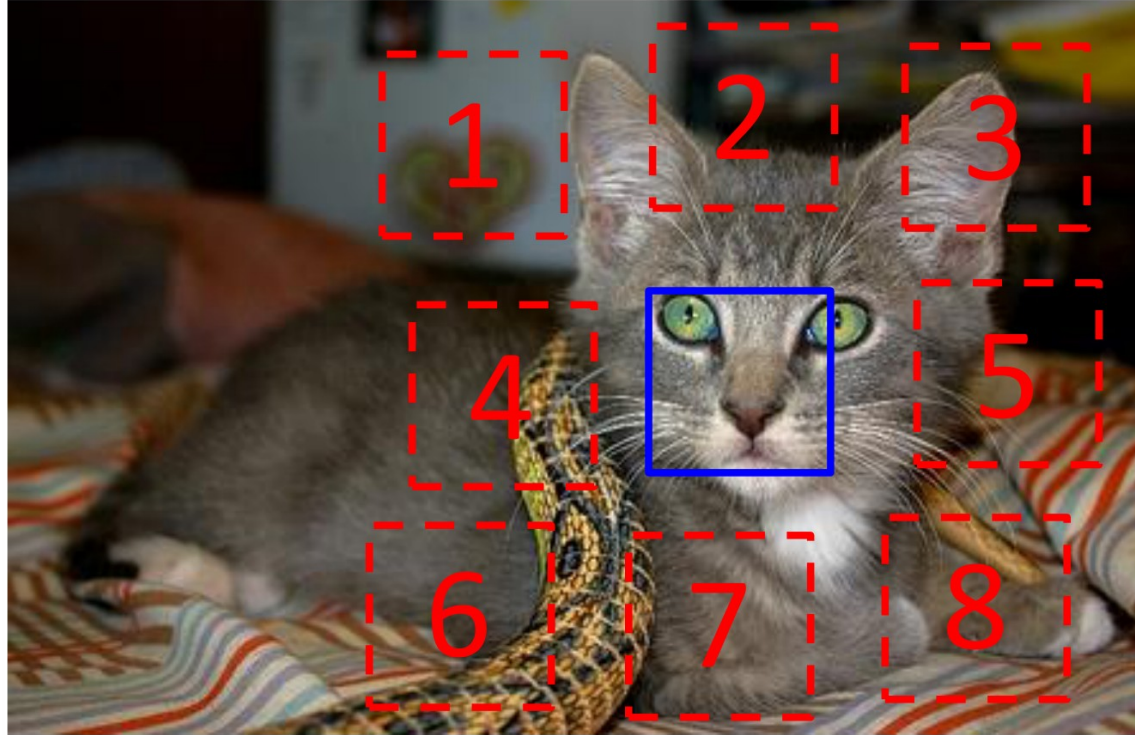
Do we really need perception?

# Shortcuts

Network will typically find the "easiest" solution, and will cheat.

We need to address all possible shortcuts and avoid them.

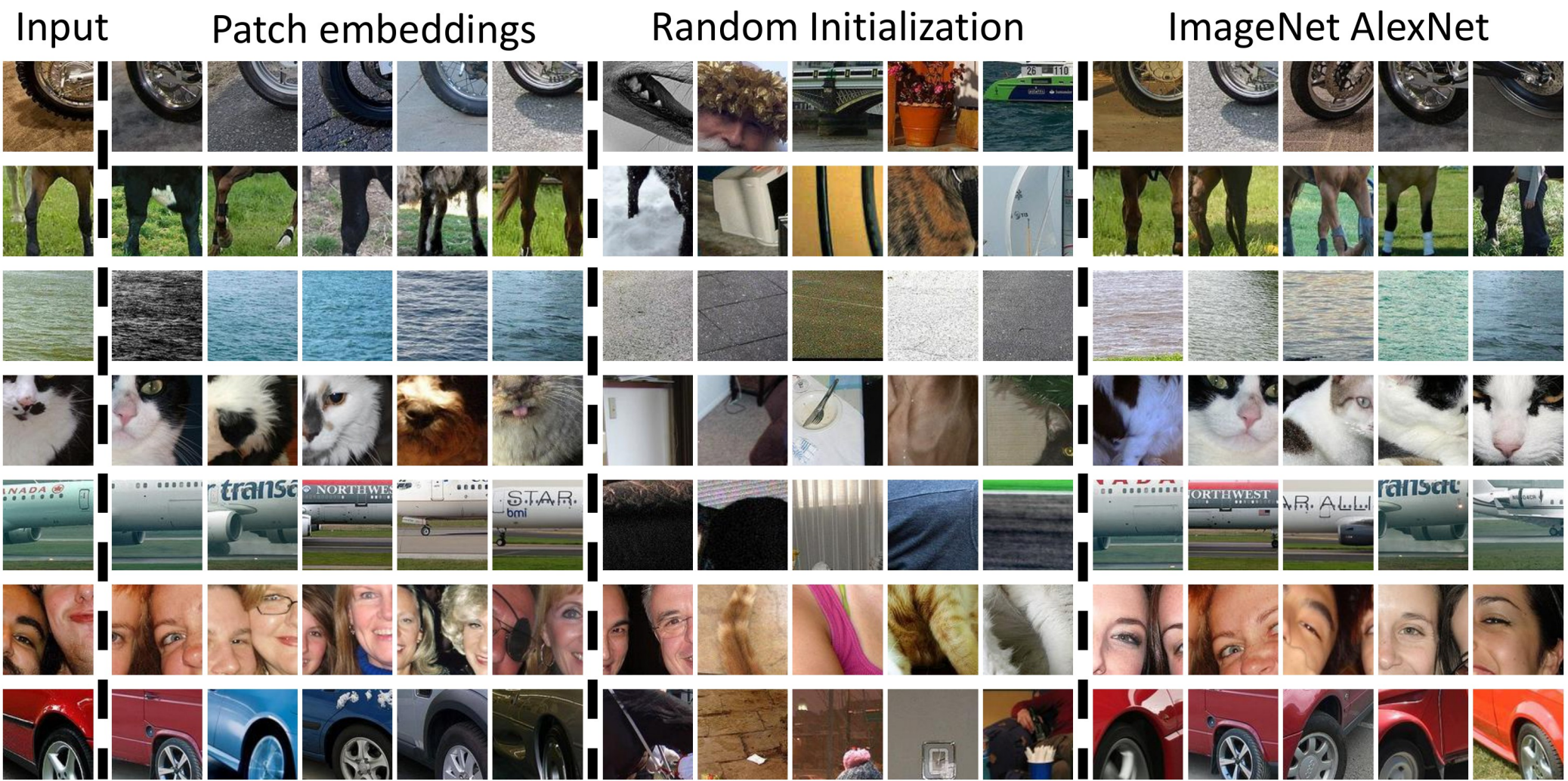
Make it so that the only way to succeed is through gaining features useful for the task.



$$X = (\text{cat face patch}, \text{cat ear patch}); Y = 3$$



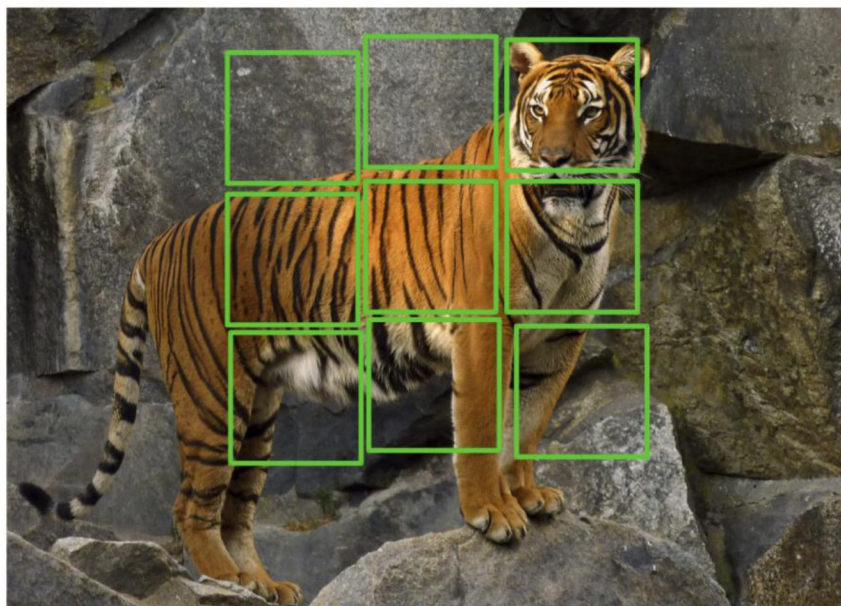
# Did it learn anything?



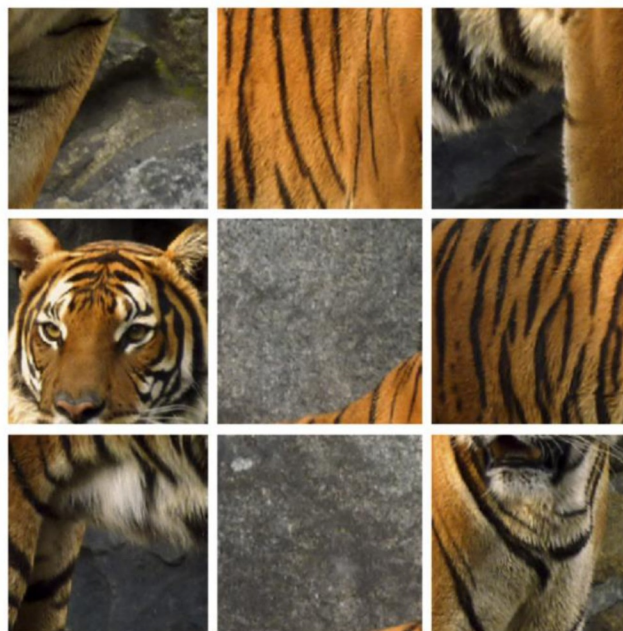


# Jigsaw puzzles

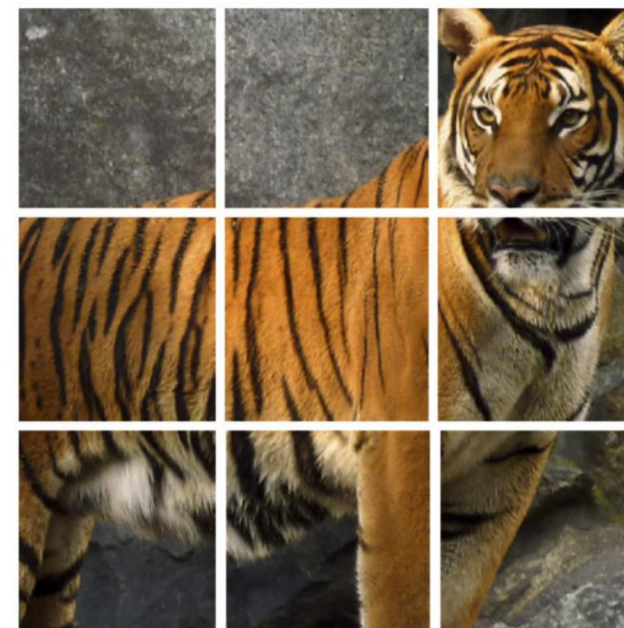
Shuffle 9 tiles, predict the permutation.  $9! = 362,880$  possible classes.



Input Image



Scrambled patches

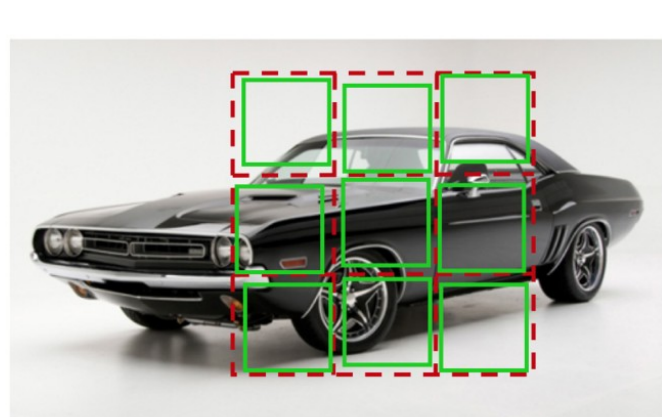


Solved image puzzle

$$9! = 362,880$$

# Jigsaw: the fine print

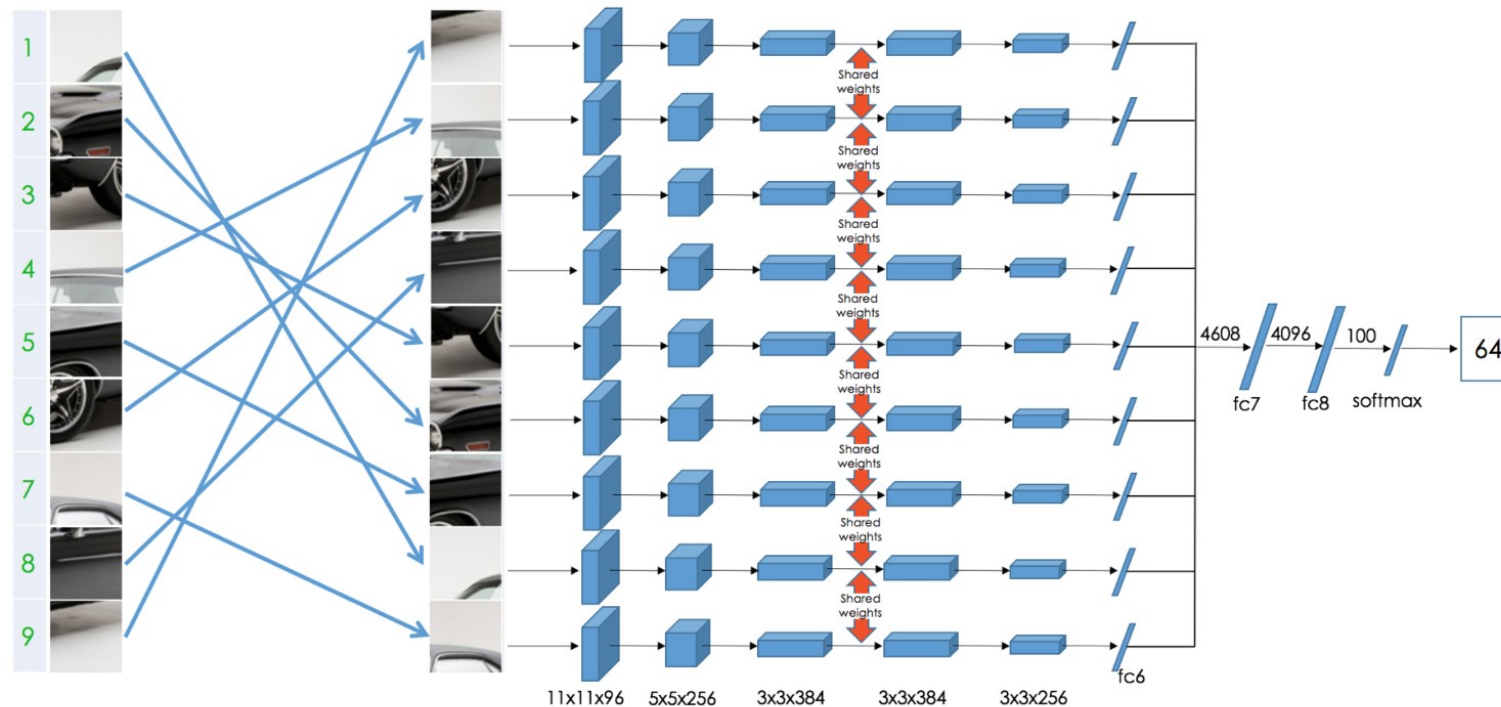
A good pretext task is neither trivial nor ambiguous.



Permutation Set

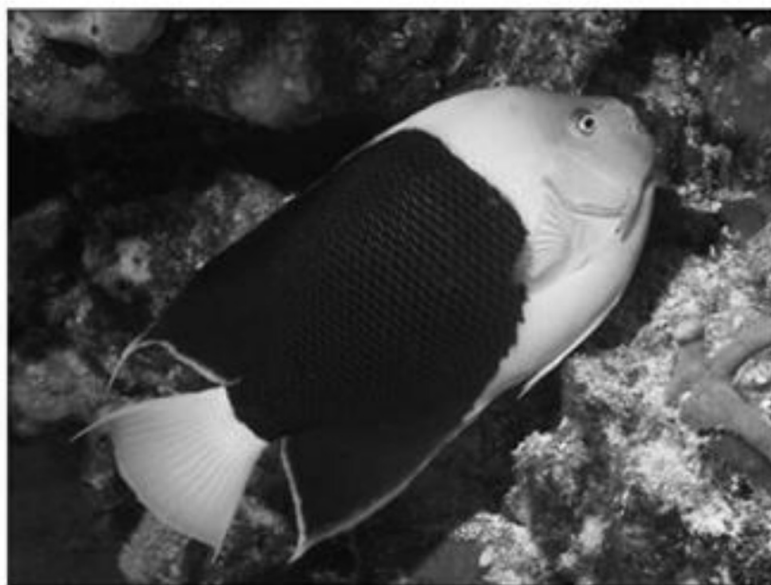
| index | permutation       |
|-------|-------------------|
| 64    | 9,4,6,8,3,2,5,1,7 |
|       |                   |
|       |                   |

Reorder patches according to the selected permutation



# Colorization

The gray world is the input. The colorful world is the label.



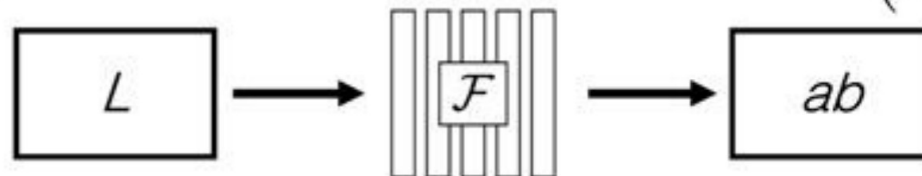
Grayscale image:  $L$  channel

$$\mathbf{X} \in \mathbb{R}^{H \times W \times 1}$$



Concatenate  $(L, ab)$  channels

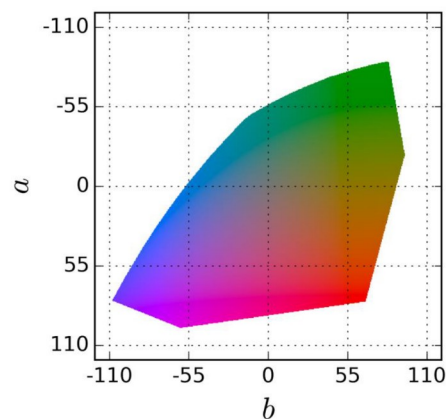
$$(\mathbf{X}, \hat{\mathbf{Y}})$$



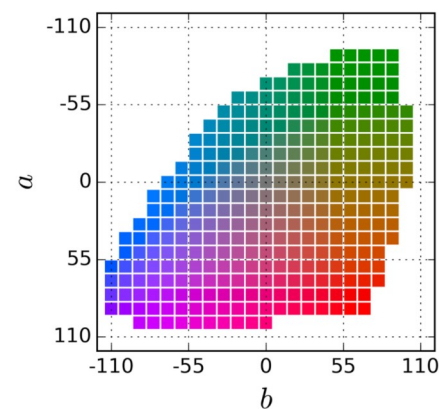


# Color is multimodal

L2 regression predicts the mean of plausible colors: gray mush. Classify quantized bins instead.



Colors in  $ab$  space  
(continuous)



Colors in  $ab$  space  
(discrete)

Ground Truth

L2 Regression

Multimodal class loss



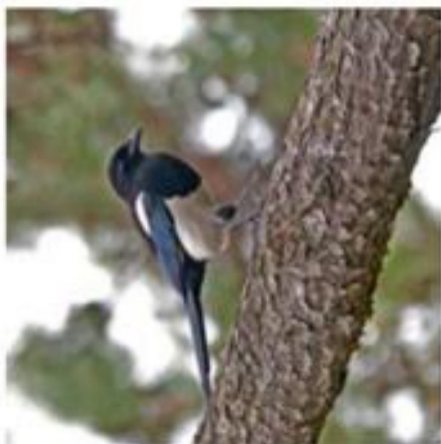
# It works



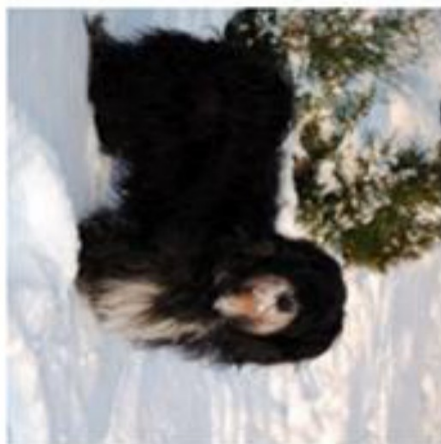


# RotNet

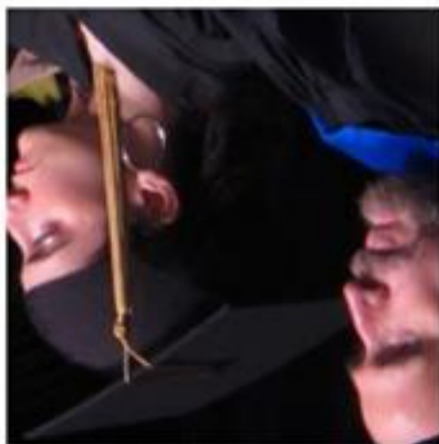
Rotate by 0, 90, 180 or 270. Predict which. That is the whole method.



90° rotation



270° rotation



180° rotation



0° rotation

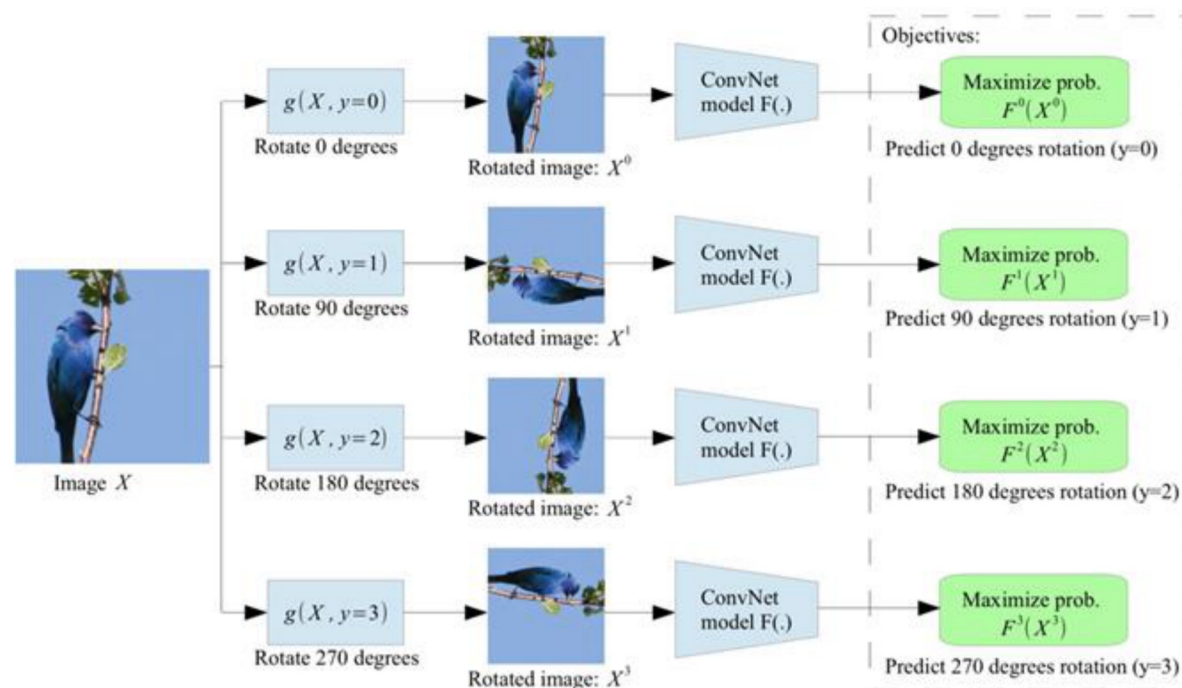


270° rotation



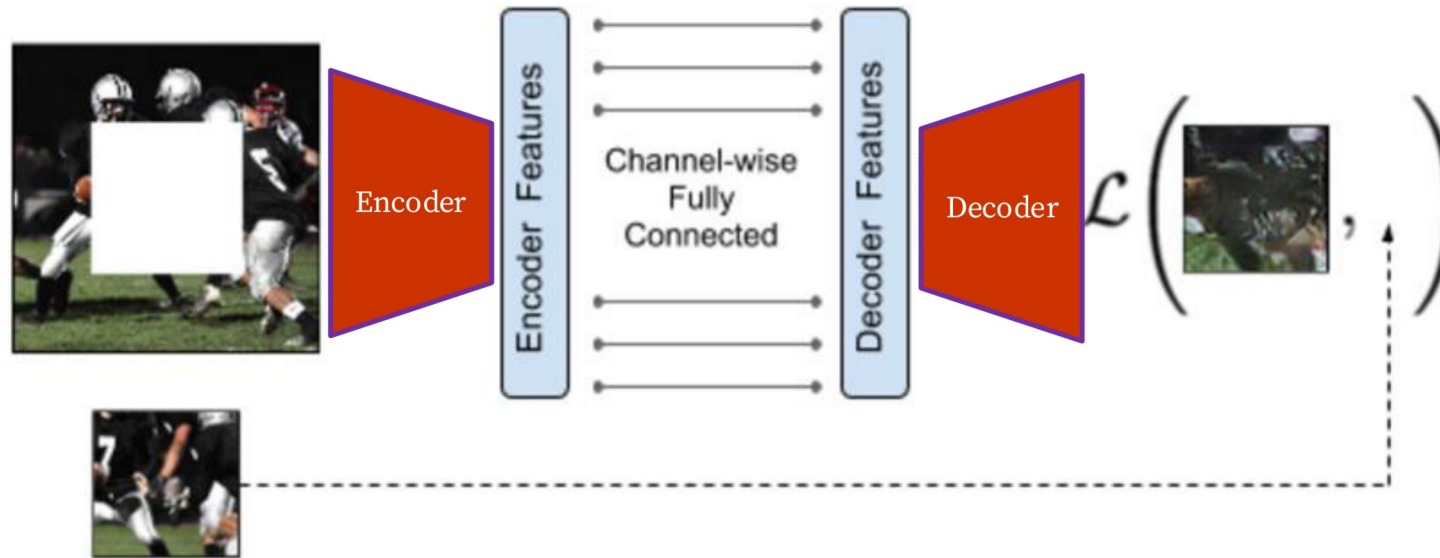
# Why rotation prediction works

To know the flamingo is upside down, you must know which way flamingos point.



# Inpainting: Context Encoders

Mask a region, regress the pixels. An encoder-decoder with a bottleneck in between.



# The loss betrays the task

Many completions are plausible. L2 predicts their mean. Adversarial loss picks a mode.



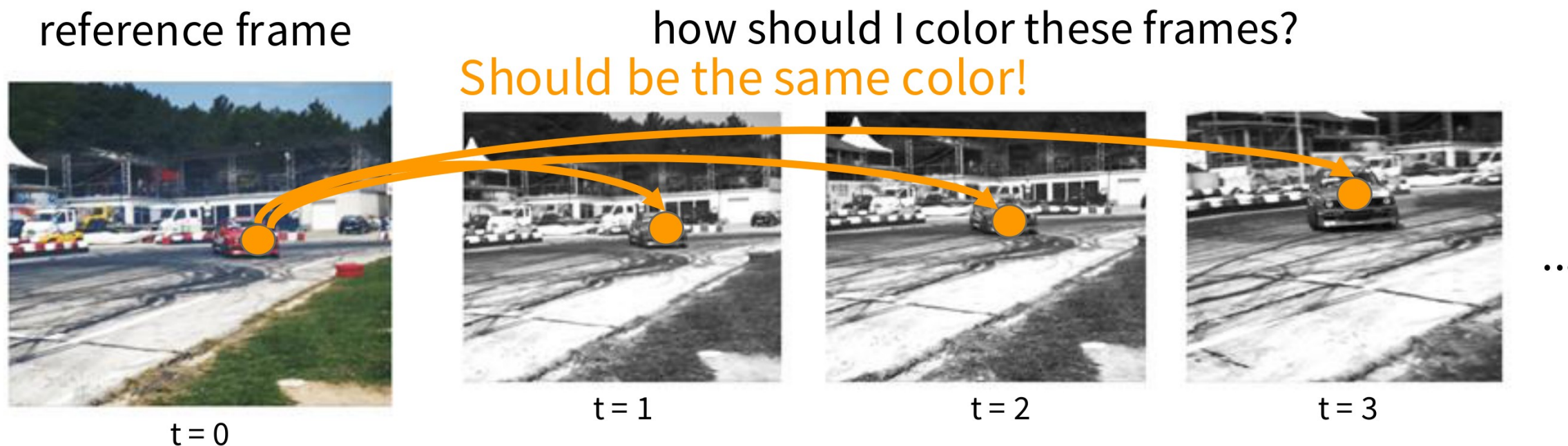
(c) Context Encoder  
( $L2$  loss)



(d) Context Encoder  
( $L2 + \text{Adversarial loss}$ )

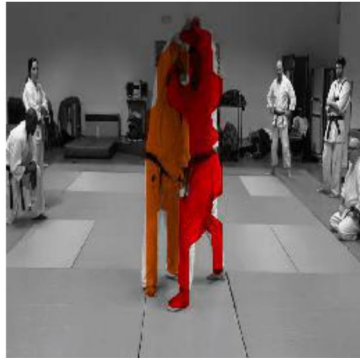
# Video as free supervision

Colorize gray frames by pointing at a reference frame.



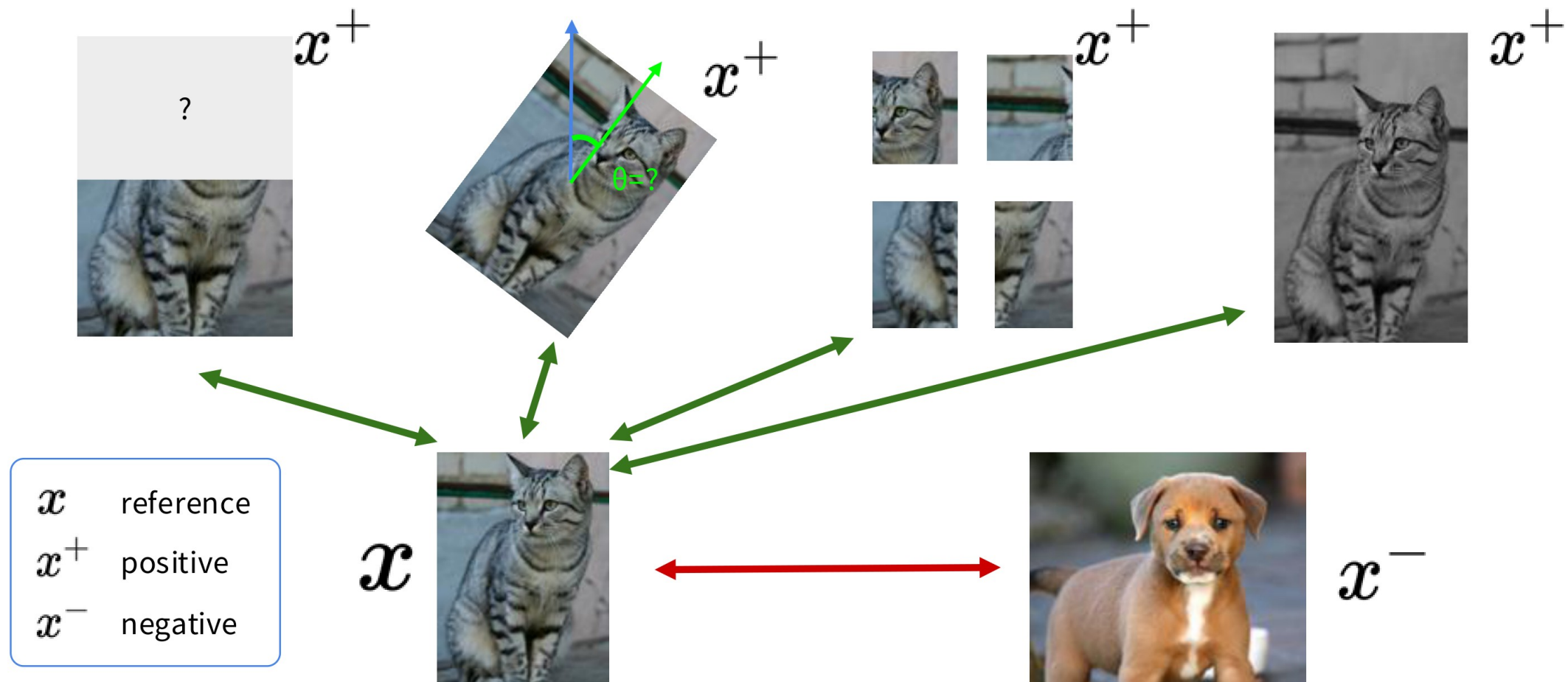


# Tracking emerges



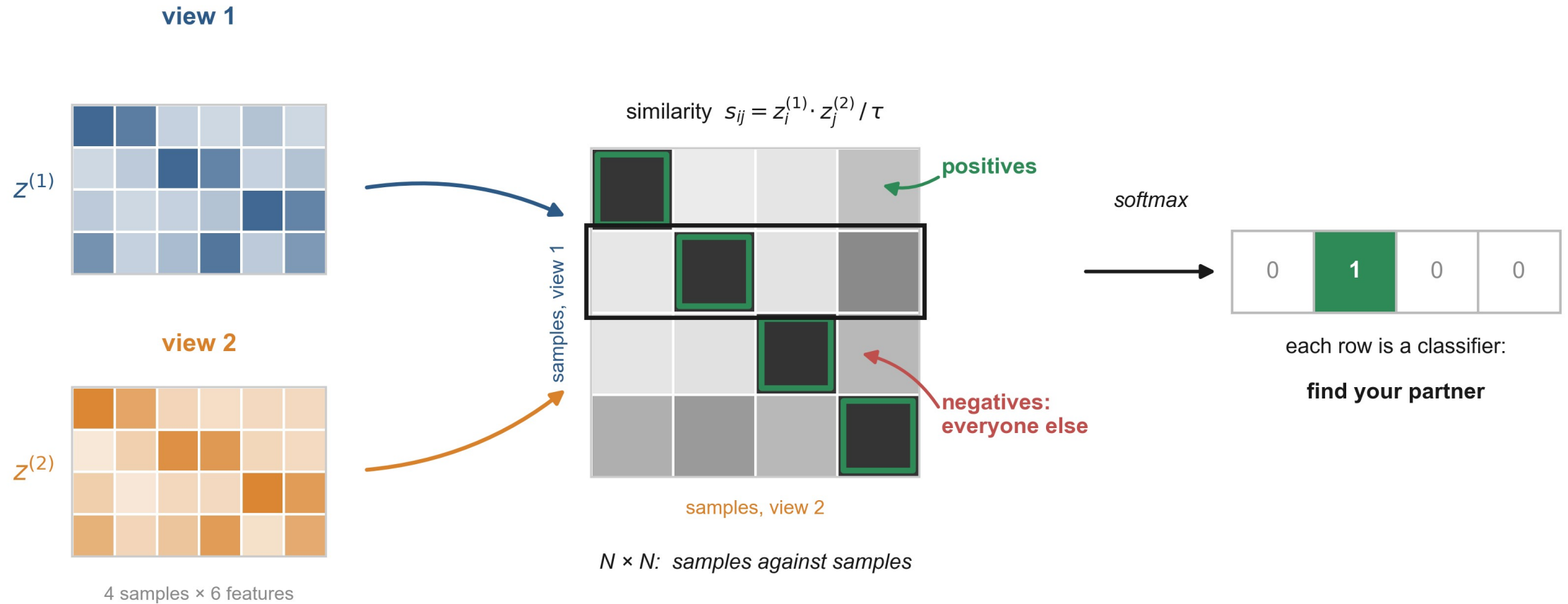
# One idea: invariance

Views of the same image attract. Different images repel.



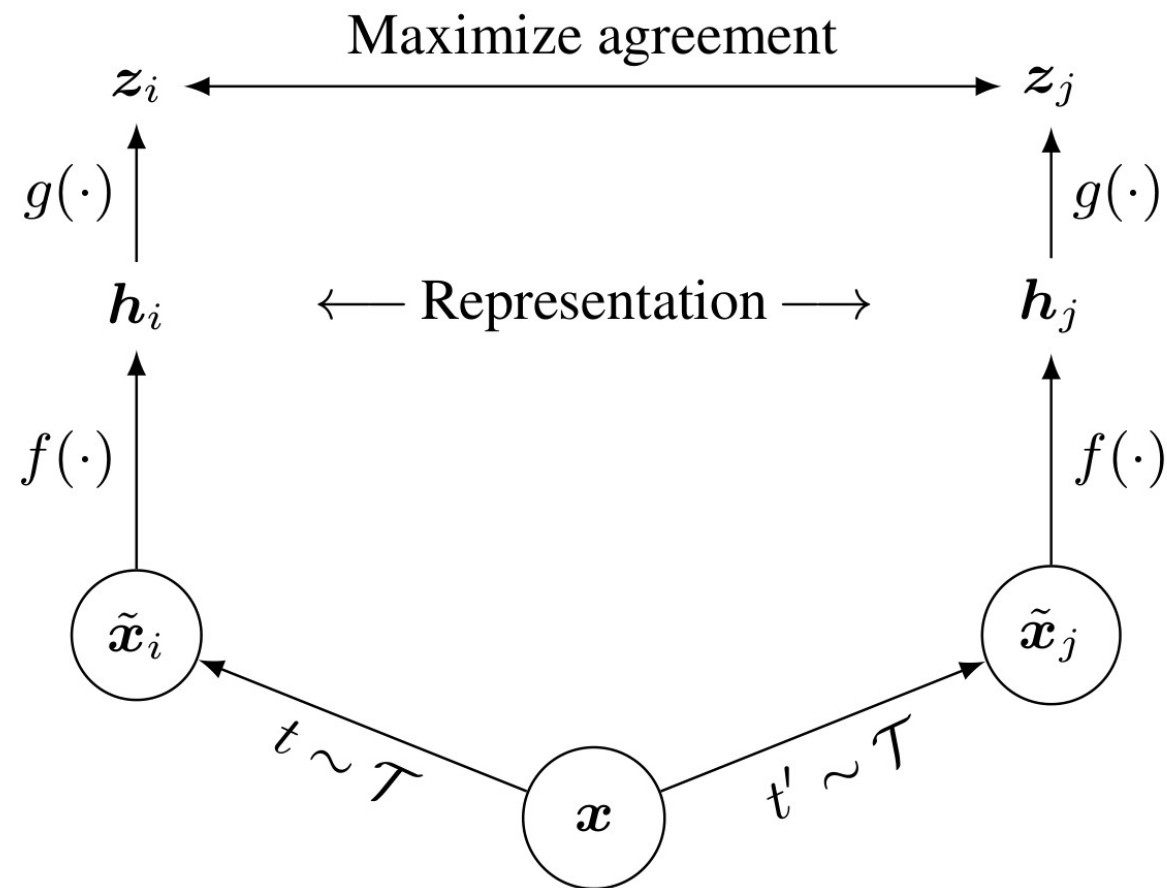
# InfoNCE

$L_i = -\log [ \exp(s_{ii}) / \sum_j \exp(s_{ij}) ]$  : an N-way softmax whose right answer is your own other view.



# SimCLR

Two augmentations, one encoder, a projection head, InfoNCE across the batch.



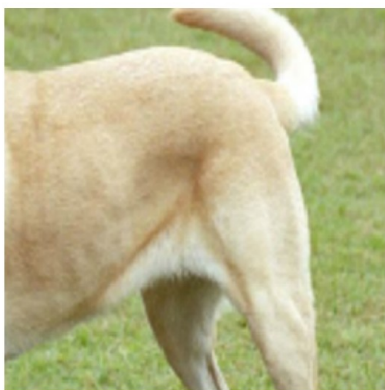


# Augmentations are the task

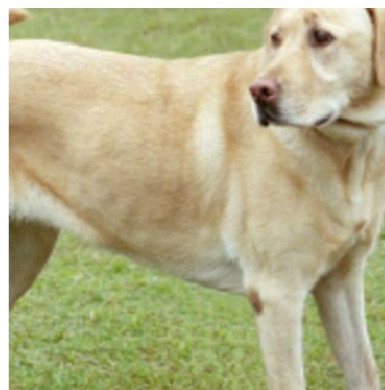
The augmentation set defines what the representation ignores. You are designing the supervision.



(a) Original



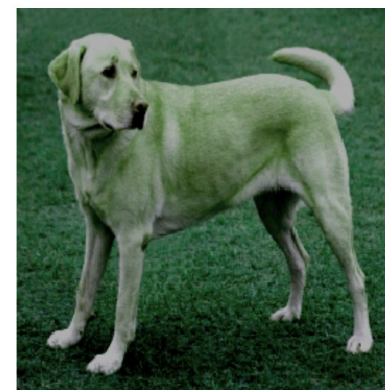
(b) Crop and resize



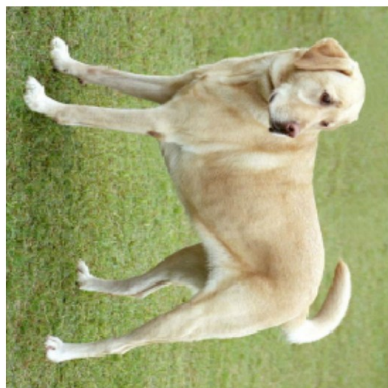
(c) Crop, resize (and flip)



(d) Color distort. (drop)



(e) Color distort. (jitter)



(f) Rotate  $\{90^\circ, 180^\circ, 270^\circ\}$



(g) Cutout



(h) Gaussian noise



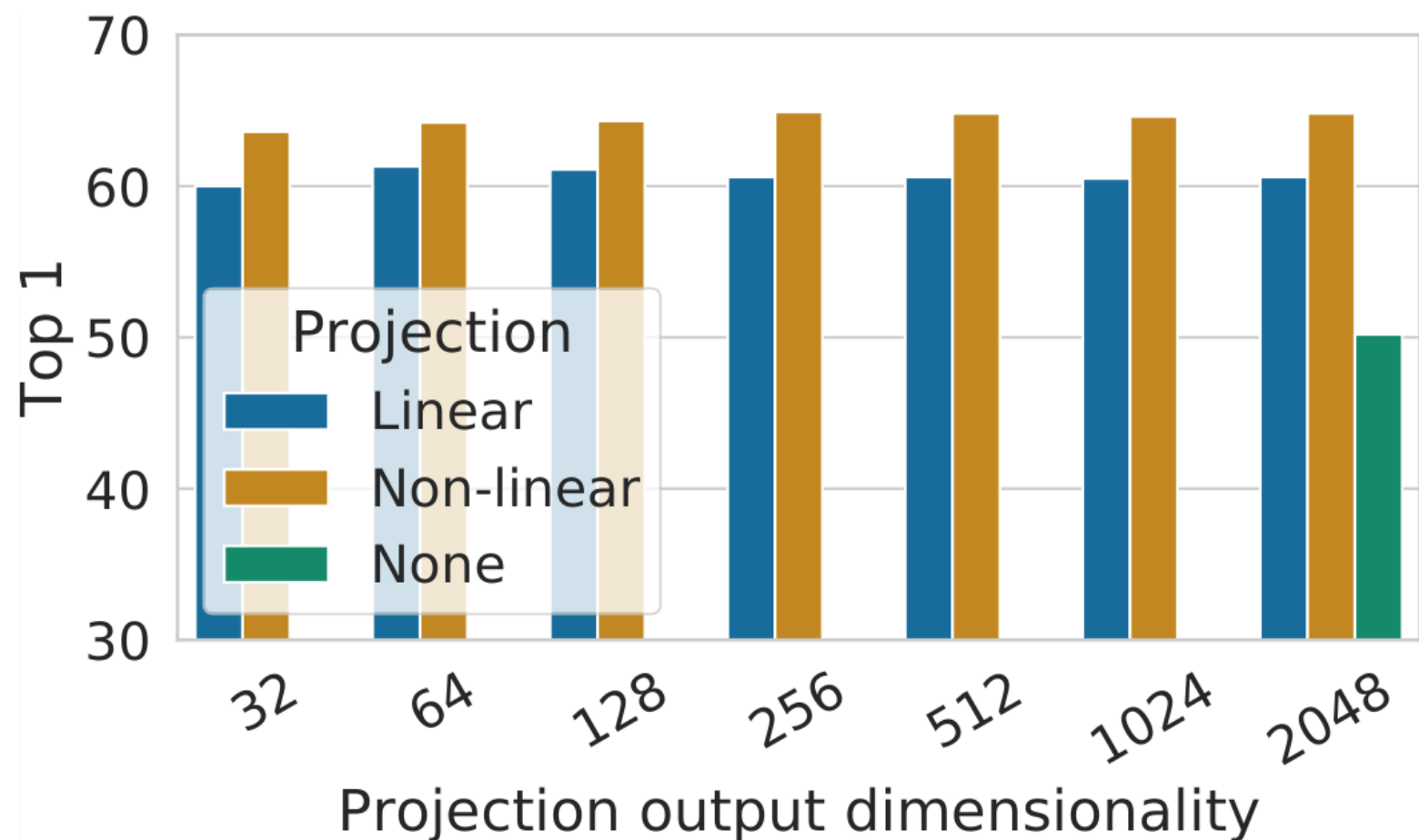
(i) Gaussian blur



(j) Sobel filtering

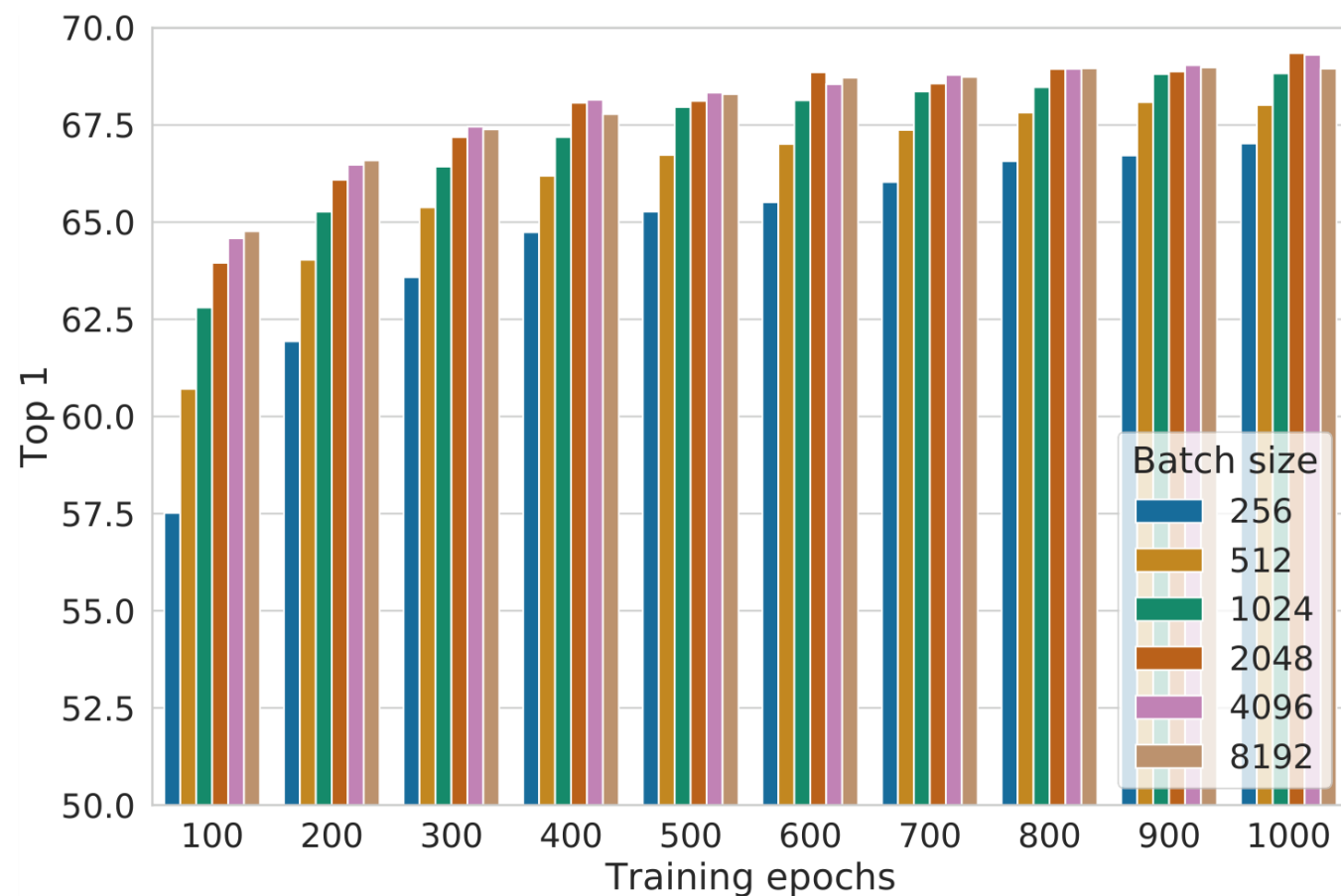
# The projection head trick

Contrast in z-space, transfer h-space. The head absorbs the invariance.



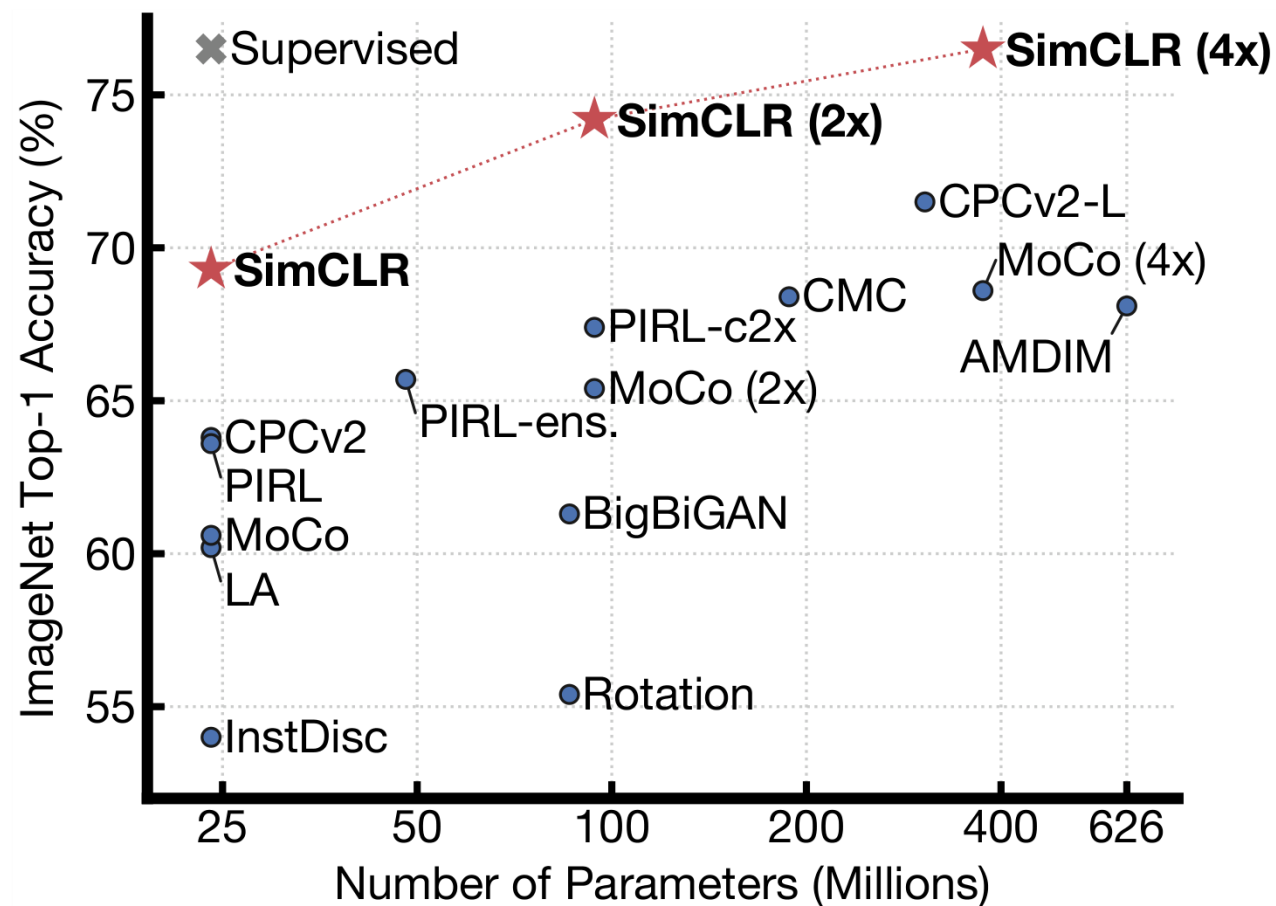
# The catch: batch size

Negatives come from the batch, so the batch must be enormous.



# It works

Linear probes approach supervised accuracy, given enough width.





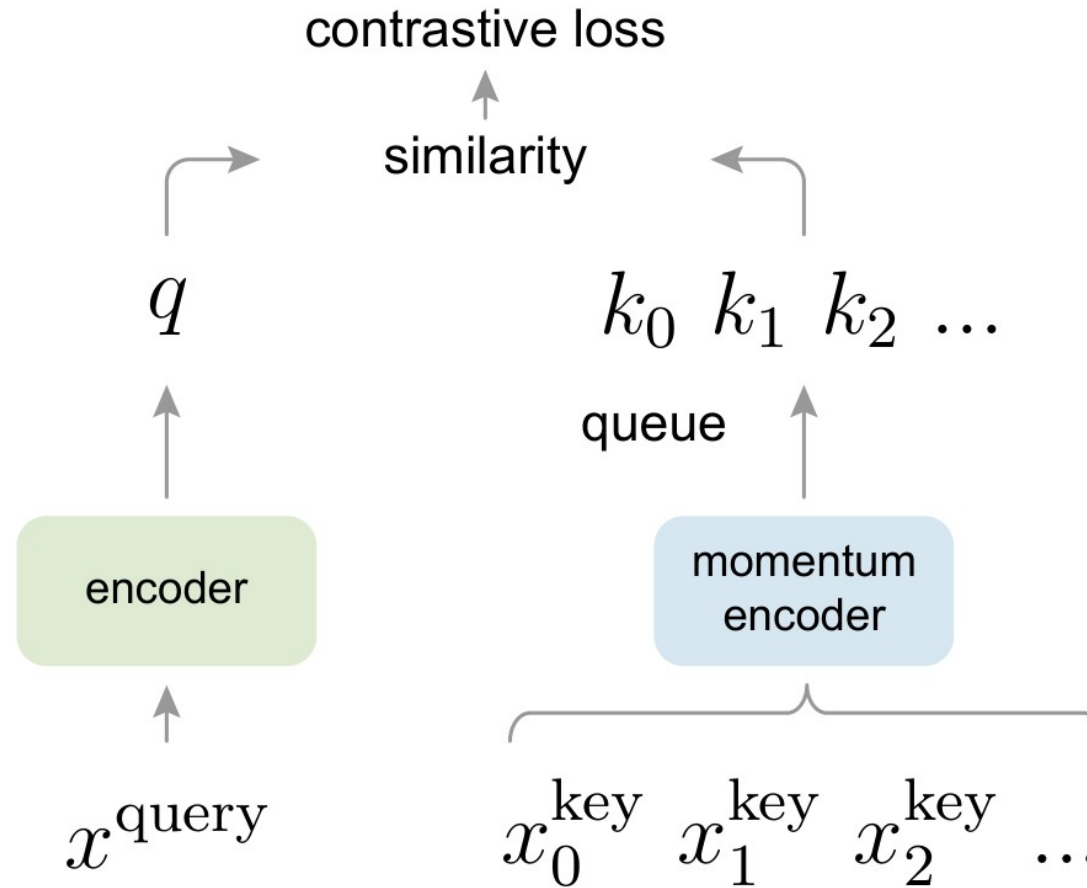
# Few labels, big win

With 1 percent of ImageNet labels, contrastive pretraining dominates.

| Method                                             | Architecture   | Label fraction |             |
|----------------------------------------------------|----------------|----------------|-------------|
|                                                    |                | 1%             | 10%         |
|                                                    |                | Top 5          |             |
| Supervised baseline                                | ResNet-50      | 48.4           | 80.4        |
| <i>Methods using other label-propagation:</i>      |                |                |             |
| Pseudo-label                                       | ResNet-50      | 51.6           | 82.4        |
| VAT+Entropy Min.                                   | ResNet-50      | 47.0           | 83.4        |
| UDA (w. RandAug)                                   | ResNet-50      | -              | 88.5        |
| FixMatch (w. RandAug)                              | ResNet-50      | -              | 89.1        |
| S4L (Rot+VAT+En. M.)                               | ResNet-50 (4×) | -              | 91.2        |
| <i>Methods using representation learning only:</i> |                |                |             |
| InstDisc                                           | ResNet-50      | 39.2           | 77.4        |
| BigBiGAN                                           | RevNet-50 (4×) | 55.2           | 78.8        |
| PIRL                                               | ResNet-50      | 57.2           | 83.8        |
| CPC v2                                             | ResNet-161(*)  | 77.9           | 91.2        |
| SimCLR (ours)                                      | ResNet-50      | 75.5           | 87.8        |
| SimCLR (ours)                                      | ResNet-50 (2×) | 83.0           | 91.2        |
| SimCLR (ours)                                      | ResNet-50 (4×) | <b>85.8</b>    | <b>92.6</b> |

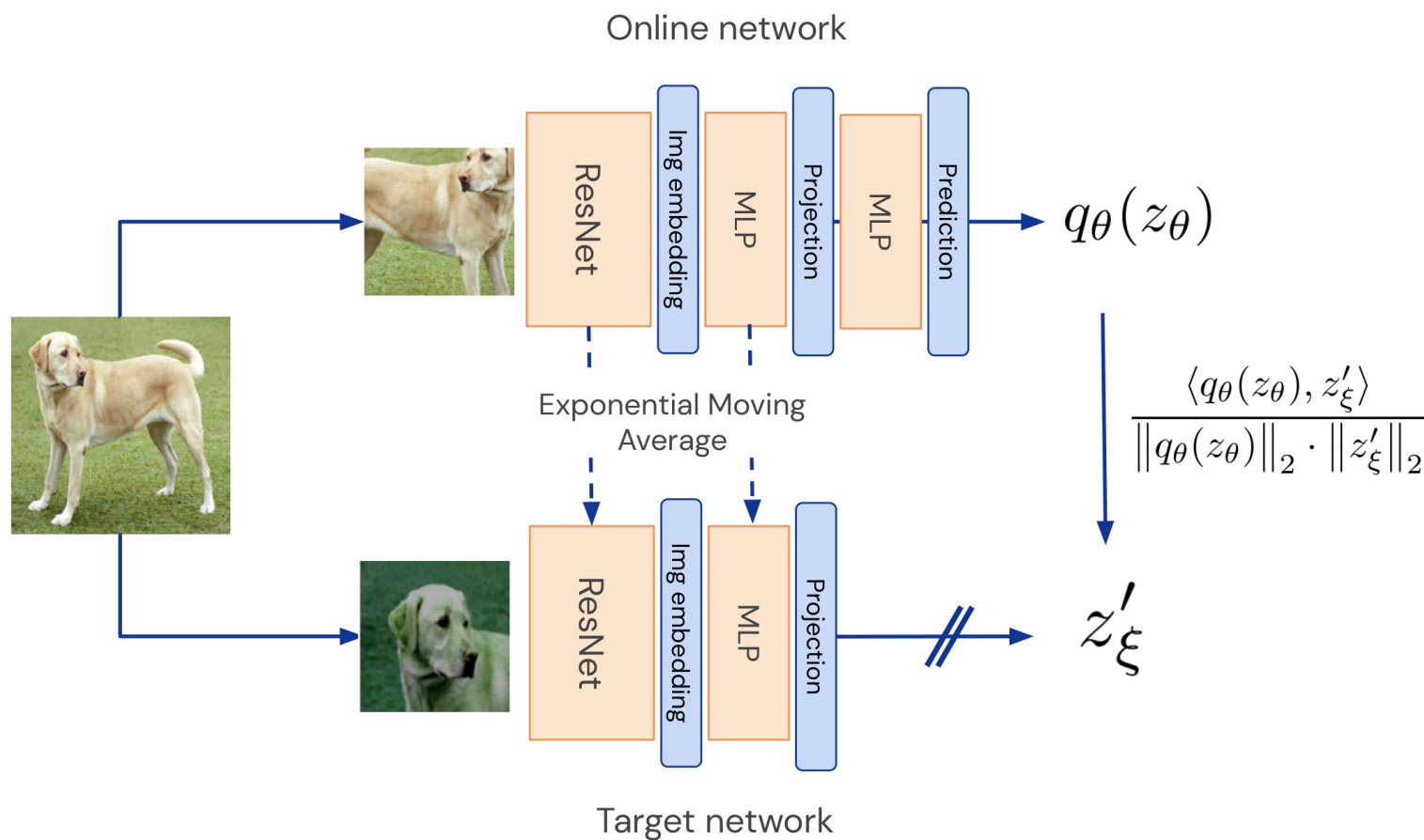
# MoCo: the other answer

Negatives from a queue. The key encoder is a slow momentum copy.



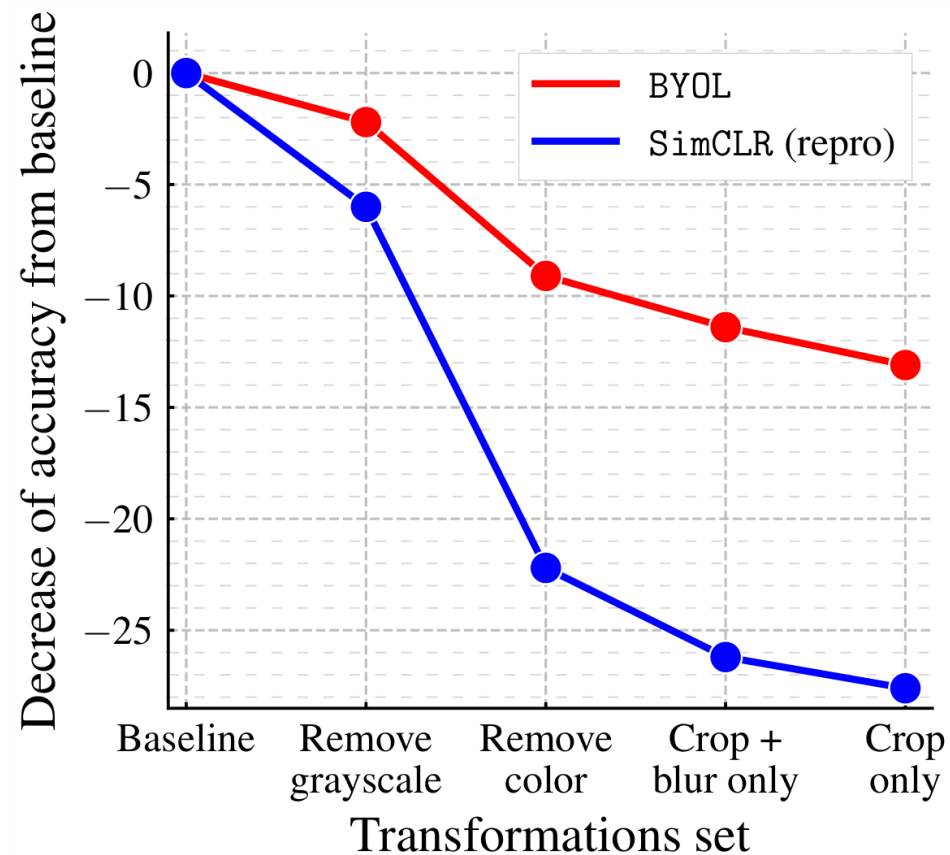
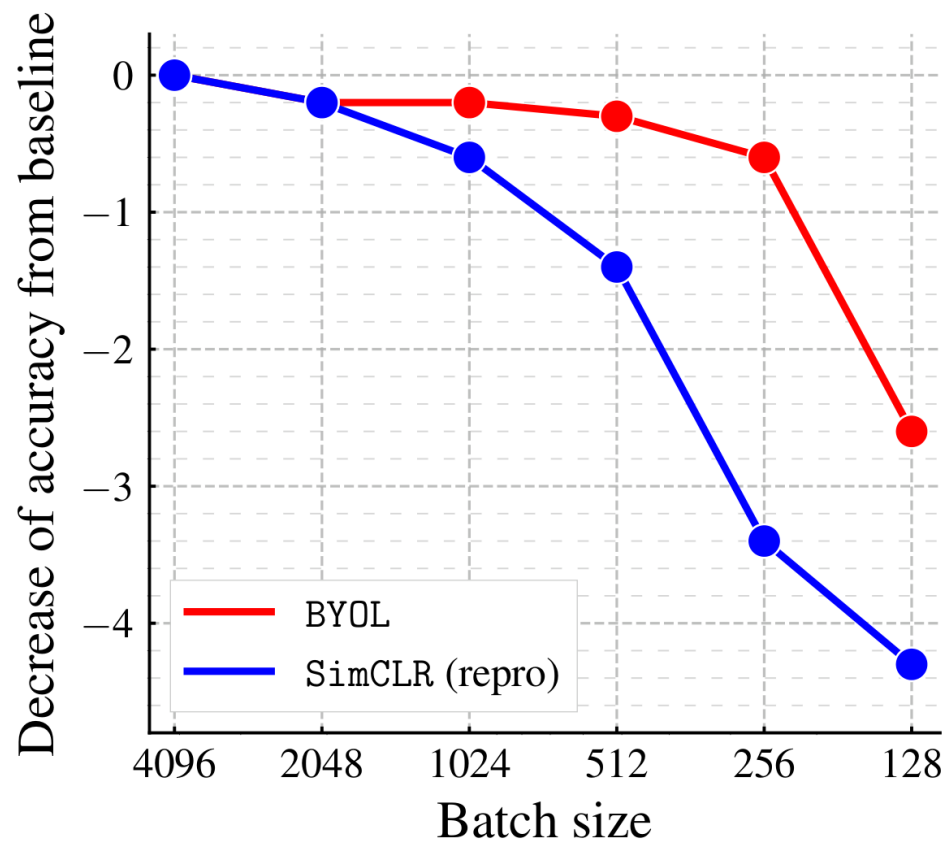
# The three amulets

EMA target, predictor on the online side only, stop-gradient. Remove one and it collapses.



# Bonus: robustness

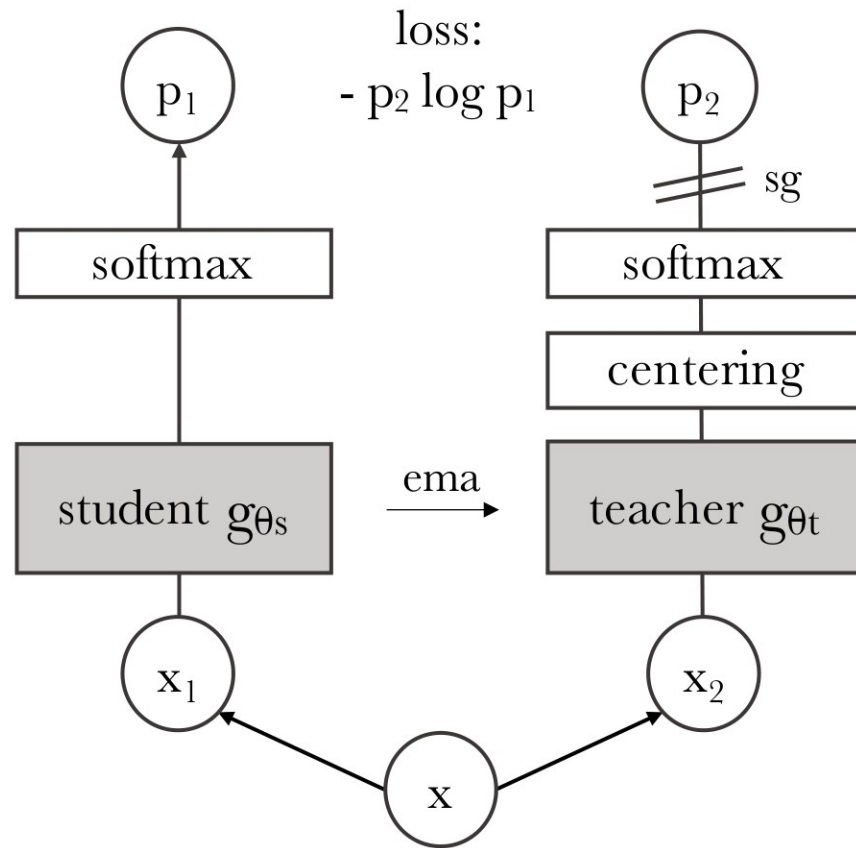
No negatives: small batches work, and the augmentation recipe matters less.





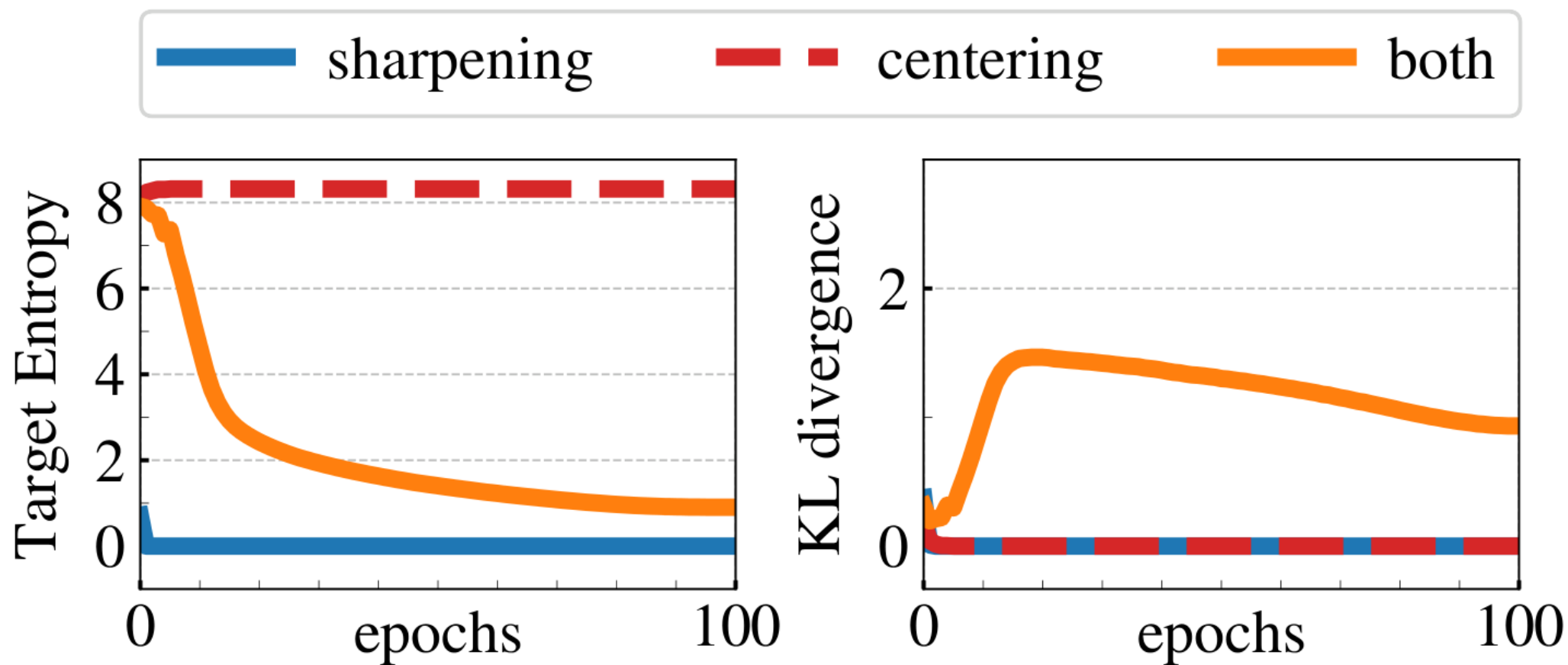
# DINO

Self-distillation: student matches the teacher's soft prototypes. Teacher is an EMA of the student.



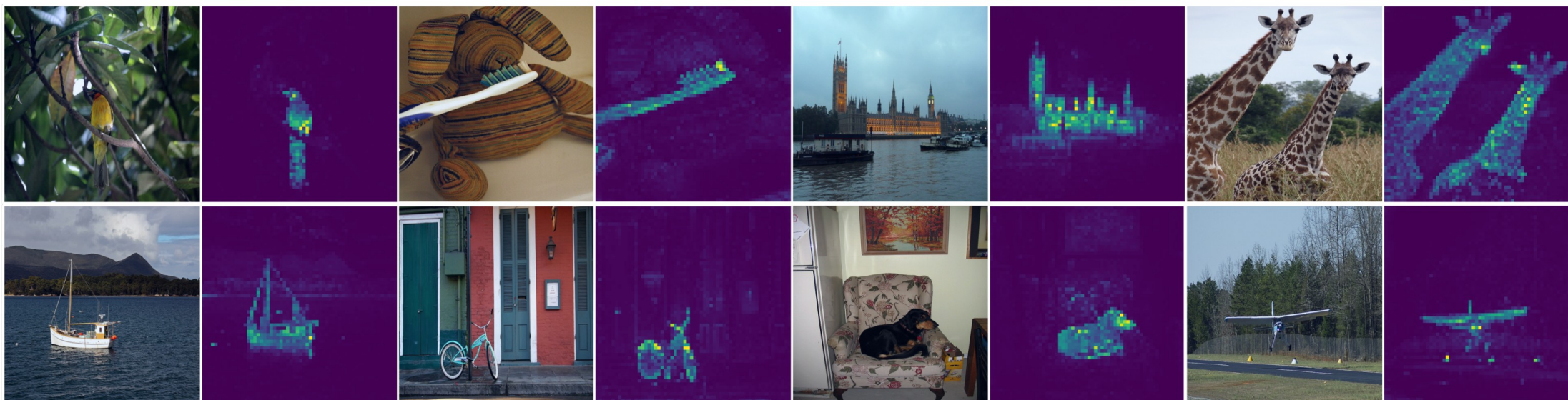
# DINO: two knobs against collapse

Centering fights one-hot collapse. Sharpening fights uniform collapse.



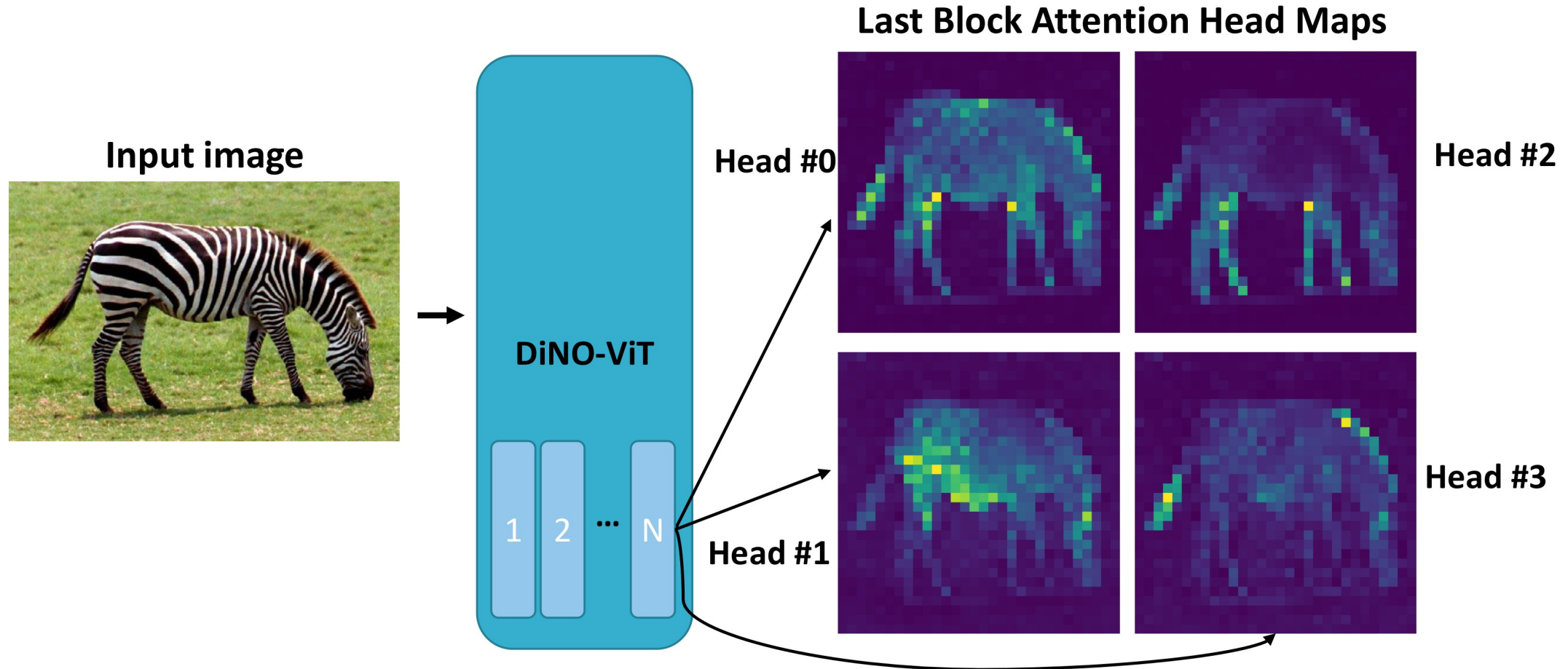
# What emerged

Class-token attention segments objects. Nobody asked for segmentation.



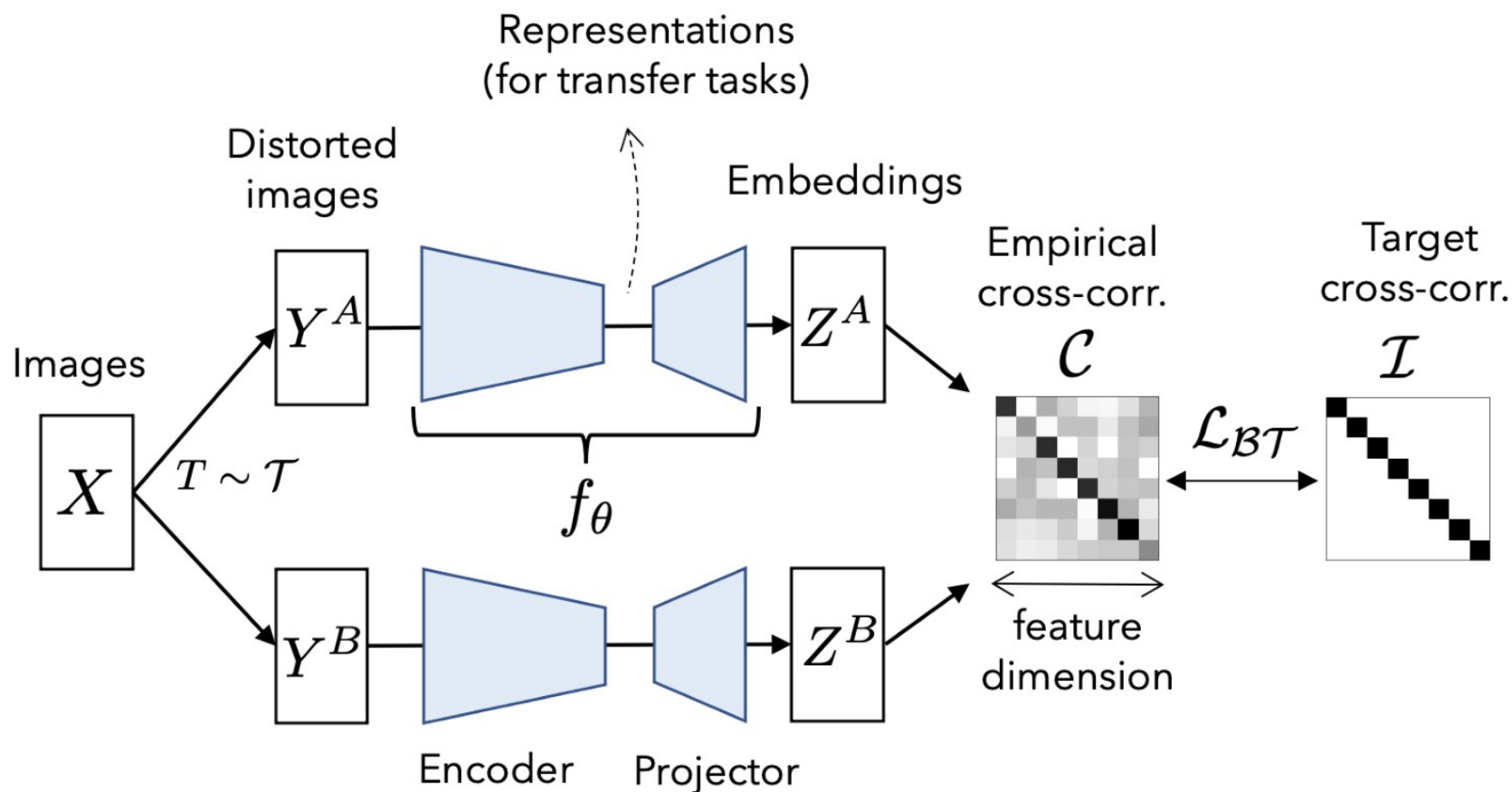
# Reading attention in practice

Per-head class-token attention, straight from the released model.



# A different cure: fix the feature space

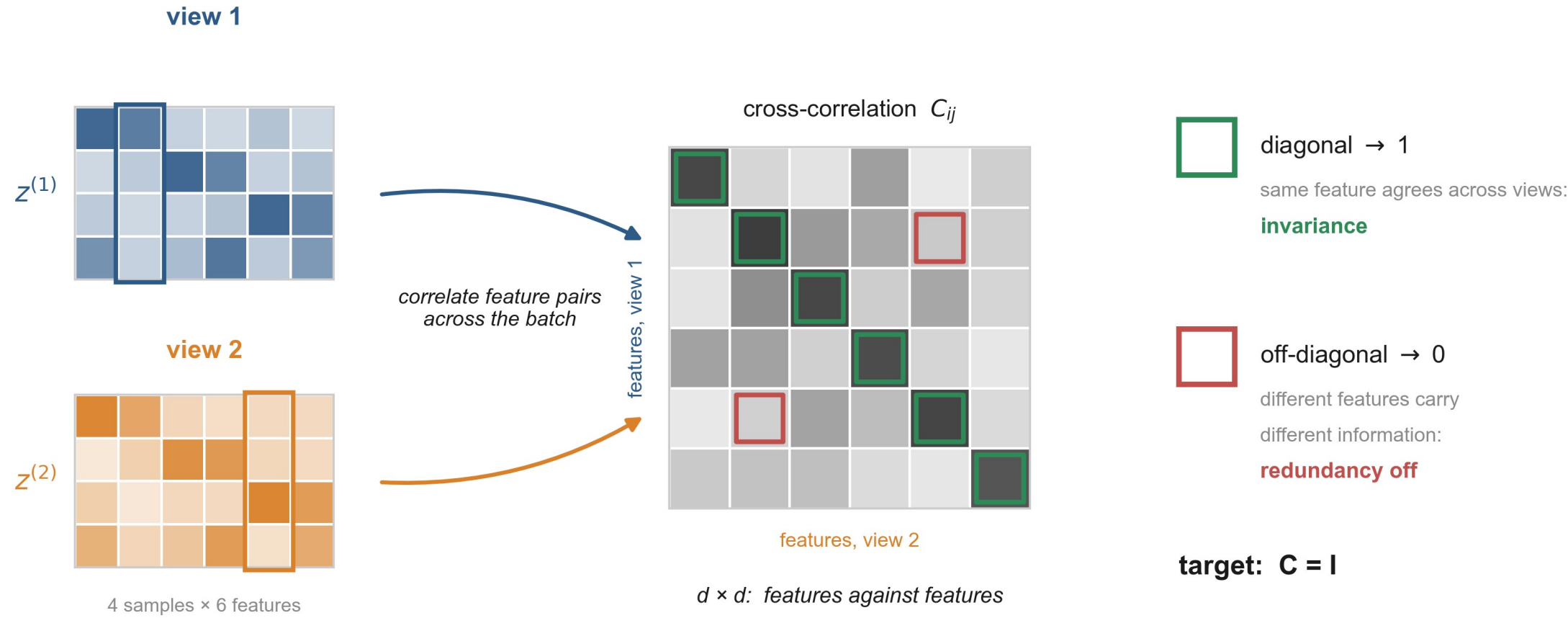
Barlow Twins: make the cross-correlation of the two views' features the identity.





# The loss

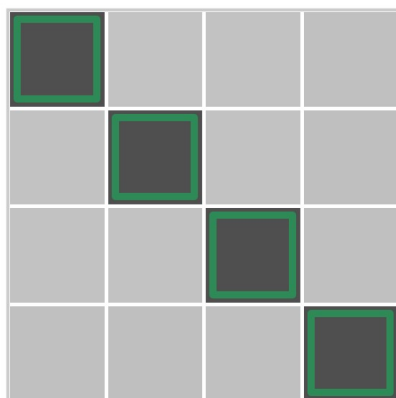
$$L_{\text{BT}} = \sum_i (1 - C_{ii})^2 + \lambda \sum_i \sum_{j \neq i} C_{ij}^2$$



# The punchline

Two matrices, one enemy: InfoNCE works the Gram matrix, Barlow Twins and VICReg work the covariance. They share eigenvalues.

InfoNCE

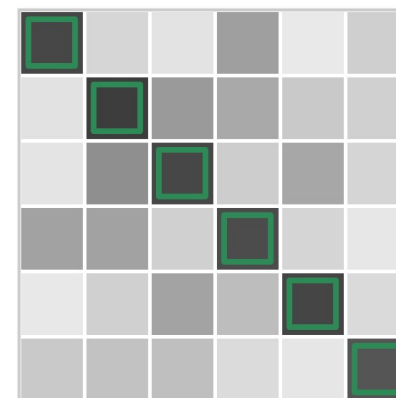


samples  $\times$  samples

*push samples apart*

*same enemy:  
collapse*

Barlow Twins, VICReg



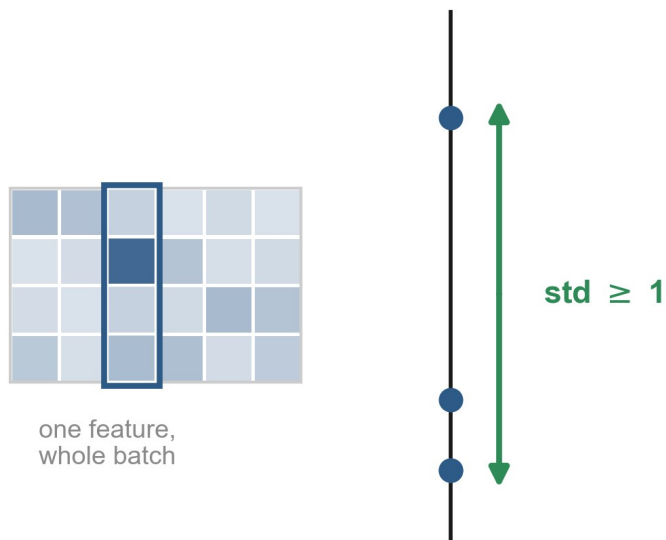
features  $\times$  features

*flatten the spectrum*

# VICReg

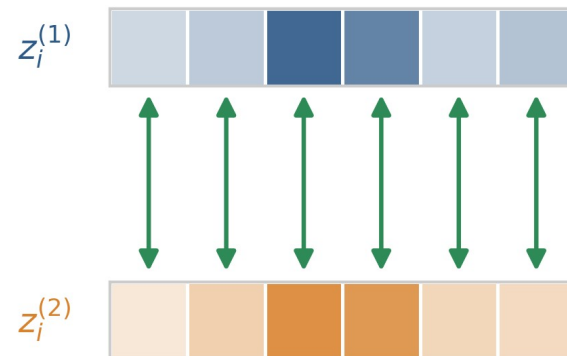
The same idea, spelled out: variance, invariance, covariance, three explicit terms.

## Variance



*every feature stays alive*

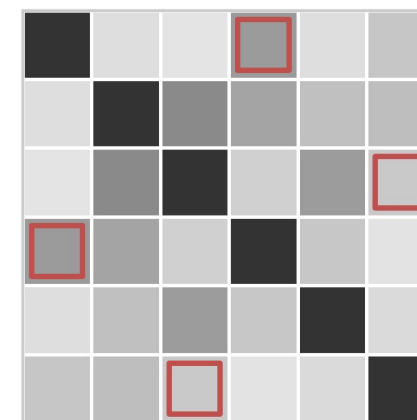
## Invariance



$$\|z_i^{(1)} - z_i^{(2)}\|^2 \rightarrow 0$$

*the two views agree*

## Covariance



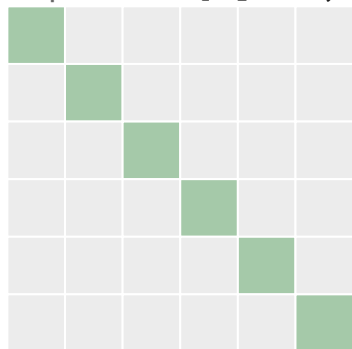
**off-diagonal  $\rightarrow 0$**

*features stay independent*



# Don't let them full ya'

feature space  $\mathbf{C} = \mathbf{Z}_1^\top \mathbf{Z}_2 / N$  ( $d \times d$ )



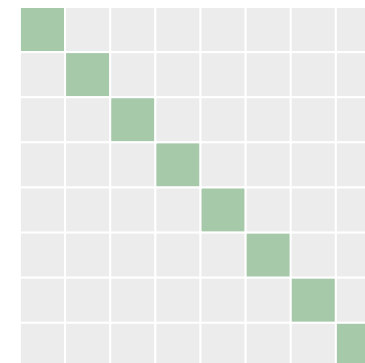
diagonal  $\rightarrow 1$  : invariance

off-diagonal  $\rightarrow 0$  : decorrelation

transpose



sample space  $\mathbf{G} = \mathbf{Z}_1 \mathbf{Z}_2^\top / N$  ( $N \times N$ )



diagonal : align positives

off-diagonal : repel negatives

$$\mathcal{L} \simeq -\frac{2}{N} \text{tr}(\mathbf{Z}_1^\top \mathbf{Z}_2) + \frac{1}{N^2} \text{tr}(\mathbf{Z}_a^\top \mathbf{Z}_b \mathbf{Z}_c^\top \mathbf{Z}_d)$$

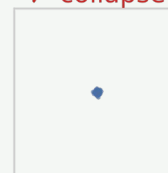
(a b c d): **Barlow 1221** · **InfoNCE 1122** (2nd order) · **VICReg 1111+2222**

shared global minimizer

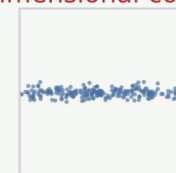
$$\mathbf{Z}_1 = \mathbf{Z}_2 \quad \forall \text{ draws}, \quad \frac{1}{N} \mathbf{Z}^\top \mathbf{Z} = \mathbf{I}$$

augmentation-invariant features, isotropic covariance;  
linear case: whitened top eigenspace of the cross-view covariance.

✗ collapse



✗ dimensional collapse

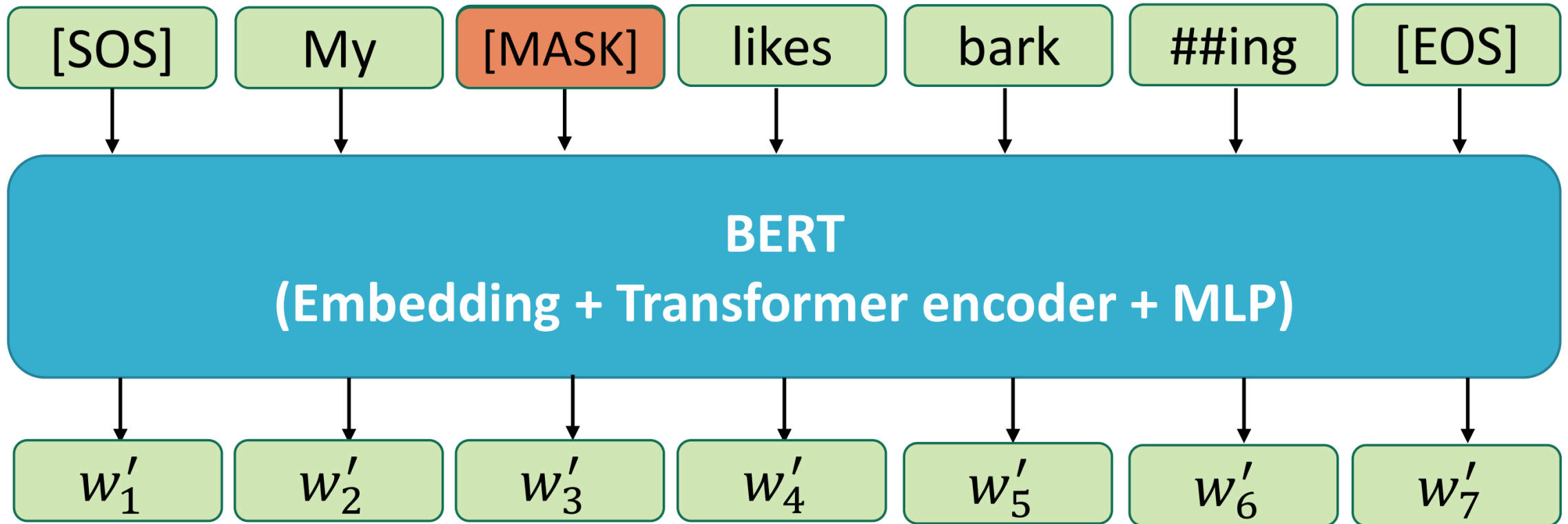


✓ whitened



# NLP solved this first

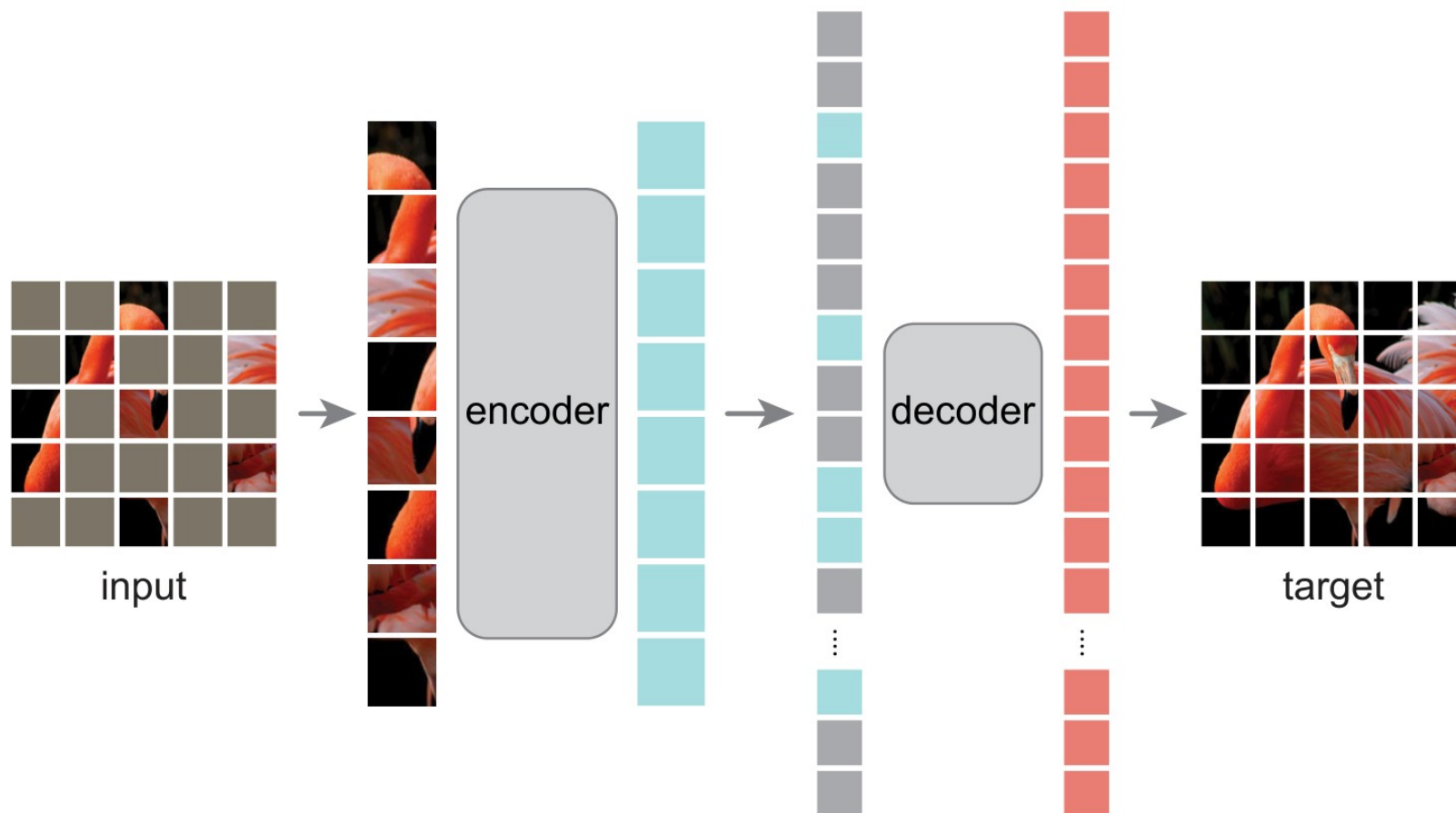
BERT, 2018: mask tokens, predict them. Pretraining ate NLP within a year.



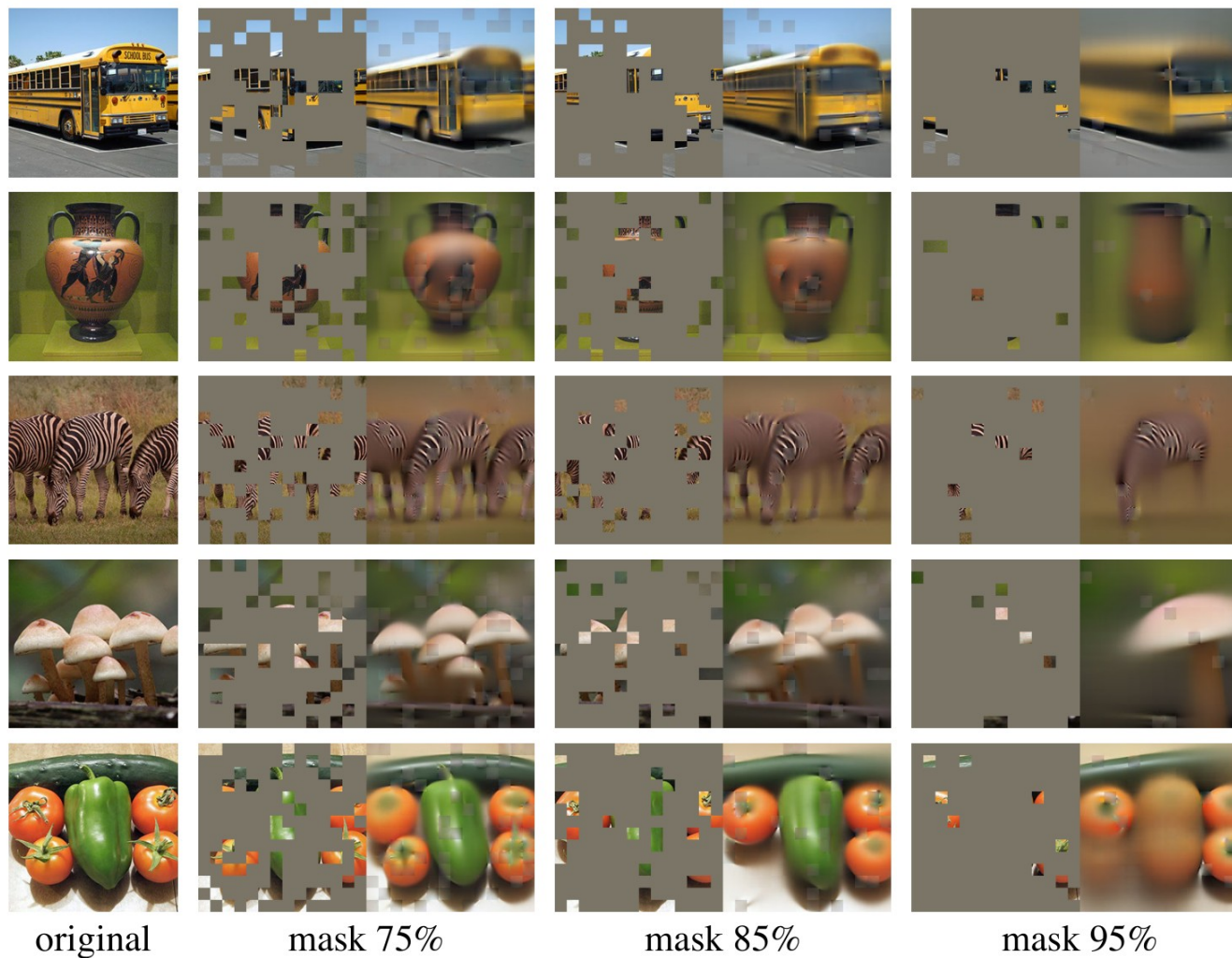


# MAE

Mask 75 percent. Encode only visible patches. A light decoder reconstructs pixels.

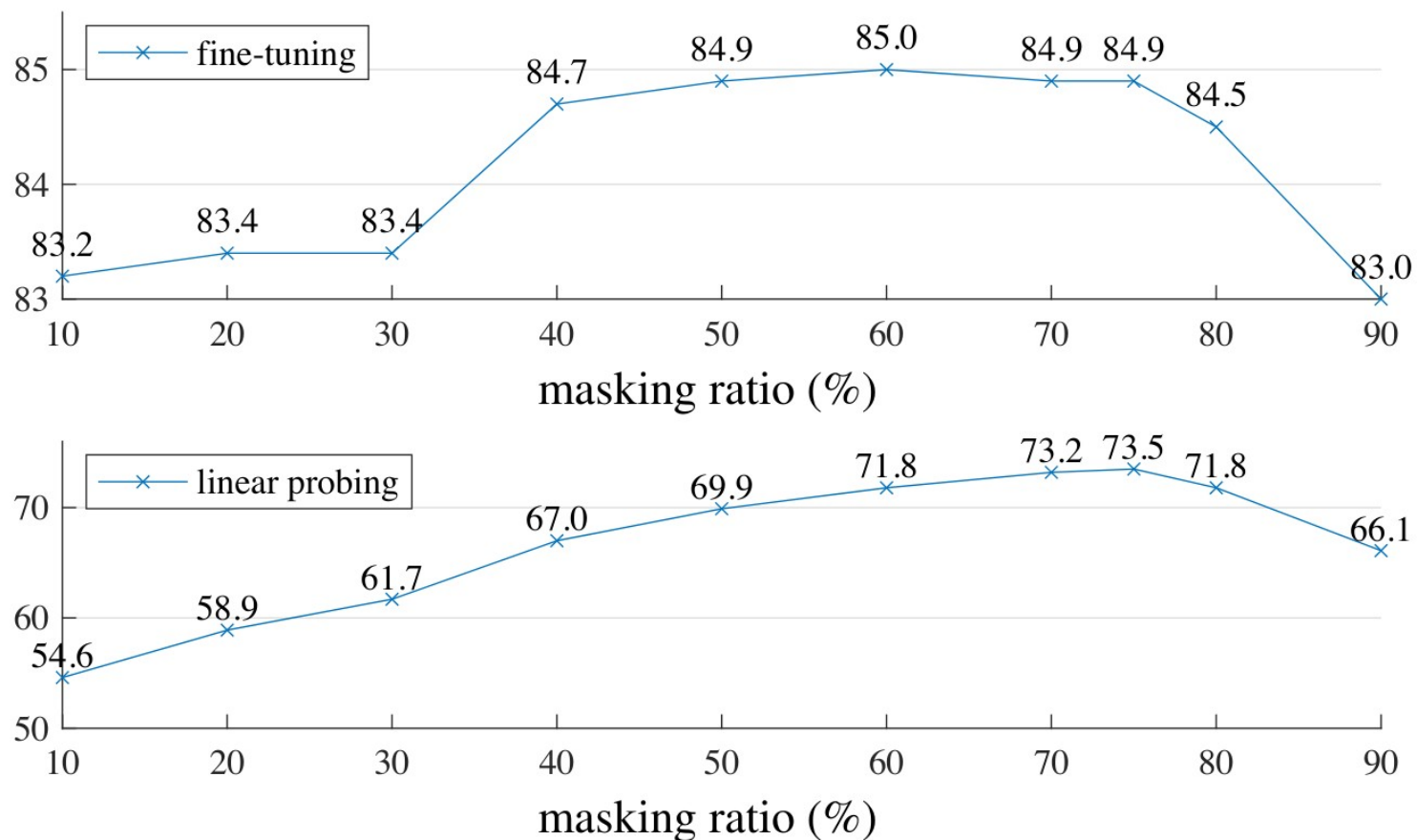


# What it reconstructs



# Why mask so much?

Images are spatially redundant. 15 percent is a texture task, 75 percent forces semantics.

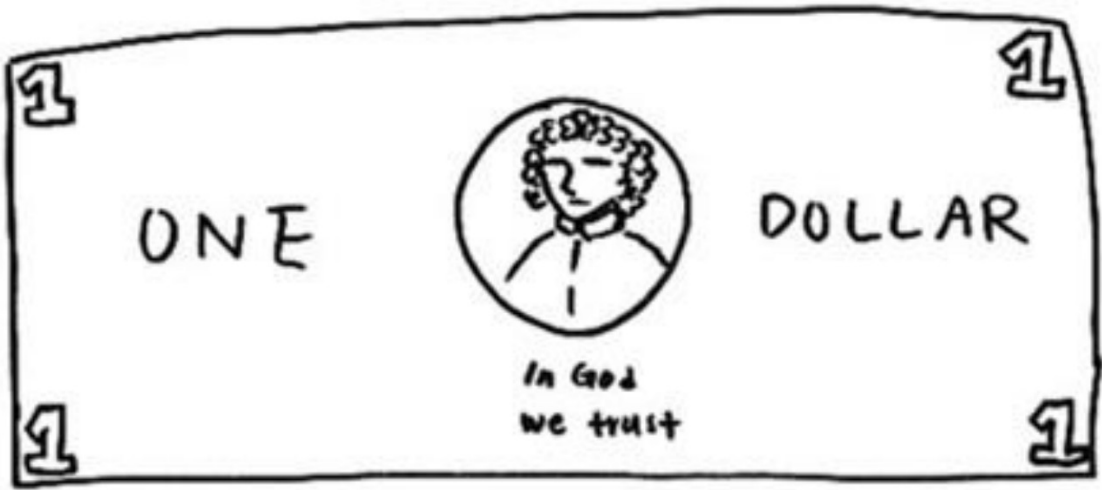


# It scales

MAE pretraining beats supervised pretraining for ViT-L and ViT-H fine-tuning.

| method             | pre-train data | ViT-B       | ViT-L       | ViT-H       | ViT-H <sub>448</sub> |
|--------------------|----------------|-------------|-------------|-------------|----------------------|
| scratch, our impl. | -              | 82.3        | 82.6        | 83.1        | -                    |
| DINO [5]           | IN1K           | 82.8        | -           | -           | -                    |
| MoCo v3 [9]        | IN1K           | 83.2        | 84.1        | -           | -                    |
| BEiT [2]           | IN1K+DALLE     | 83.2        | 85.2        | -           | -                    |
| MAE                | IN1K           | <u>83.6</u> | <u>85.9</u> | <u>86.9</u> | <b>87.8</b>          |

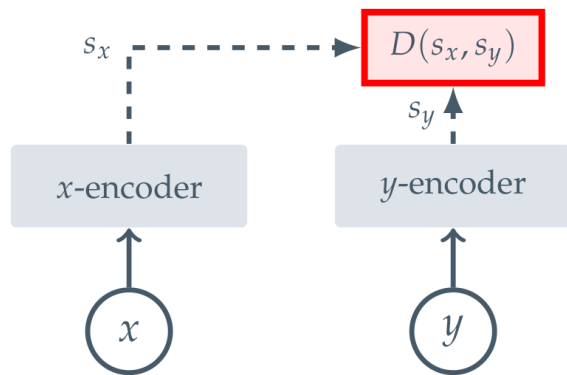
# Draw a dollar bill



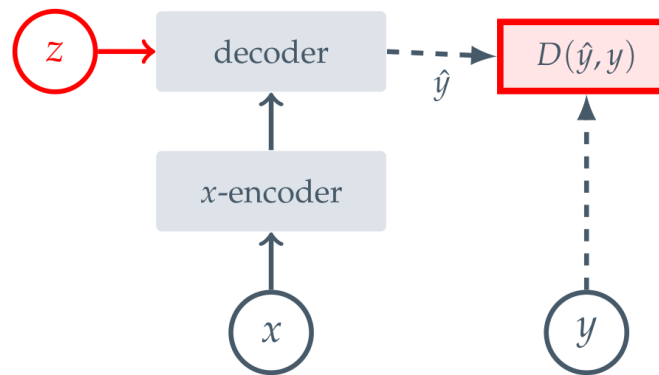


# Three architectures

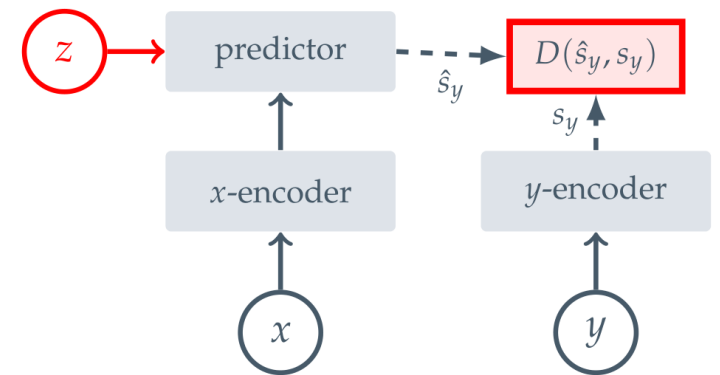
Joint-embedding, generative, and joint-embedding predictive.



(a) **Joint-Embedding Architecture**



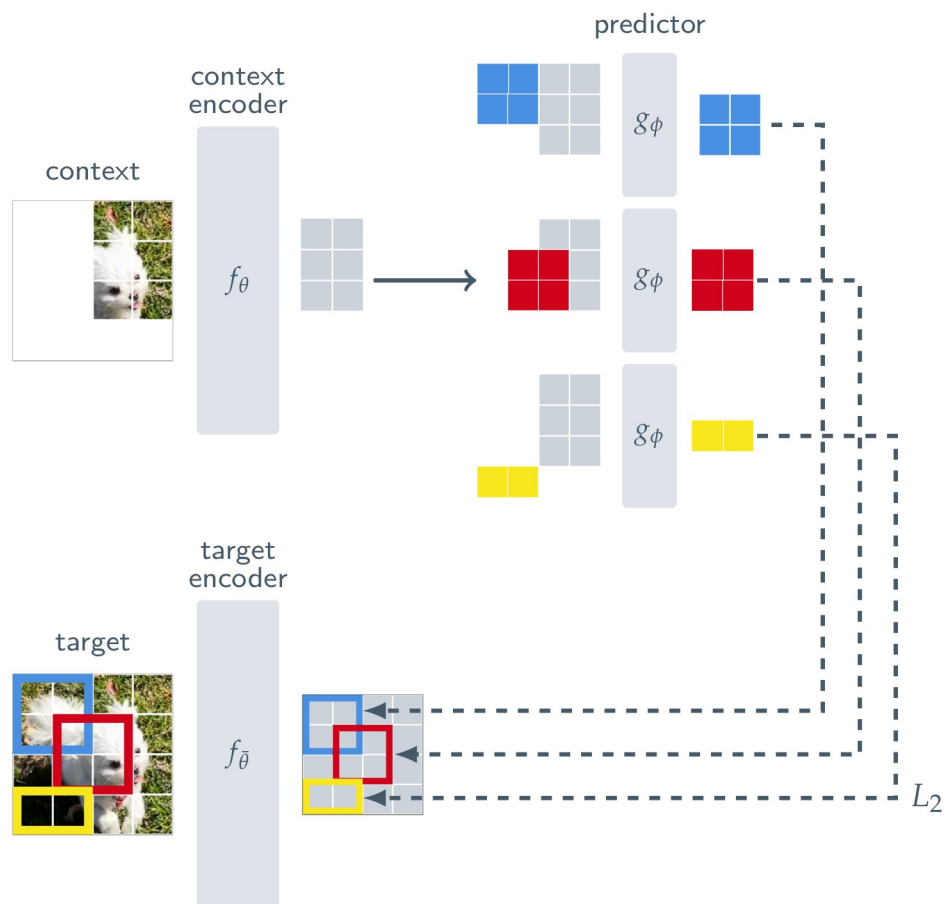
(b) **Generative Architecture**



(c) **Joint-Embedding Predictive Architecture**

# I-JEPA

Context encoder, predictor, EMA target encoder. No augmentations, no pixels.



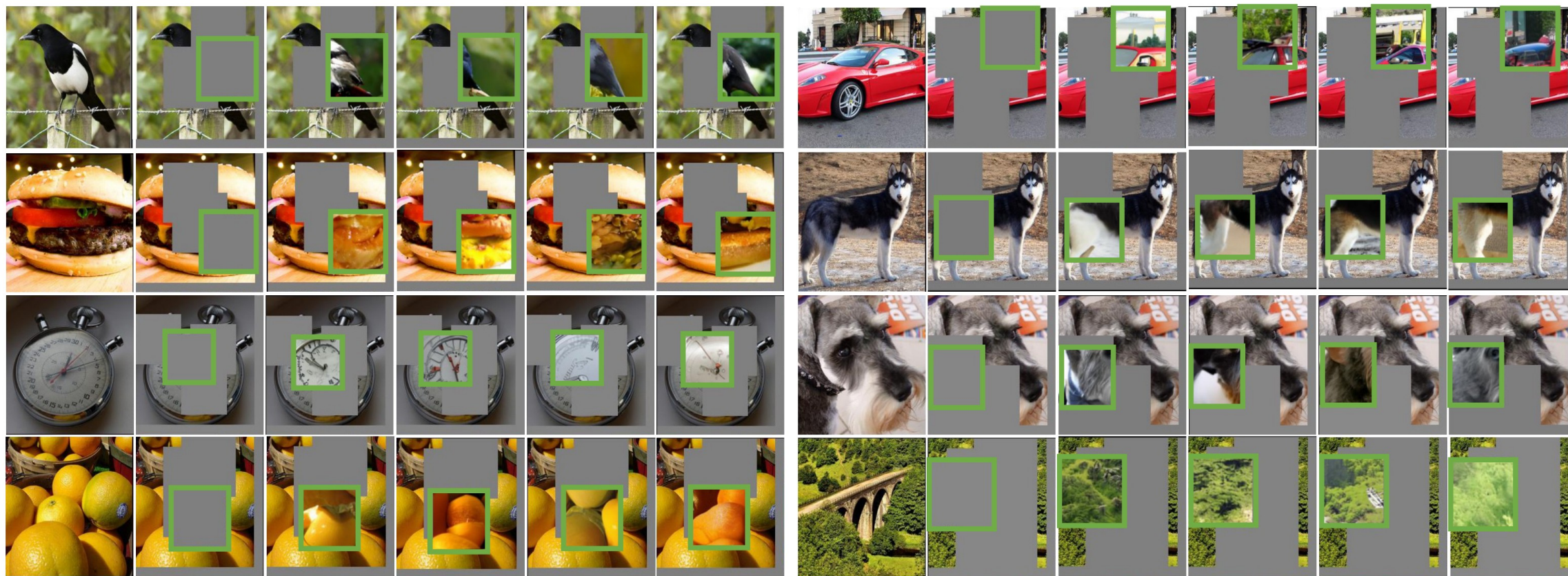
# The masking is the curriculum

A large informative context predicts multiple reasonably-sized targets.



# The predictor is a small world model

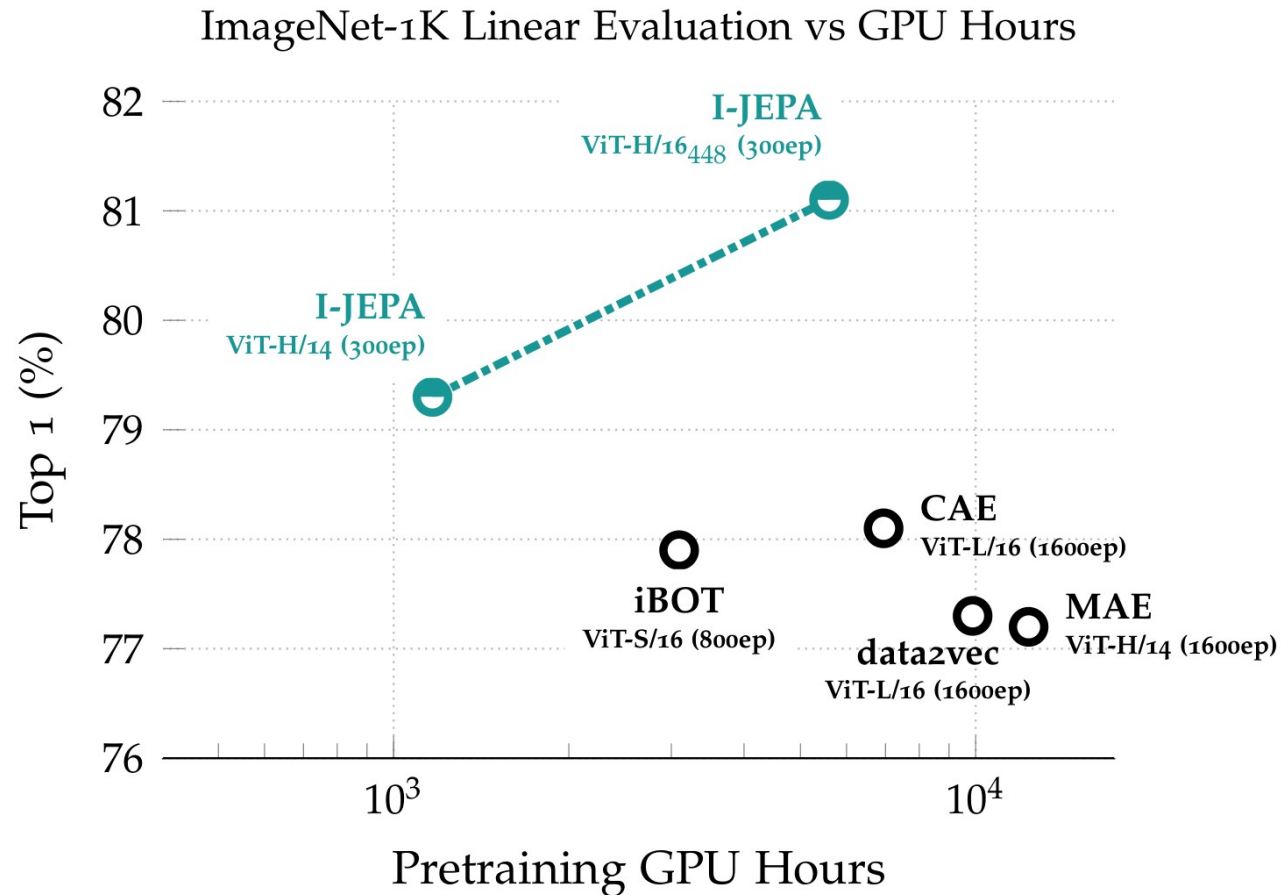
Decode its outputs: it knows where things are and their pose, not their texture.





# And it is efficient

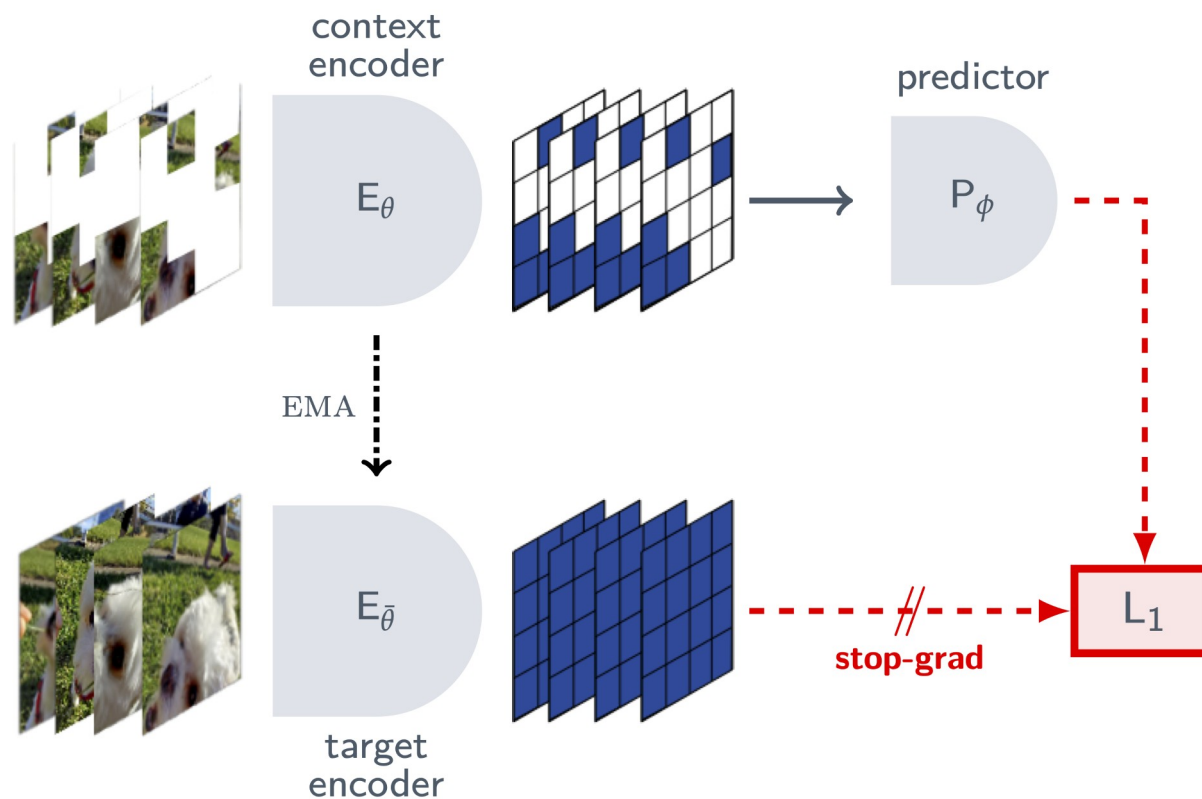
Better linear probes than MAE, at a fraction of the pretraining compute.





# V-JEPA: add time

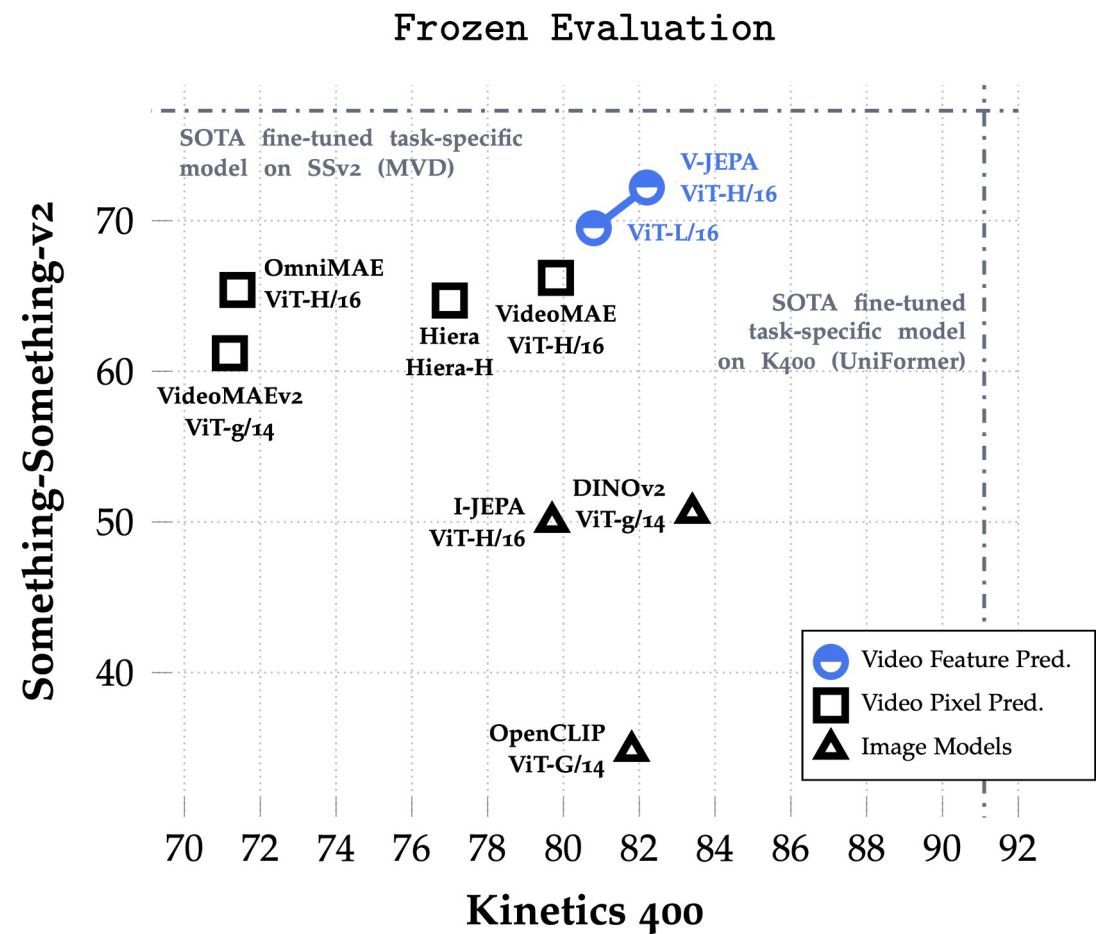
The same recipe on video, with spatiotemporal masking.



**V-JEPA**

# Feature prediction wins

Frozen evaluation: V-JEPA beats pixel-prediction video models, especially on motion.



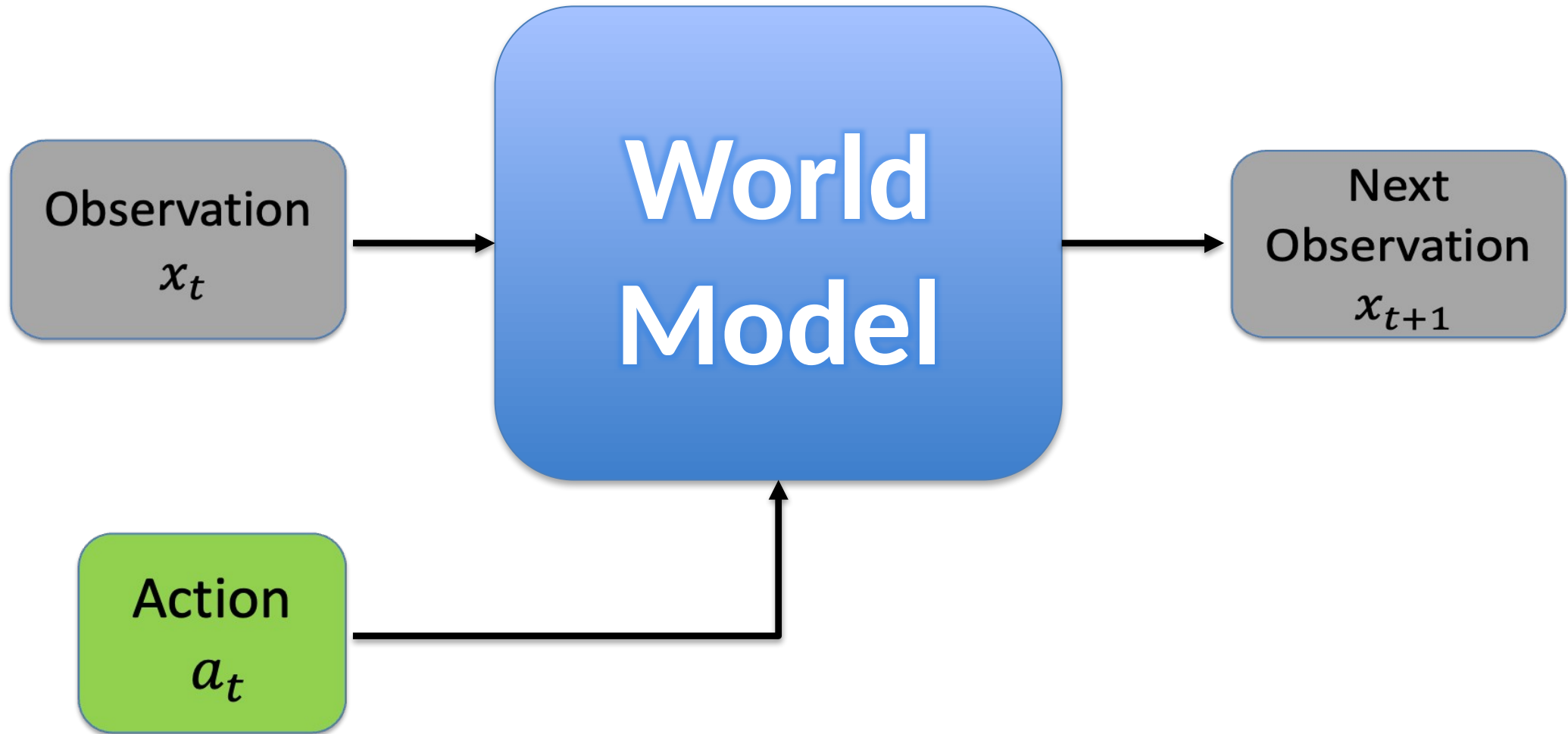
# Peeking into the latents

A frozen decoder probe renders the predictions: coherent objects and motion, fuzzy textures.

Frozen

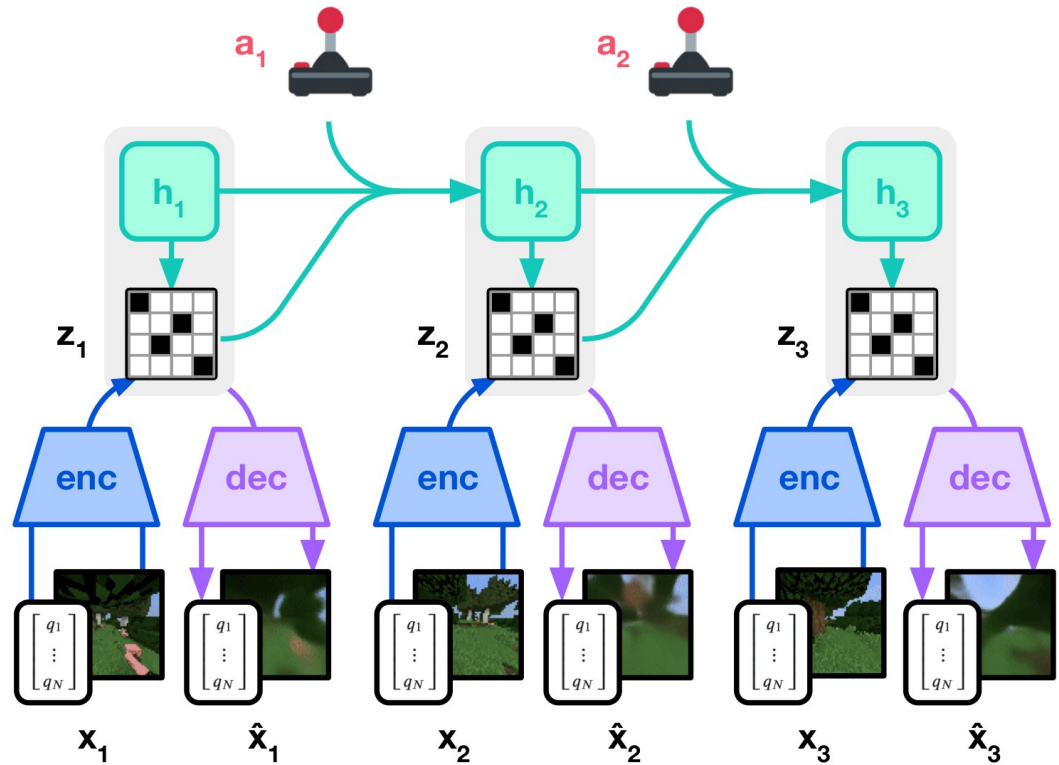


# World models

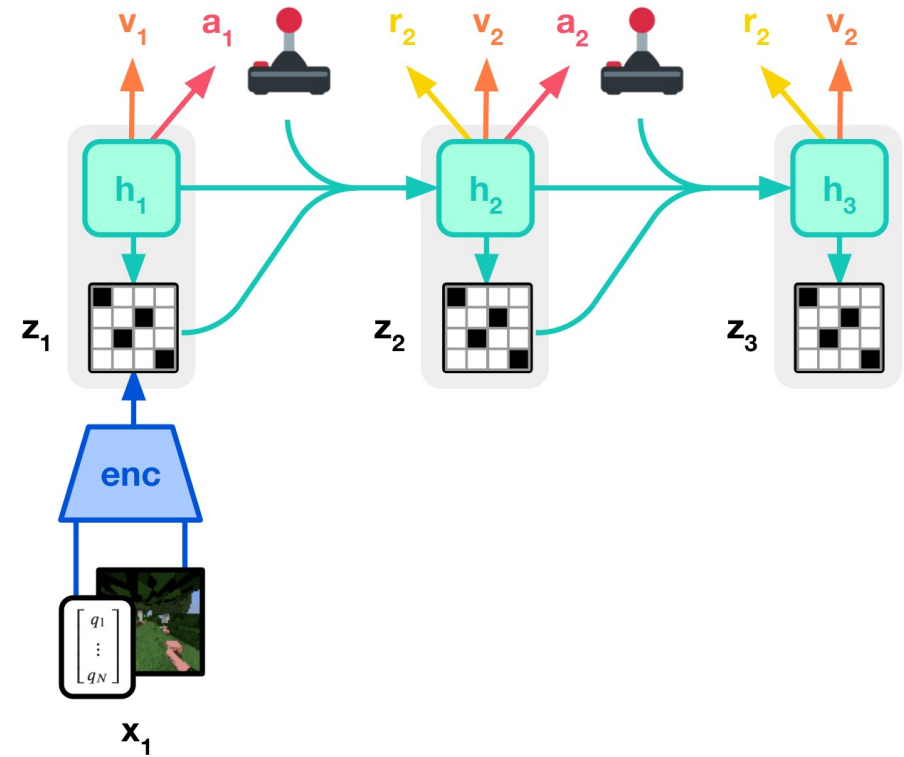


# Dreamer: acting in imagination, at scale

Learn the world model from replay, train actor and critic purely on imagined rollouts.



(a) World Model Learning

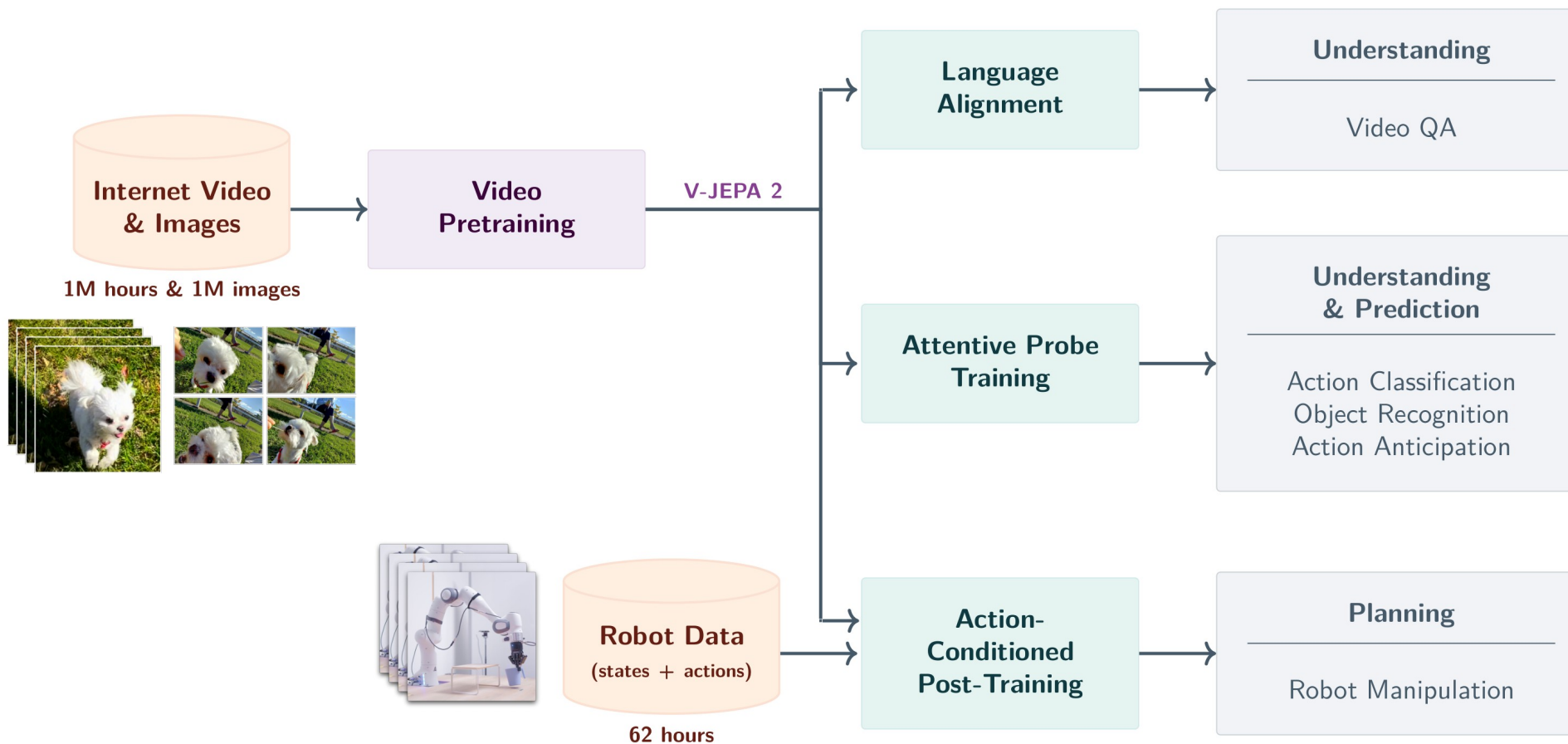


(b) Actor Critic Learning



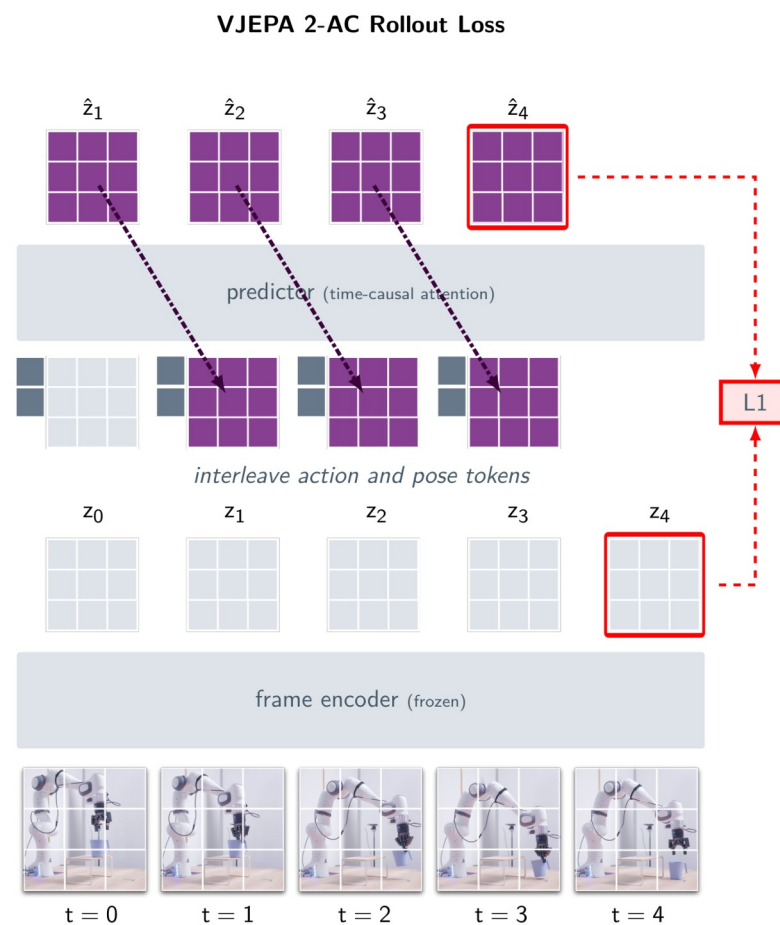
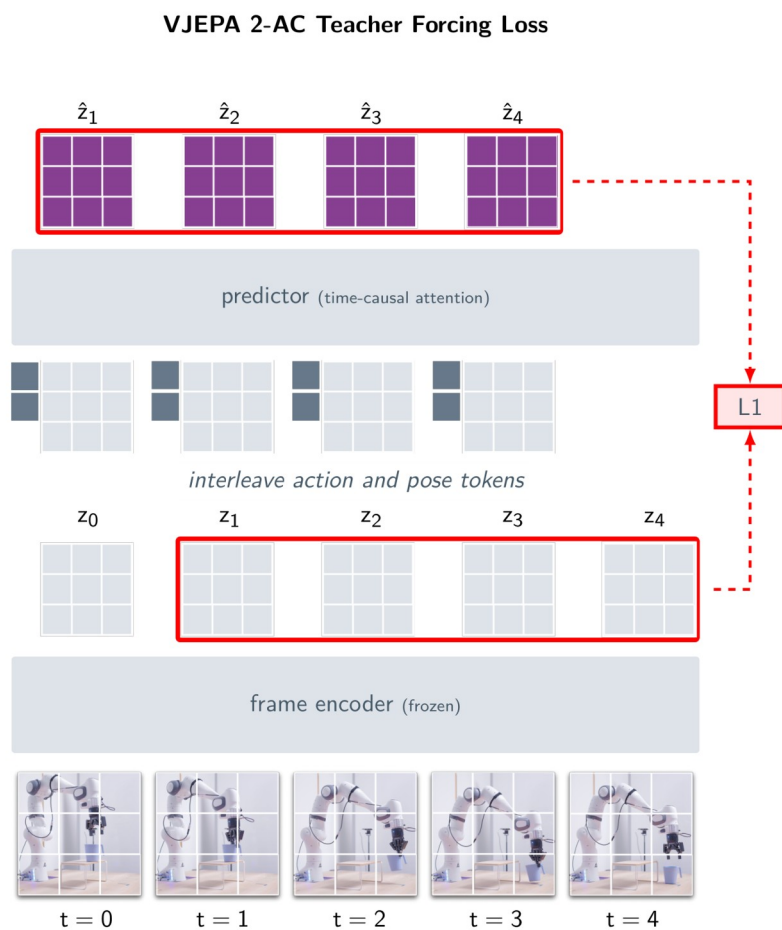
# V-JEPA 2

1M hours of video. Understanding, prediction, and now planning.



# V-JEPA 2-AC

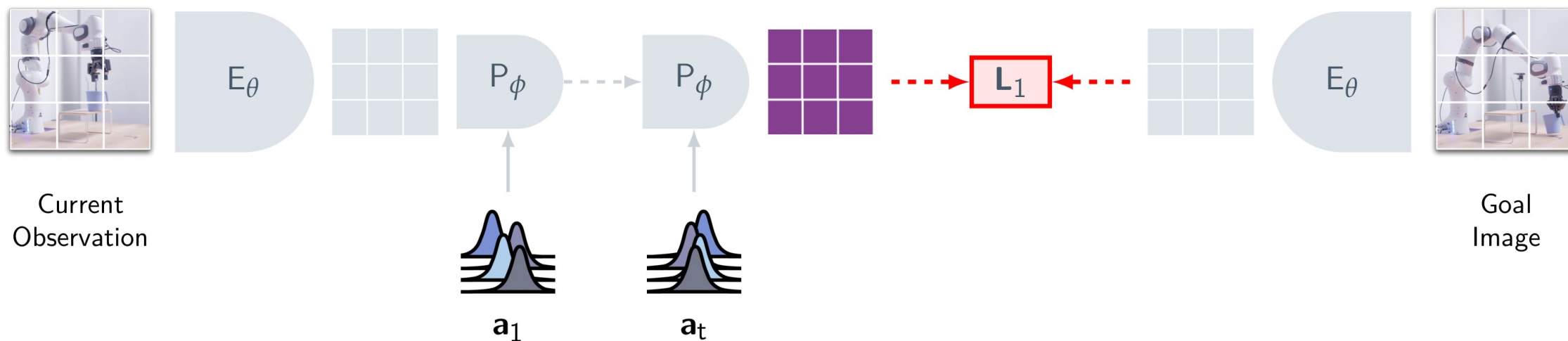
Post-train an action-conditioned predictor on 62 hours of robot data.



# Planning is energy minimization

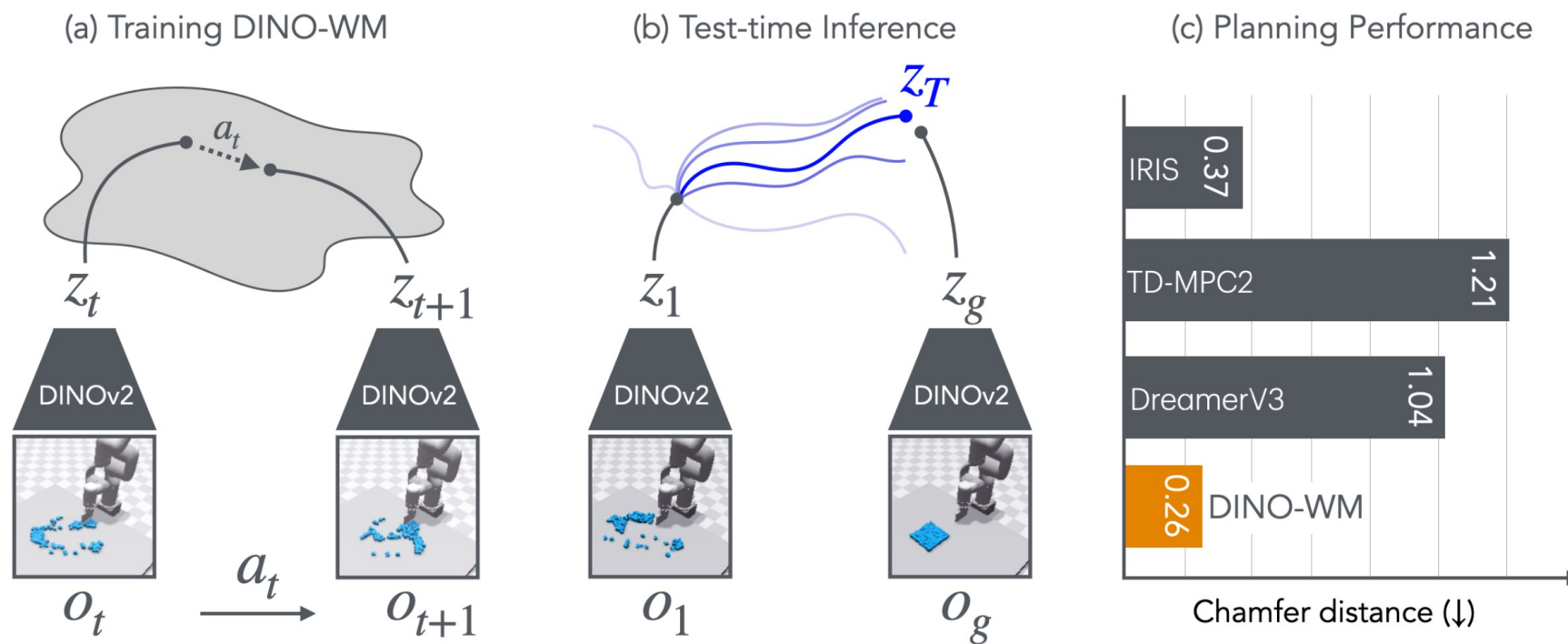
Roll out latent futures for candidate actions. Pick the sequence that lands nearest the goal encoding.

## VJEPA 2-AC Model Predictive Control



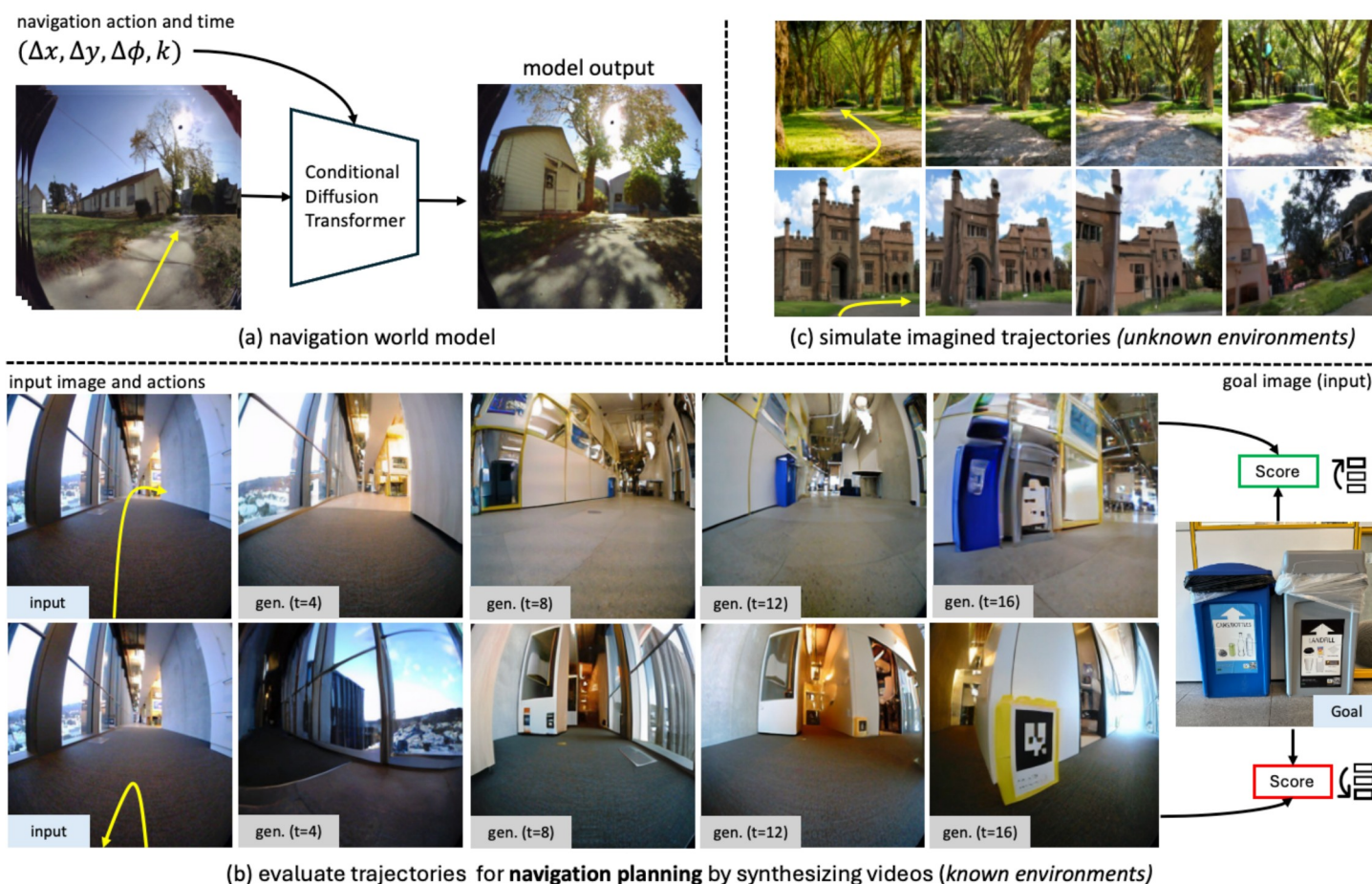
# DINO-WM: world models from frozen features

Freeze DINOv2, learn only the dynamics on its patch features, plan by optimizing actions.



# The other branch: generative world models

Navigation World Models: a diffusion transformer imagines the video of a candidate trajectory, then scores it.





# The world-model landscape, 2026

## Latent-state world models

Dreamer, V-JEPA 2-AC, DINO-WM: predict features, plan cheaply, nothing for a human to watch.

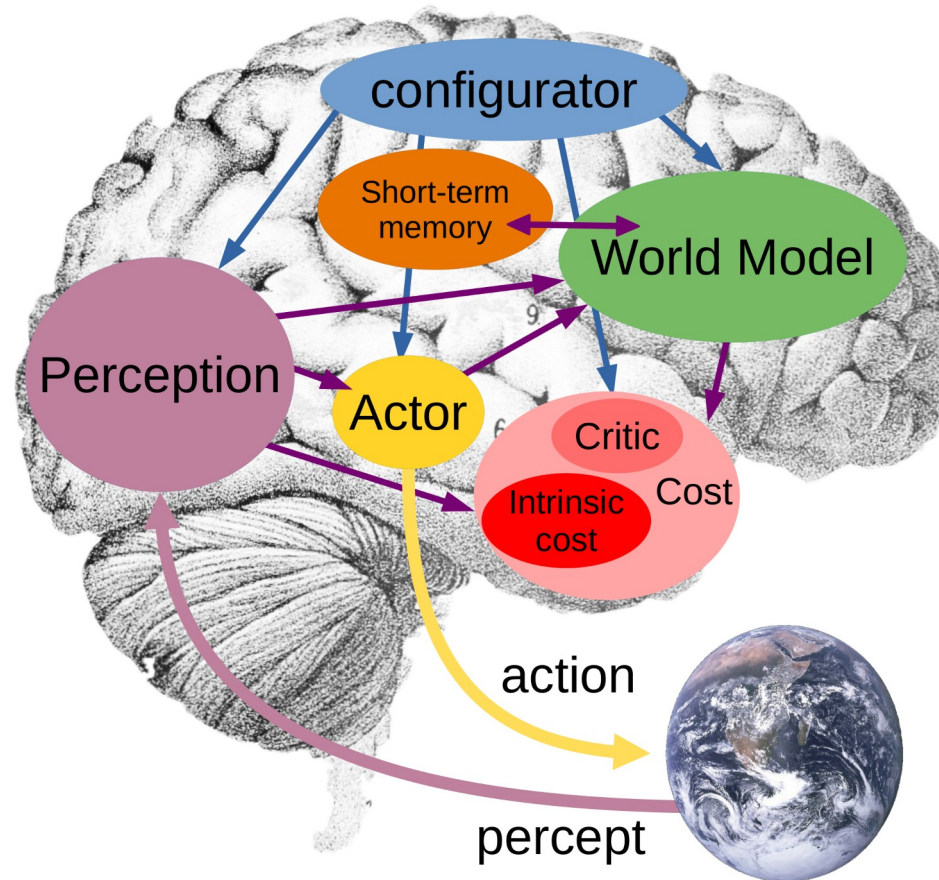
## Generative video world models

NWM, Genie 3, Cosmos: predict pixels with diffusion or autoregression, inspectable and even playable futures, at heavy compute per rollout.

Same job: imagination for agents. The field has not picked a winner.

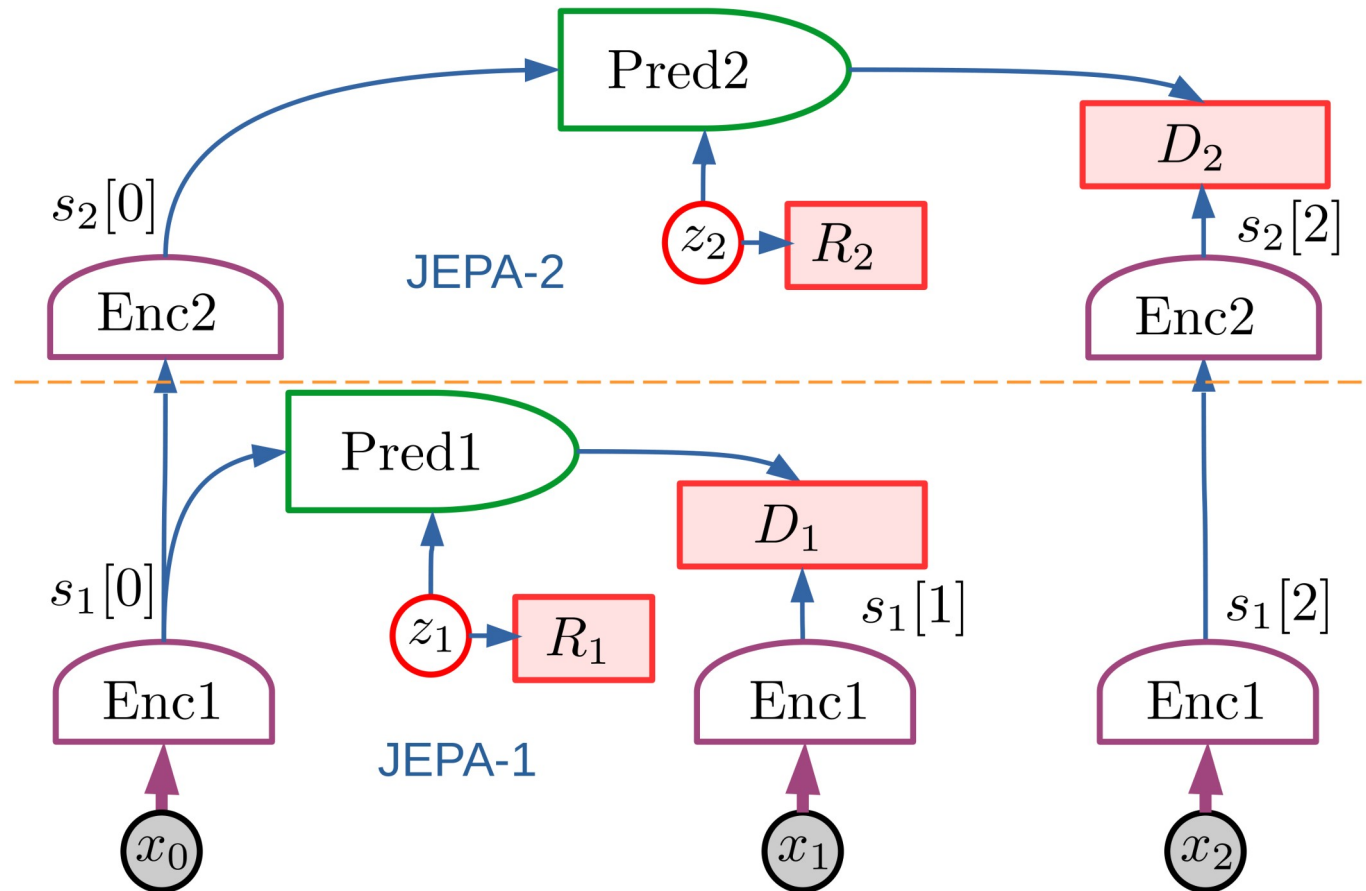
# The bigger blueprint

A cognitive architecture around a predictive world model.



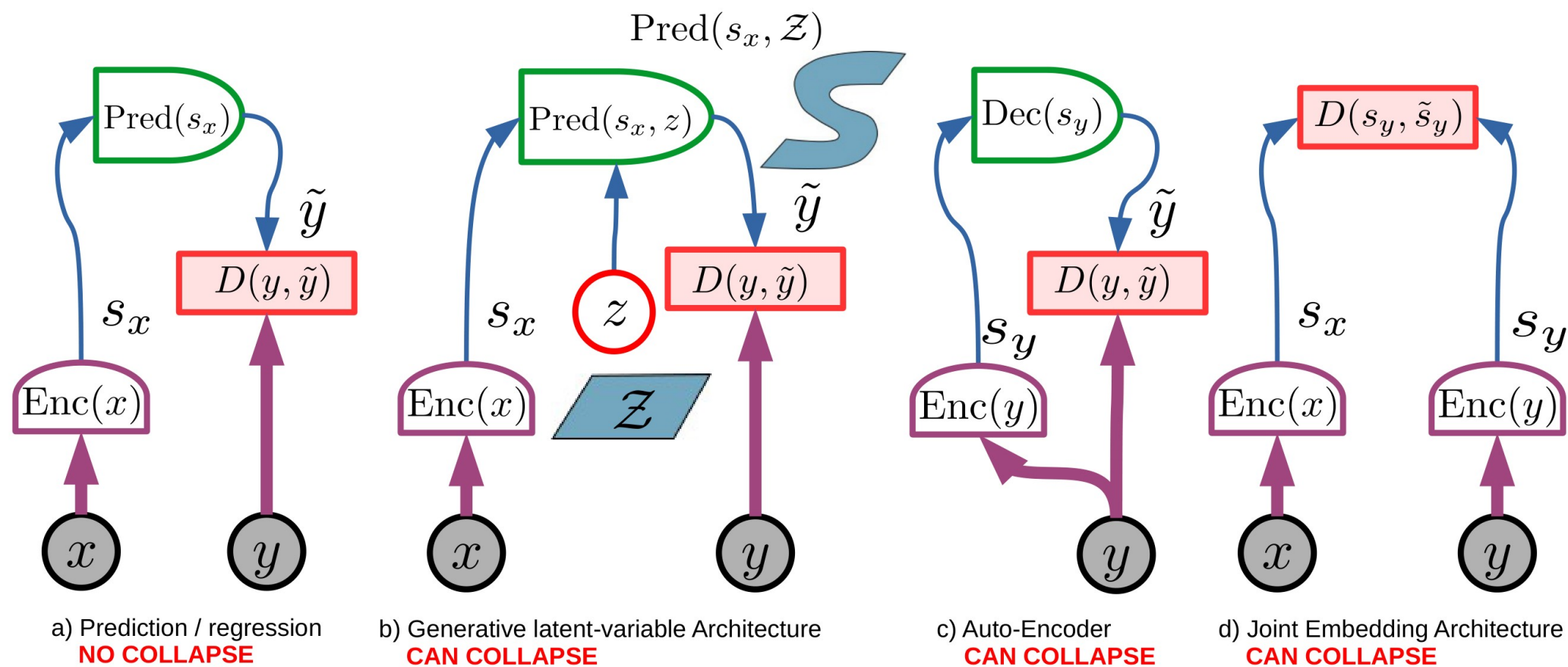
# H-JEPA: hierarchy

Predict at multiple abstraction levels and timescales.



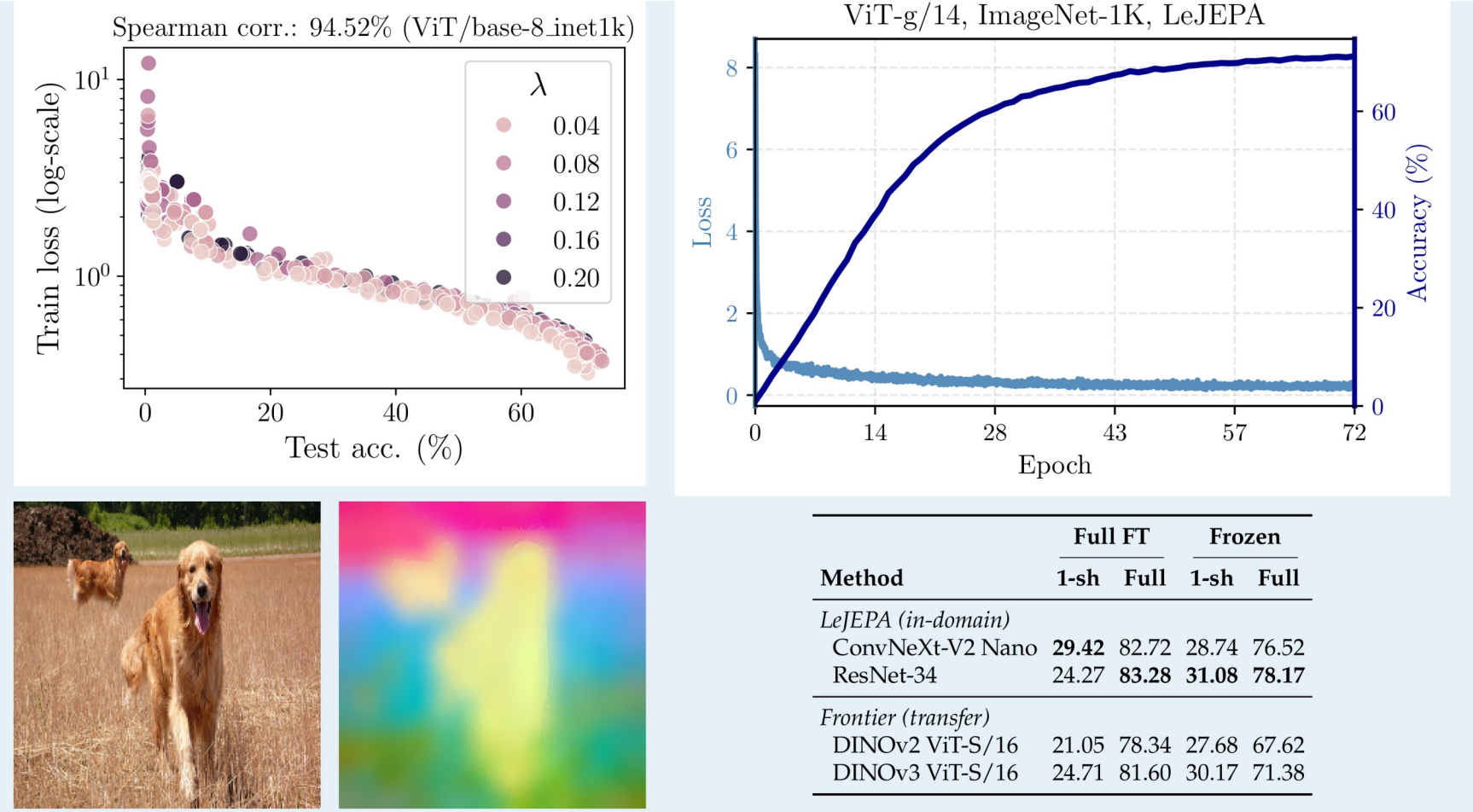
# Act III, revisited

The position paper's collapse taxonomy: the same spectral story.



# LeJEPA

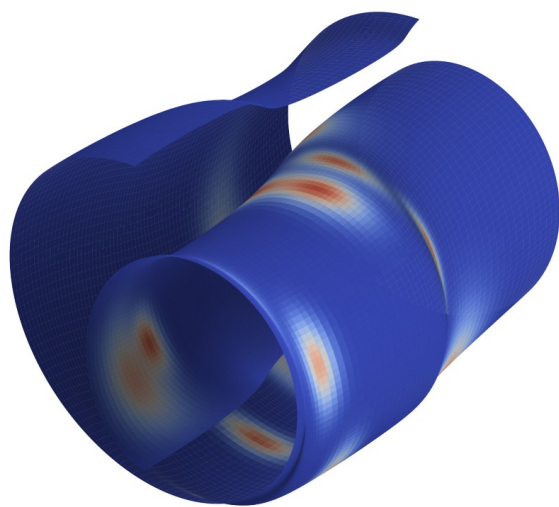
What should the embedding distribution be? Isotropic Gaussian, provably.



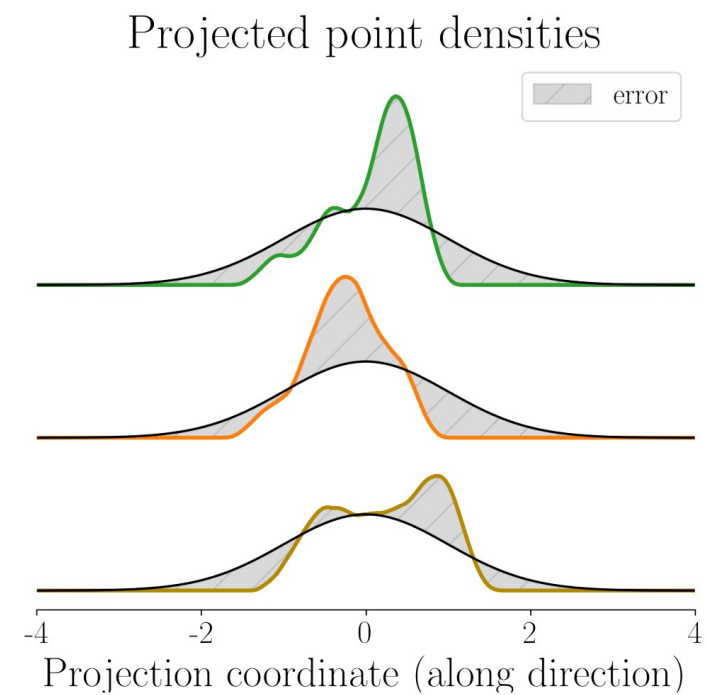
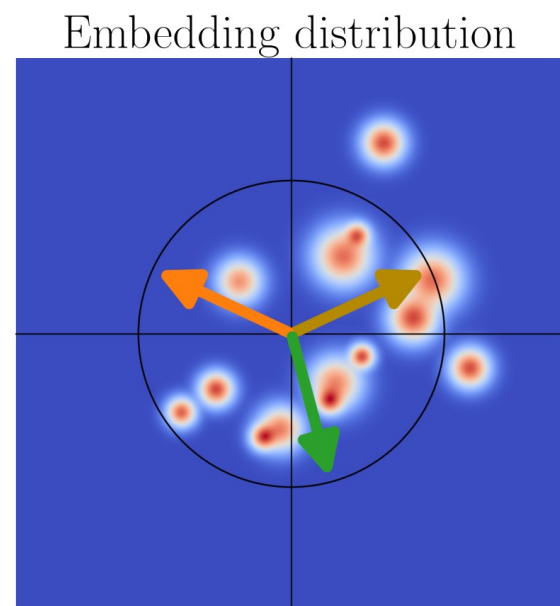


# SIGReg

A statistical test as a loss: match the embedding to isotropic Gaussian along random slices.



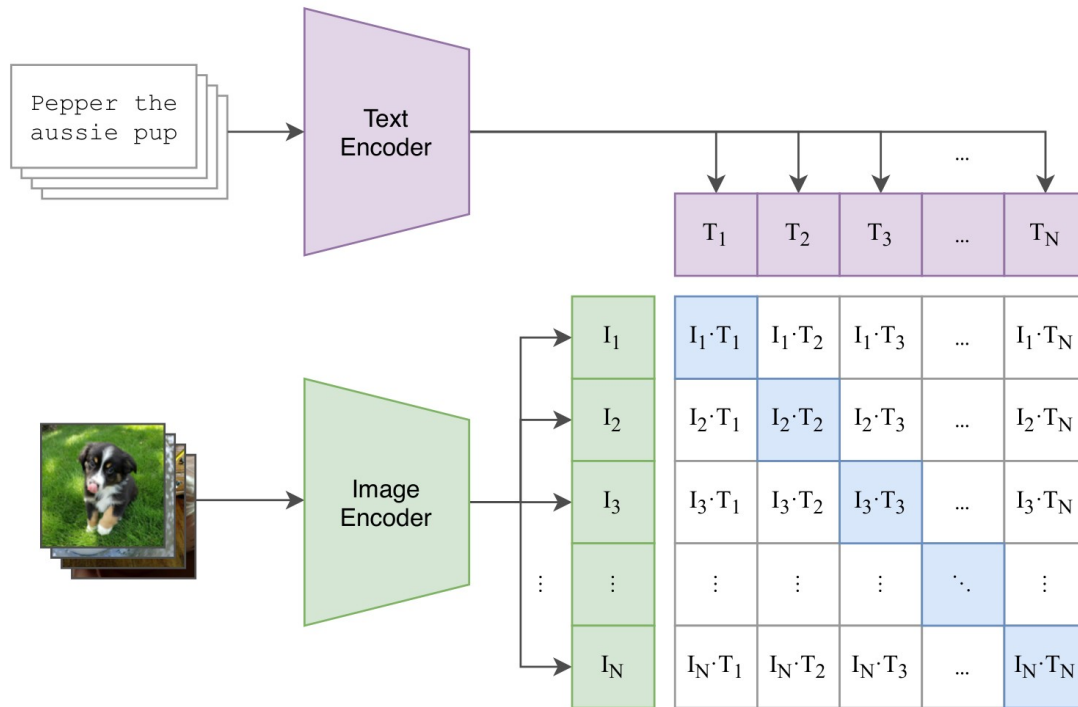
$f_{\theta}$   
 $\rightarrow$



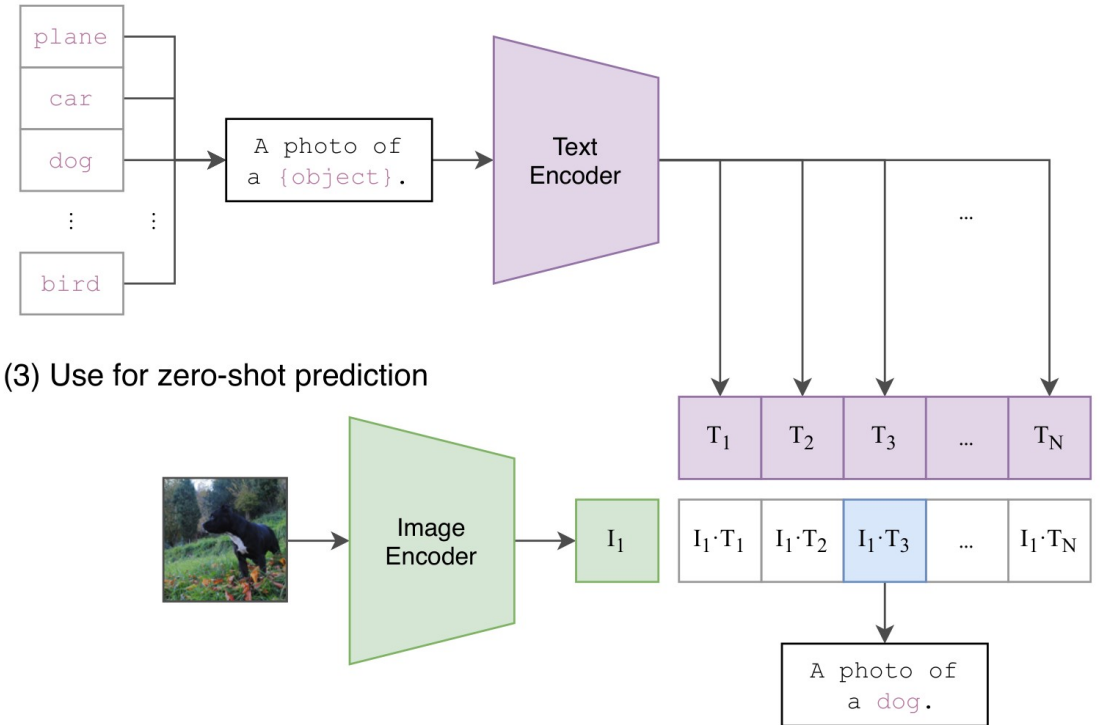
# CLIP

Contrastive learning where the second view is a caption. Language becomes the augmentation.

## (1) Contrastive pre-training



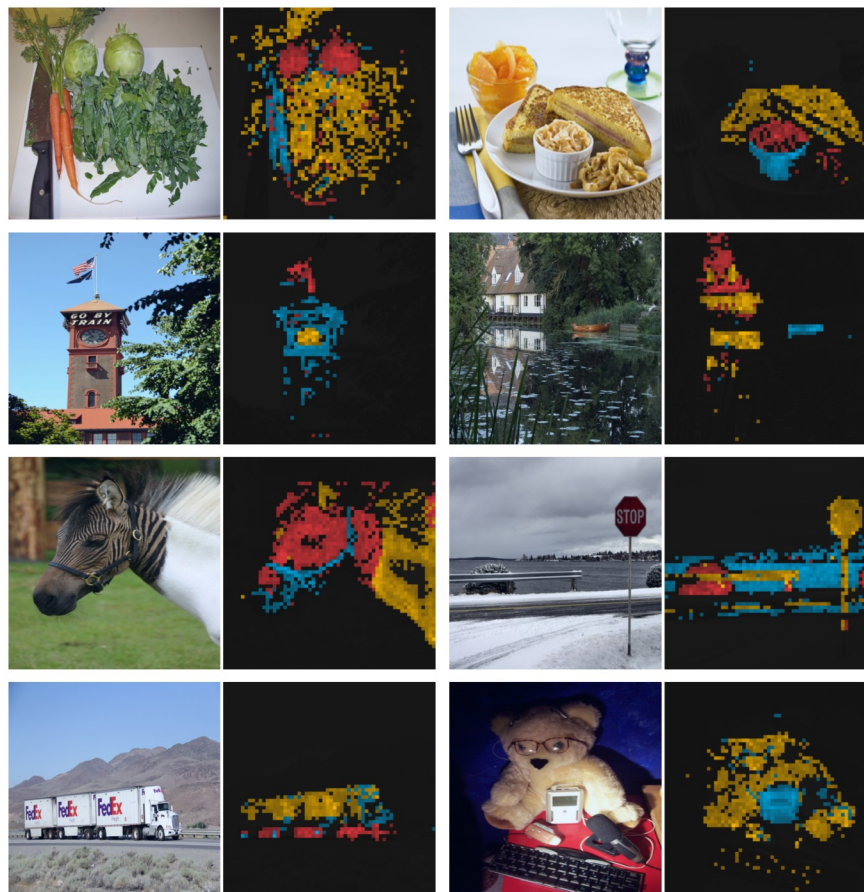
## (2) Create dataset classifier from label text



## (3) Use for zero-shot prediction

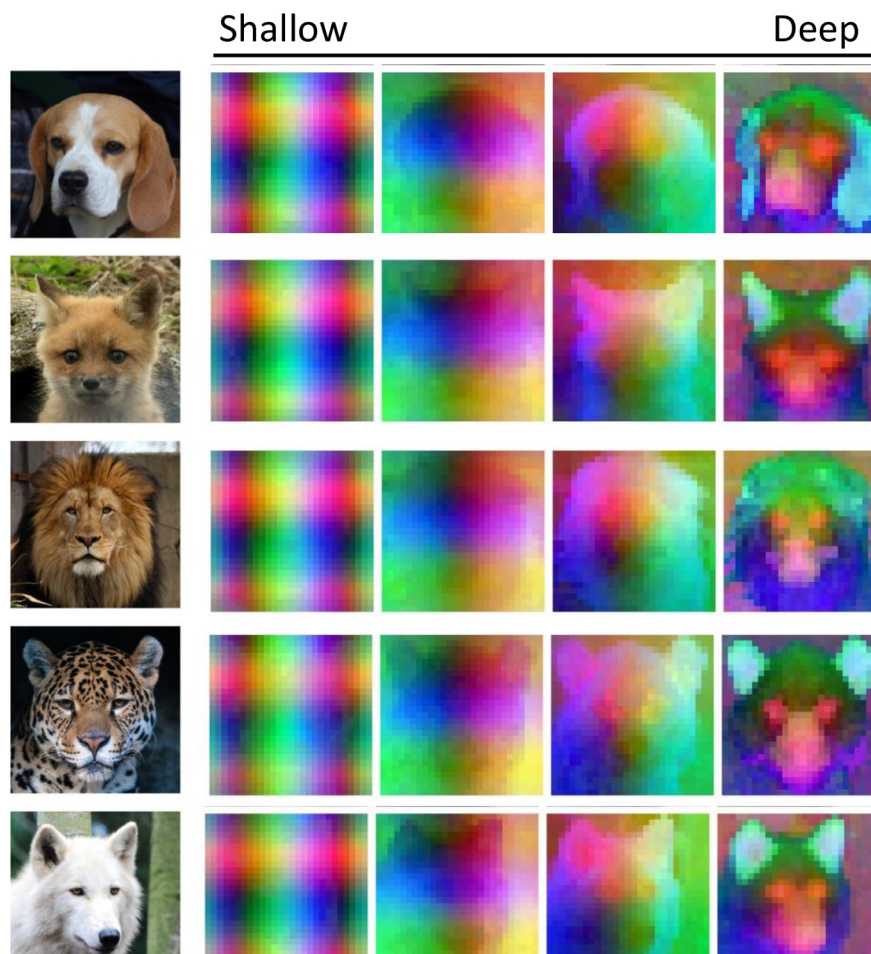
# The DINO payoff

Attention heads decompose objects into parts. Trained with zero labels.



# Features know parts

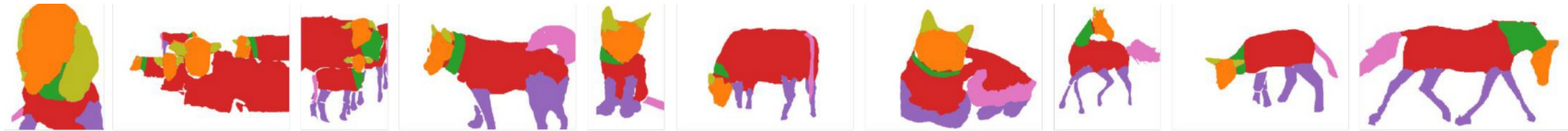
PCA on DINO-ViT features: components align with object parts, across instances.





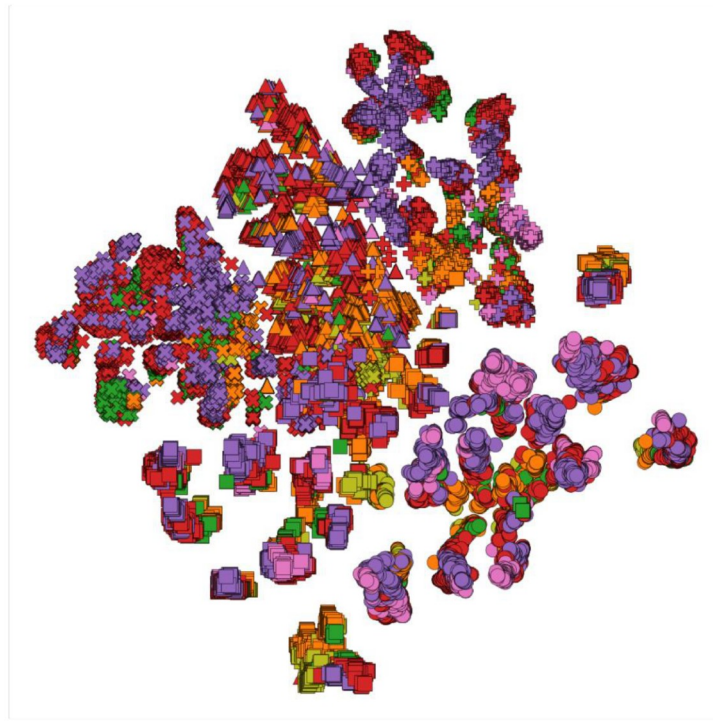
# Parts cluster across species

t-SNE of DINO part features: heads with heads, torsos with torsos, cats with dogs.

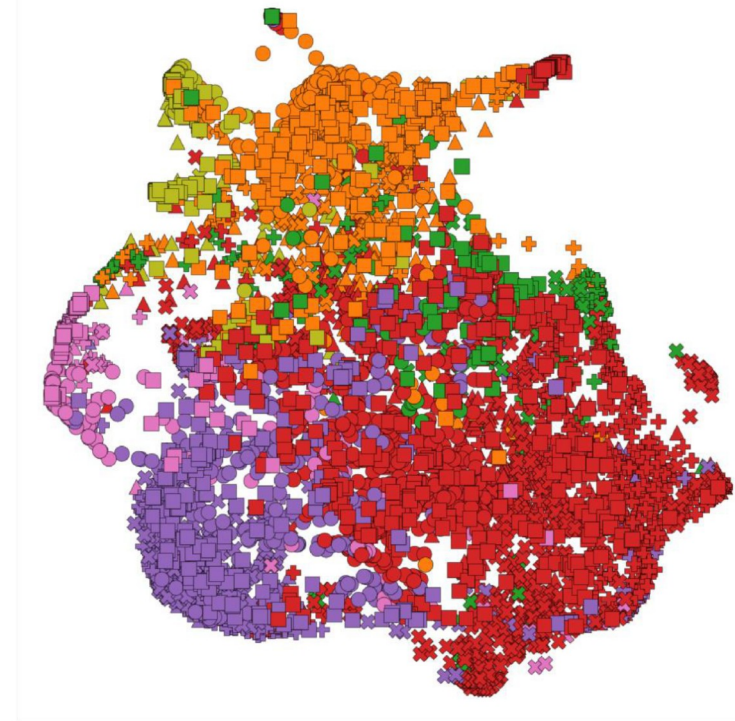


(a) Sample images and ground truth parts

**Torso**  
**Neck**  
**Head**  
**Ears**  
**Tail**  
**Limbs**  
○ Cat  
□ Dog  
+ Horse  
× Sheep  
△ Cow



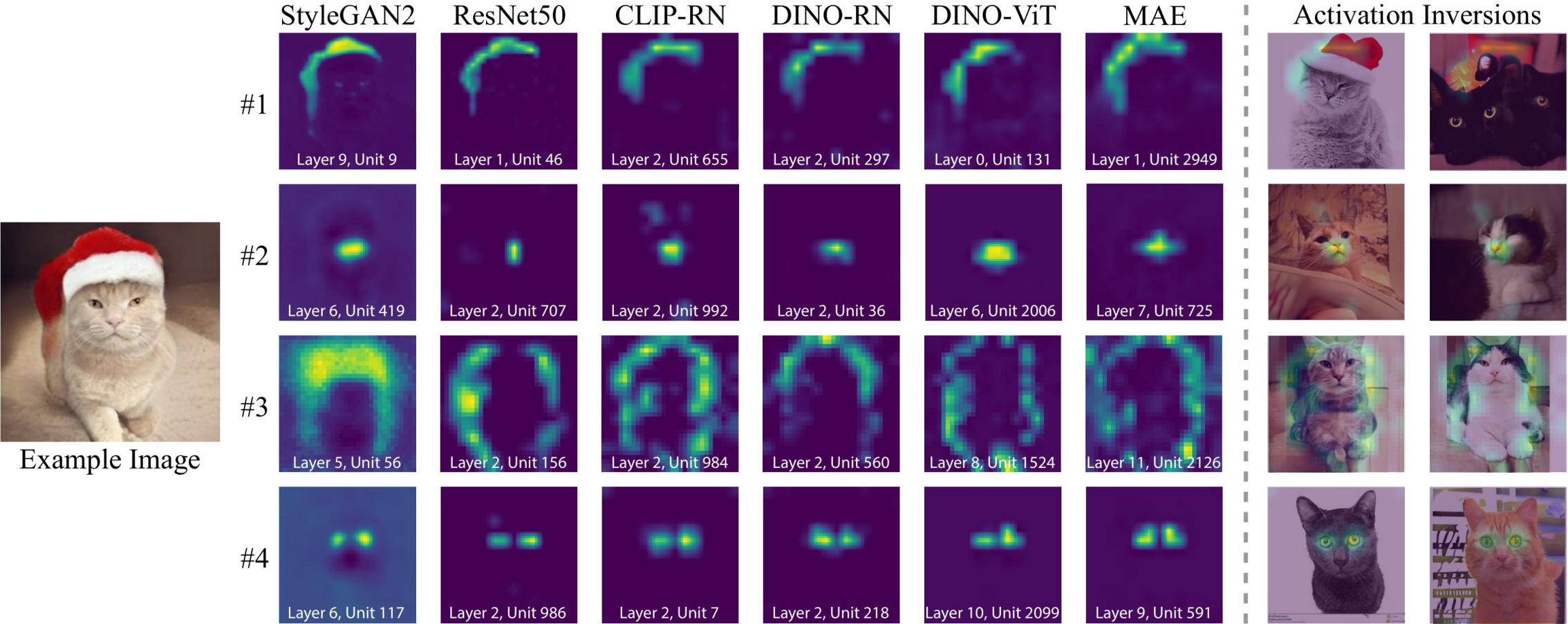
(c) Supervised ViT



(b) Self-Supervised ViT (DINO-ViT)

# Rosetta Neurons

The same visual concepts emerge as units across GANs, DINO, CLIP, MAE.





# Rosetta Neurons

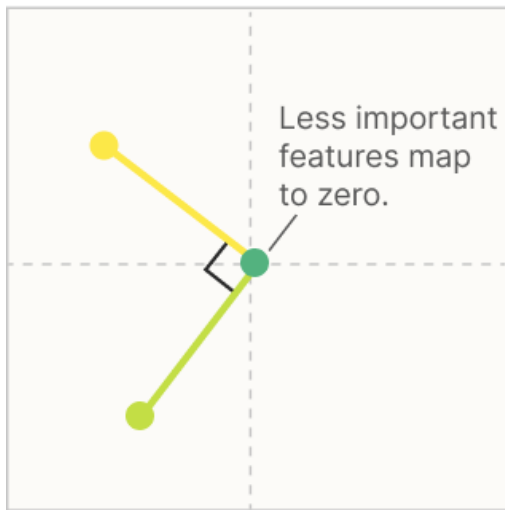


# Superposition

Networks pack more features than they have dimensions, tolerating interference.

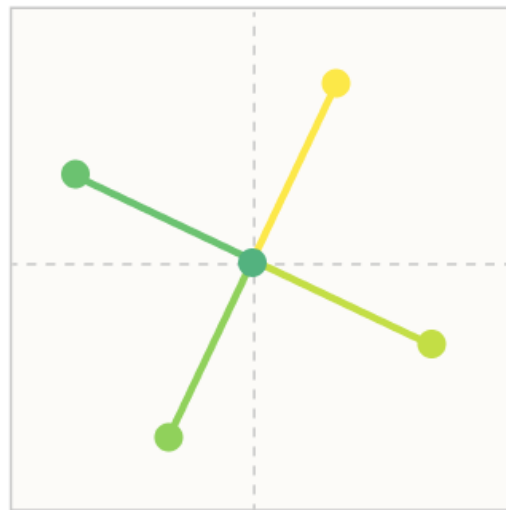
**As Sparsity Increases, Models Use “Superposition” To Represent More Features Than Dimensions**

Increasing Feature Sparsity →



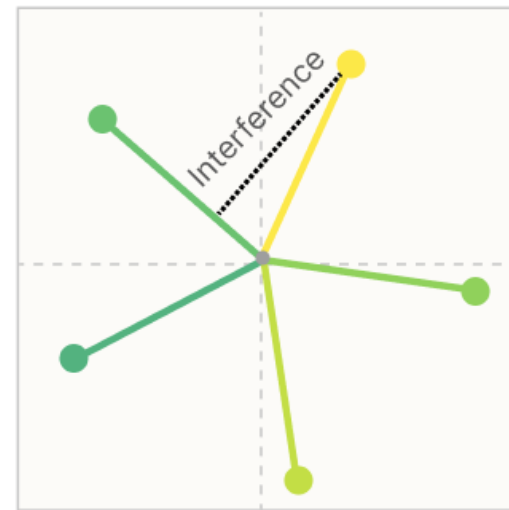
## 0% Sparsity

The two most important features are given **dedicated orthogonal dimensions**, while other features are **not embedded**.



## 80% Sparsity

The four most important features are represented as **antipodal pairs**. The least important features are **not embedded**.



## 90% Sparsity

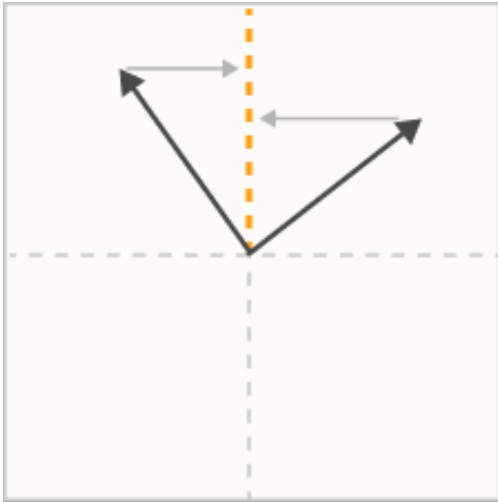
All five features are embedded **as a pentagon**, but there is now “positive interference.”

### Feature Importance

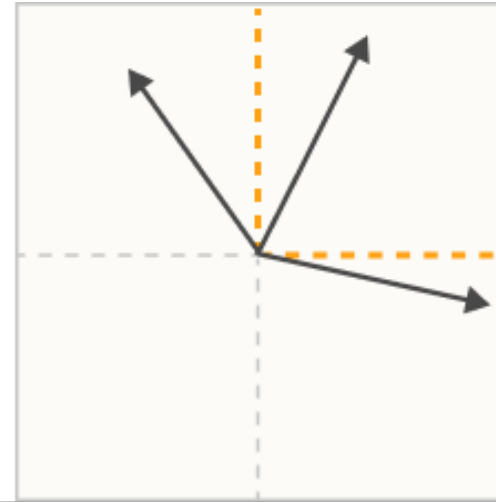
- Most important
- Medium important
- Least important

# Polysemanticity

One neuron, many concepts. The unit of meaning is a direction, not a neuron.



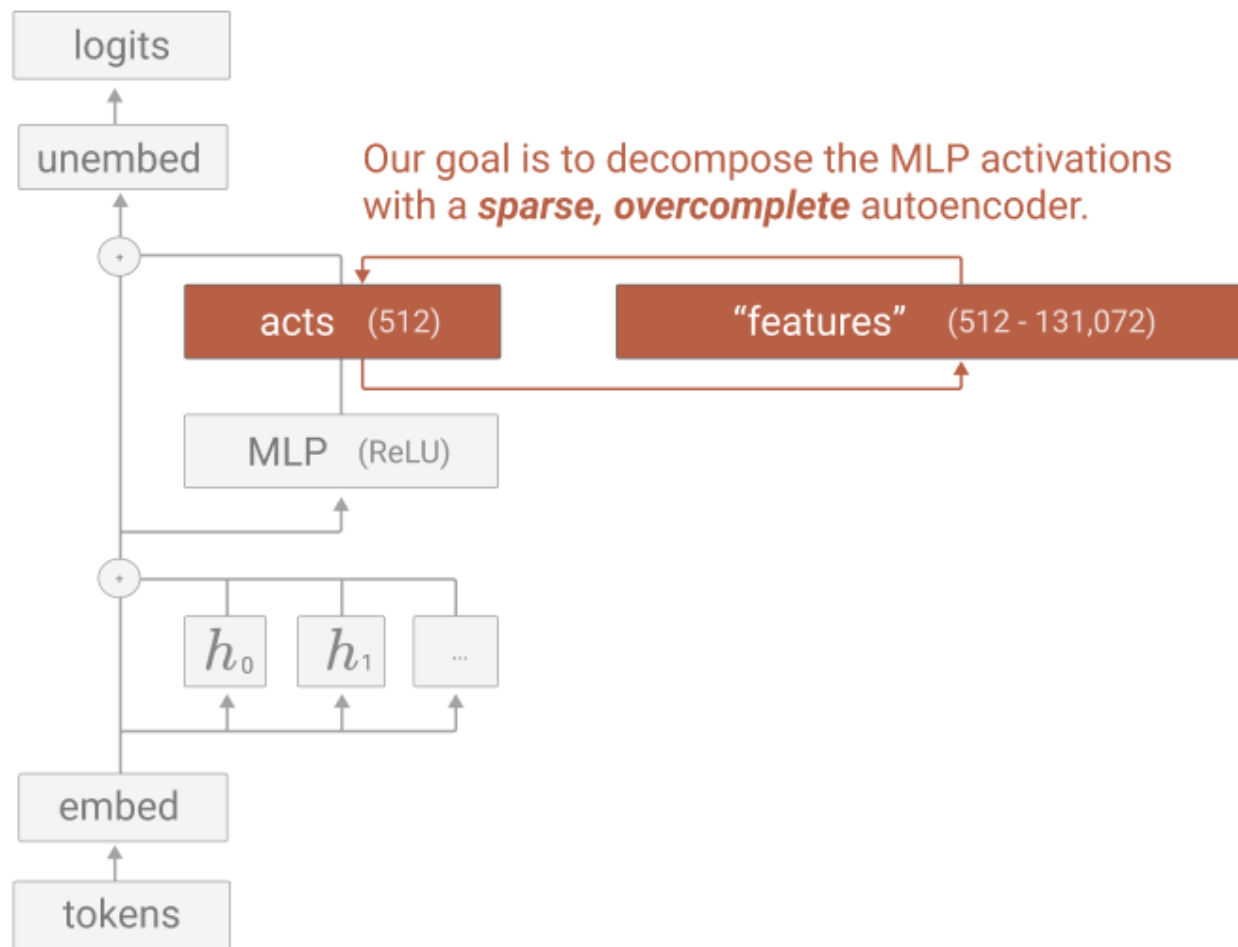
**Polysemanticity** is what we'd expect to observe if features were not aligned with a neuron, despite incentives to align with the privileged basis.



In the **superposition hypothesis**, features can't align with the basis because the model embeds more features than there are neurons. Polysemanticity is inevitable if this happens.

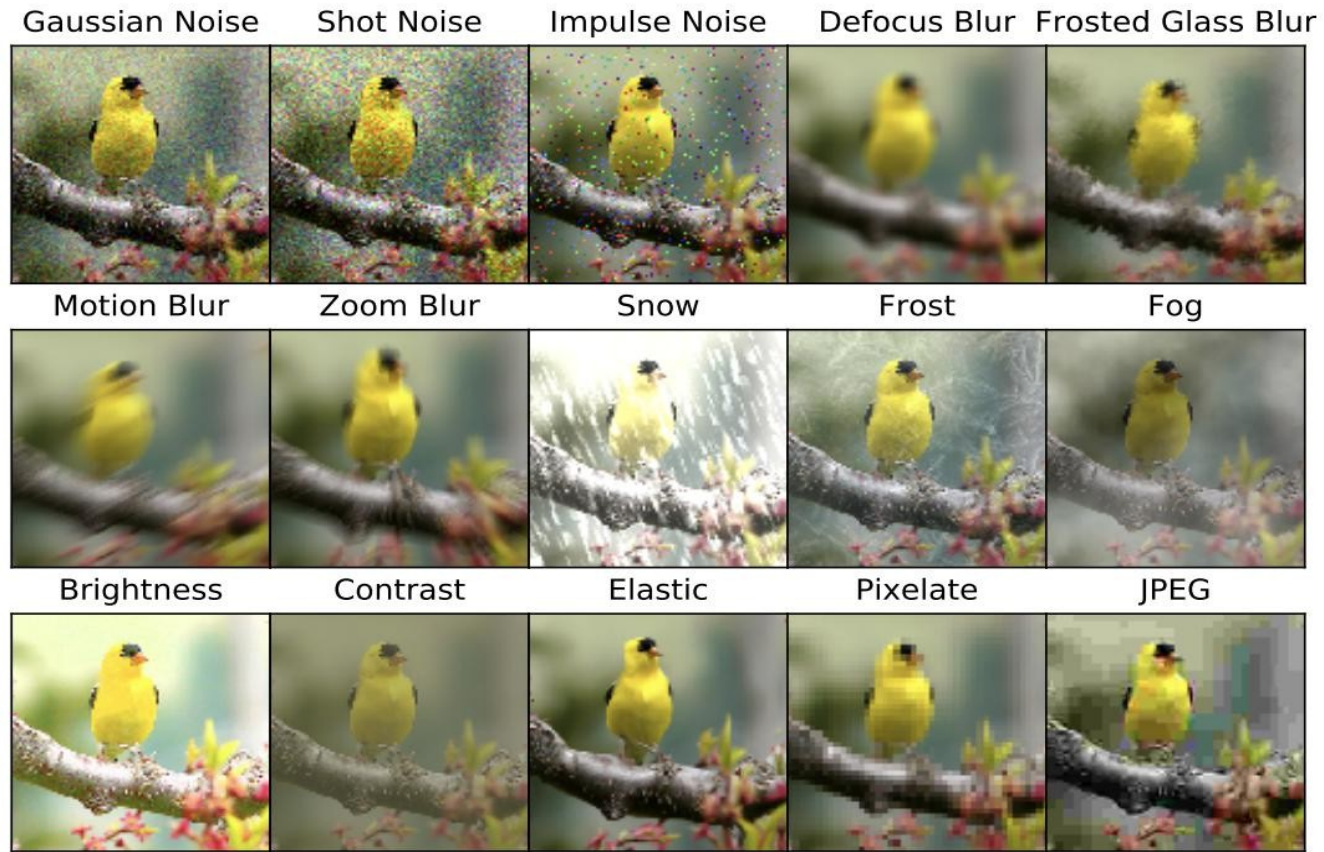
# Sparse autoencoders

Decompose activations into a sparse dictionary of mostly-monosemantic features.



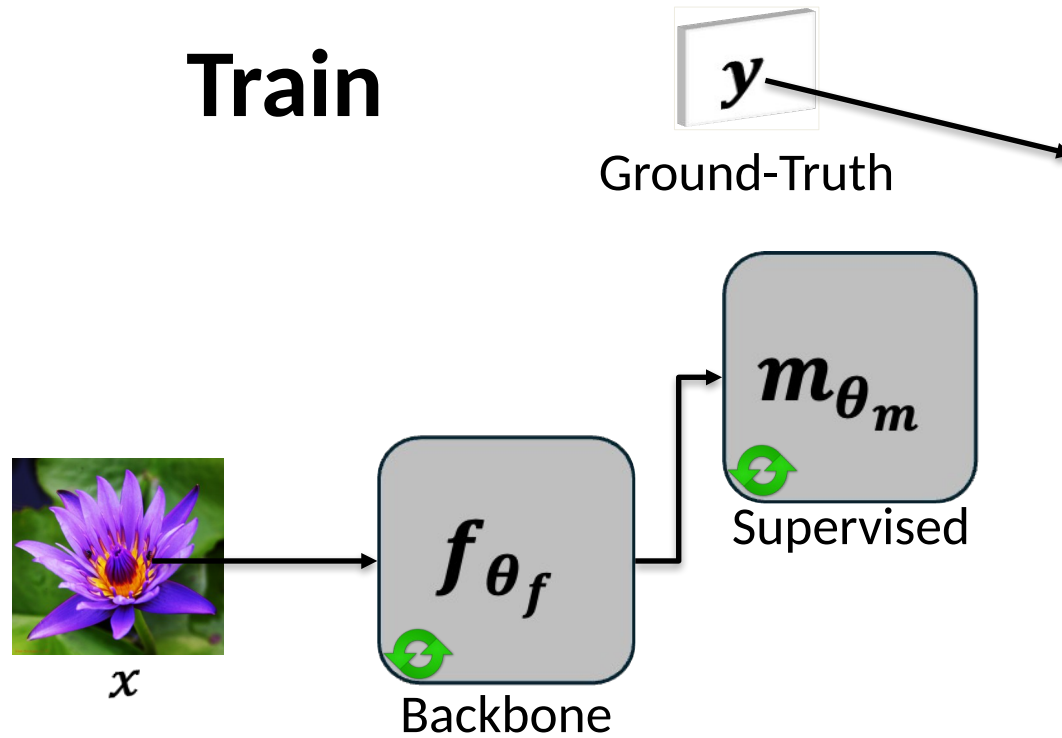
# OOD Test Data

## Corrupted images

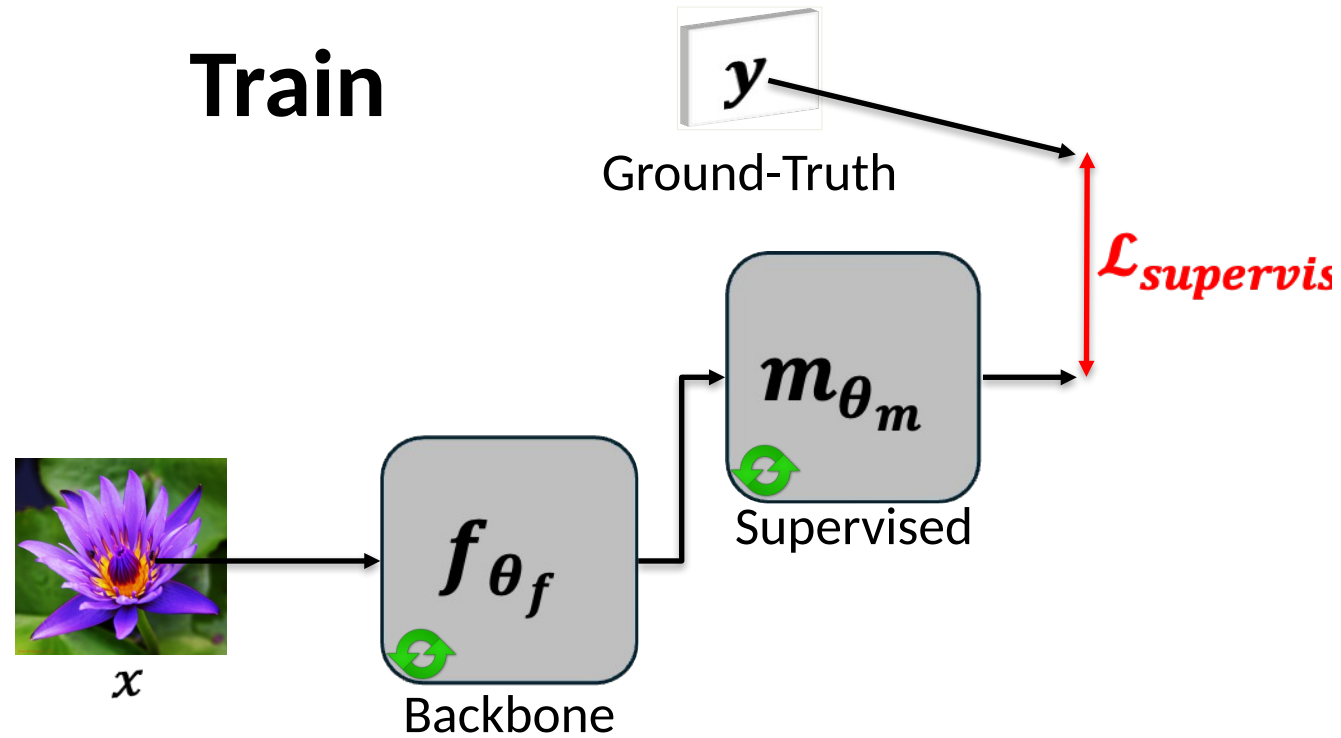




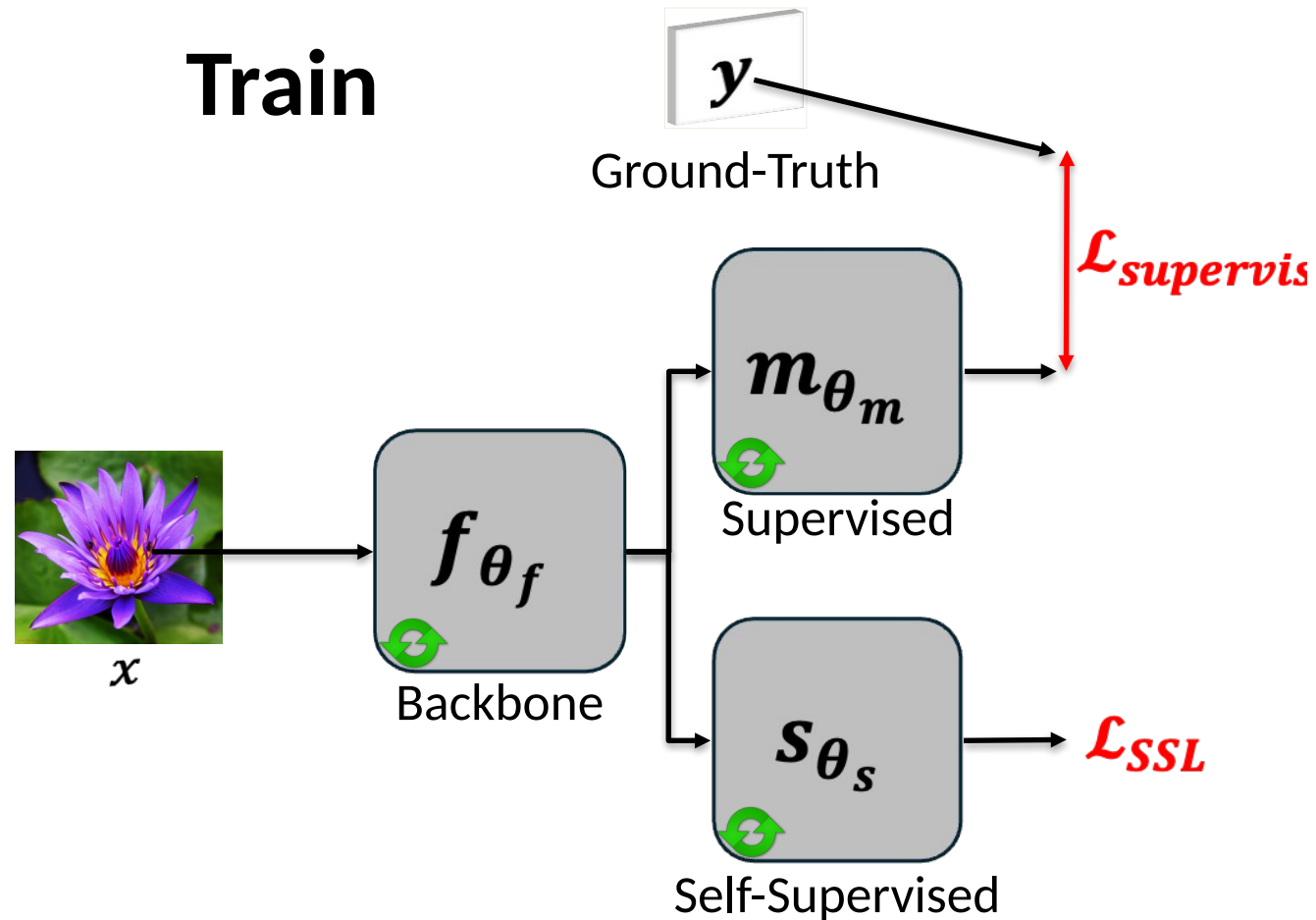
# Test-Time Training (TTT)



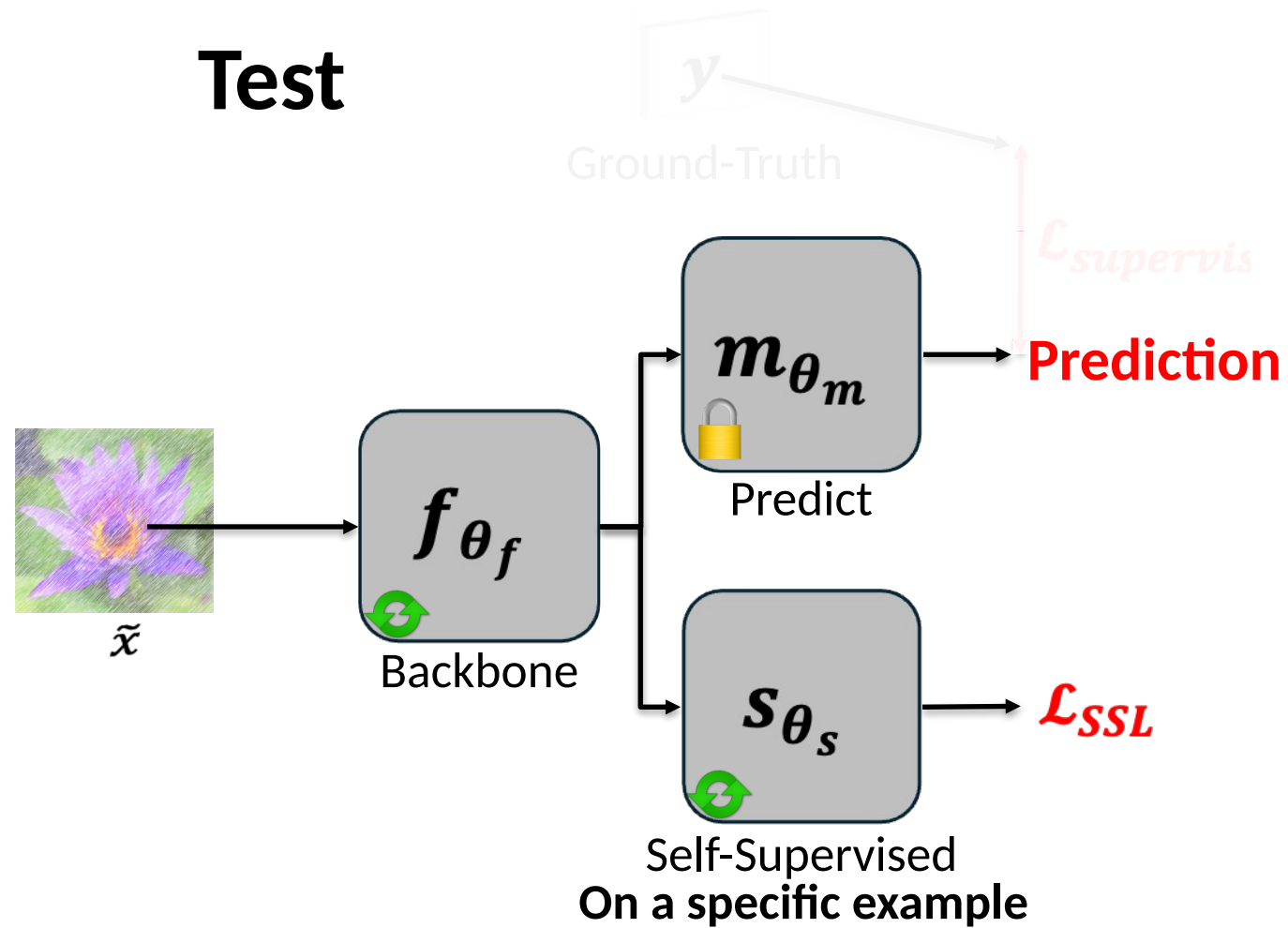
# Test-Time Training (TTT)



# Test-Time Training (TTT)



# Test-Time Training (TTT)



# Thank you!

**nature**

EDITORIAL | 22 October 2025

**What makes PhD students happy?  
Good supervision**

How I was supervised:

